

Do Global Stocks Outperform US Treasury Bills?*

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Abstract

We study compound returns to nearly 62,000 global common stocks during the 1990 to 2018 period, documenting that the majority, 56% of US stocks and 61% of non-US stocks, underperform one-month US Treasury bills over the full sample. Focusing on aggregate shareholder wealth creation measured in US dollars, we find that the top-performing 1.3% of firms account for the \$US 44.7 trillion in global stock market wealth creation from 1990 to 2018. Outside the US, less than one percent of firms account for the \$US 16.0 trillion in net wealth creation. These results highlight the practical implications of the fact that the distribution of long-run stock returns is strongly positively skewed.

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1. Introduction

Finance theory implies that assets subject to positive systematic risk should earn a risk premium in the form of an expected or mean rate of return that exceeds the return to risk-free investments. However, it has been documented that stock returns are positively skewed, particularly when returns are compounded over longer time horizons.¹ Positive skewness in a distribution most often implies that the mean of the distribution exceeds the median, and that the mean can therefore be misleading as an indication of outcomes to more typical observations. The prior work focusing on stock return skewness has mainly studied US markets. Here, we examine compound return data for nearly 62,000 global common stocks traded on public markets over the 1990 to 2018 period. Our key finding is that while the cross-sectional mean stock return is indeed positive in all 42 countries we examine, returns to the majority of global stocks fall short of returns to one-month US Treasury bills over matched time horizons.²

We primarily focus on long-horizon stock returns measured by compounding monthly returns inclusive of dividends. The resulting “buy-and-hold” return represents the outcome to a hypothetical investor who makes an initial investment at the beginning of the sample period, but otherwise does not trade except to reinvest dividends in additional shares of the same stock. We find that only 40.5% of global common stocks, including 43.7% of US stocks and 39.3% of non-US stocks, have full-sample buy-and-hold return that exceeds the accumulated return to one-month US Treasury Bills over matched time horizons.

While buy-and-hold returns are simple to interpret and represent outcomes to a viable trading strategy that can be followed by some investors, they do not measure the experience of stock investors in

¹ See, for example, Farago and Hjalmarsson (2019), Bessembinder (2018), Fama and French (2018), Albuquerque (2012), Heaton, Poulson and Witte (2017), and Simkowitz and Beedles (1978).

² We focus on returns measured in US dollars to provide a common yardstick that can be compared across stocks traded in currencies with differing inflation rates. The comparison to the short-term US Treasury rate reflects the fact that this is often viewed as the best available proxy for the “risk-free” interest rate envisioned by theorists.

aggregate. Since secondary market trading is always zero-sum across buyer and seller, investors in aggregate do not reinvest dividends through their secondary market trading. Further, investors in aggregate do fund new equity issuances and receive the proceeds of stock repurchases. We therefore also measure returns to shareholders that include net equity issuances and dividends, without reinvestment in stocks. This aggregate return measure has been referred to (e.g., Dichev, 2007) as the “dollar-weighted” return, and is computed as the internal rate of return (IRR) of the cash flows paid or received by investors in aggregate. We find that the dollar-weighted return exceeds the Treasury-bill return over the matched time horizon for an even lower proportion of stocks, 39.1% globally and 37.5% for non-US stocks.

The finding that most stocks underperform US Treasury bills does not contradict the evidence (see, for example, Dimson, Marsh, and Staunton, 2002) that returns to broad stock *markets* handily outperform returns on Treasury instruments. We document cross-stock *mean* buy-and-hold returns in our global sample of 0.96% at the monthly horizon, 14.11% at the annual horizon, 95.62% at the decade horizon, and 260.42% for the full sample period, each of which greatly exceeds the Treasury bill return at the matched horizon. Indeed, by our calculations, stocks traded in the public global markets created over \$US 44.7 trillion in shareholder wealth during the twenty-nine-year sample period, over and above outcomes that would have prevailed if the invested capital had earned US Treasury-bill returns. The distinction between the positive return premium (in excess of Treasury bills) for the overall stock markets and the negative premium for most individual stock returns is attributable to strong positive skewness in returns to individual stocks, particularly at longer horizons. Simply put, the positive mean excess return for the broad stock market is driven by very large returns to a relative few stocks, not by positive excess returns to typical stocks.

To further assess the importance of the positive skewness in the distribution of global stock returns, we assess the extent to which stock market wealth creation is concentrated. To do so, we measure wealth creation for each stock as the difference between the December 2018 US dollar value of investors’ actual investments in the stock and the value that would have obtained if the same capital

investments had earned one-month US Treasury bill returns. Summing across the 61,100 firms that issued common stock contained in the January 1990 to December 2018 sample, we calculate net global stock market wealth creation of \$US 44.74 trillion, measured as of December 2018. Focusing only on the 23,905 firms with positive value creation, we calculate gross stock market wealth creation of \$US 66.57 trillion dollars. That is, the majority (37,195, or 60.9%) of firms generated negative wealth for their investors, reducing full sample wealth by \$US 21.83 trillion.

We show that the five firms (0.008% of the total) with the largest wealth creation during the January 1990 to December 2018 period (Apple, Microsoft, Amazon, Alphabet, and Exxon Mobile) accounted for 8.27% of global net wealth creation and 5.56% of global gross wealth creation. The best performing 306 firms (0.5% of total) accounted for 73.03% of global net wealth creation and 49.08% of global gross wealth creation. The best performing 811 firms (1.33% of total) accounted for all net global wealth creation, and 67.20% of gross global wealth creation.

Bessembinder (2018) studied data from the CRSP database (which focuses on US stock markets) pertaining to the 1926 to 2016 period, documenting that most US-traded common stocks provided long-term buy-and-hold returns less than Treasury-bill returns, and showing wealth creation among US common stocks to be quite concentrated. Except for Norang and Agustsson (2018), a Masters' thesis focused on Norwegian stocks, researchers have not to our knowledge assessed whether the key results of his study hold for non-US stock markets. Here, we show that Bessembinder's results are not US-centric, and in fact are stronger outside the US. The present sample includes 44,476 non-US stocks over the 1990 to 2018 period. Of these, only 39.3% generated buy-and-hold returns measured in US dollars that exceed one-month US Treasury bill returns over matched horizons. Buy-and-hold returns to non-US developed market stocks more frequently (40.0%) outperform US Treasury bills as compared to emerging market stocks (37.6%). The percentage of stocks with buy-and-hold returns that exceed one-month US Treasury bills is greater than fifty in only seven of the forty-two sample countries.

Outside the US, the 111 (0.25% of total) top-performing firms accounted for 34.18% of gross wealth creation and over two thirds (69.20%) of net wealth creation. The 221 top-performing non-US

firms (0.50% of total) accounted for 46.32% of gross non-US wealth creation and over nine tenths (93.79%) of net non-US wealth creation. The 441 top-performing non-US firms (1% of total) accounted for 59.94% of gross and 121.37% of net wealth creation by non-US firms. That is, less than one percent of non-US firms in the sample accounted for all net stock market wealth creation outside the US during the January 1990 to December 2018 sample period.

We assess empirically the determinants of cross-country variation in the proportion of stocks that outperform US Treasury bills and in the degree to which wealth creation is concentrated in a relative few stocks. Perhaps surprisingly, the degree to which monthly returns to individual stocks in a country are skewed has limited explanatory power for these outcomes. Rather, the average standard deviation of monthly returns to the individual stocks in a country has significant explanatory power. Higher stock return volatility is associated with a lower proportion of stocks outperforming Treasury bills and greater concentration of wealth creation. These results are consistent with the simulations reported by Bessembinder (2018) and the theoretical predictions of Farago and Hjalmarsson (2019) that the positive skewness of long horizon stock returns is primarily attributable to the effects of compounding, and that long horizon skewness will be greater if short horizon returns are more volatile.

The results reported here are important from a number of perspectives. Most empirical analyses of stock markets consider unconditional or (by use of regression specifications) conditional arithmetic means of returns measured over short (e.g., monthly) horizons. However, the investment horizons of individuals or fund managers (particularly pension funds) can stretch to decades and differ across investors. The results here show that the properties of stock returns compounded over long horizons differ substantially from those of short horizon returns, particularly in terms of the degree of return skewness and the proportion of returns that exceed various benchmarks. These results are even more pronounced for non-US as compared to US stocks. Investors may rationally prefer the positive skewness that is present in compound returns but is obscured by a focus on short horizon returns.

The results also raise questions regarding the evaluation of portfolio and investment manager performance. Two frequently used measures include versions of Jensen's alpha and the Sharpe ratio,

typically implemented using monthly returns. The former is a conditional arithmetic mean, while the latter is the ratio of the arithmetic mean return to the variance. As such, neither allows for the fact that the arithmetic mean return observed for a given sample overstate outcomes to buy-and-hold investors (since for any given stock or portfolio it exceeds the geometric mean, which when compounded, gives the actual buy-and-hold return in the same sample) or the empirical fact that stock returns are positively skewed, especially at longer horizons.

The results are also relevant to the debate regarding active stock selection vs. the passive holding of broadly diversified stock indices. The results here show that the wealth created by stock market investing is largely attributable to positive outcomes to a relatively few stocks. For those investors without a comparative advantage in identifying the few stocks that will create the most wealth (or in selecting a manager with the ability to do so) and without a substantial preference for positive skewness, the results reinforce the desirability of investing in a broad passive index. On the other hand, for the (presumably few) investors with the appropriate comparative advantage, the results highlight the degree to which successful stock selection can enhance wealth.

Finally, the finding that net wealth creation over a twenty-nine-year period is concentrated in only 1.3% of firms (less than one percent of firms outside the US) presents a challenge to capital market and industrial organization theorists. Can existing models of firm dynamics, including entry, growth, mergers, competition, and failure account for the observed degree of concentration in wealth creation, or are new models required?

2. Sample and Measures Employed

2.1 Data sources and sample overview

We identify securities as common stocks using methods described in detail in the Data Appendix. The data required to compute monthly returns, market capitalization, and trading volume for US stocks is obtained from CRSP, and for non-US stocks from the Compustat Global and Compustat North America

databases. Our study includes forty-one countries, comprised of the forty countries with the largest average GDP during the sample interval, except that we exclude Iran since return data is available for only ten years, and include Singapore and New Zealand due to their relative economic prominence. Many common stocks are listed and traded in more than one country. To avoid double counting, we assign each common stock to a single “home” country, as described more fully in the Data Appendix.

Our sample includes twenty-five developed and sixteen developing economies. In addition, we compute outcomes for 195 firms that are traded in the US as American Depository Receipts (ADRs), but do not have during the period of ADR trading publicly listed primary common stock data in Compustat.³ We categorize these “homeless” ADRs as a separate country, and hence refer to outcomes across forty-two countries. The countries included in the sample represent approximately 87% of global stock market capitalization as of the end of 2018.

We begin our study as of January 1990 (as Compustat coverage is thin prior to this date) or at the first date when monthly return data is available and end the study at December 2018.⁴ The CRSP and Compustat data pertains to publicly listed stocks. Our study should therefore be viewed as summarizing return outcomes and wealth creation in the publicly accessible stock markets. We do not capture the pre-IPO experience of private (e.g., venture capital, private equity, and founder) investors, or returns from the IPO price to the first end-of-month price contained in the databases. To ensure a common yardstick for firms traded in multiple currencies, returns, market capitalizations, and trading volumes for non-US stocks are all converted to US dollars. We exclude stocks listed on minor stock exchanges, where an

³ Prominent examples include Alibaba Group and Baidu, Inc., Chinese firms whose only public trading is through ADRs.

⁴ While the twenty-nine-year sample we employ is very informative regarding the positive skewness of stock returns and the implications thereof, some results are necessarily sample specific. For example, the Nikkei Index of Japanese stocks reached its all-time high on December 29, 1989, immediately prior to our sample period; measures of long-term returns to many Japanese stocks would have been more favorable had our sample started at an earlier date.

exchange is deemed to be minor if its share of own-country trading volume (measured in US dollars) during the sample period is less than two percent.

Stocks are tracked through time based on the CRSP PERMNO variable (for US stocks) and the Compustat GVKEY and IID variables (for non-US stocks). We compute returns separately by share class for firms with multiple classes. We also compute separate return series for Chinese stocks that are traded in Hong Kong as H-shares and in China as A-shares. However, dollar wealth creation is aggregated to the firm level by summing across share classes, based on the PERMCO variable for firms contained in the CRSP database and the Compustat GVKEY variable for other firms.

Visual examination indicates that the data for non-US stocks contain occasional but substantial data errors. Prior authors have addressed this problem by either excluding or Winsorizing extreme observations.⁵ While these methods may be adequate for studies that consider returns to value-weighted portfolios, our focus is on the distribution of long horizon returns to a broad cross-section of individual stocks. While we also eliminate from the sample some observations that are likely to reflect potentially influential errors, we attempt to retain large but accurate observations and to repair some data errors, e.g., those that result from an erroneous temporary shift of the decimal. Our filtering and correction algorithms are described in the Data Appendix. After implementing these filters the sample contains 7.68 million monthly observations on 61,981 stocks issued by 61,100 firms. Among the monthly data retained in the sample, slightly less than one percent of the firm/months contain an observation (return or market capitalization) that has been corrected by our algorithms.

Table 1 lists the countries included in the study, along with descriptive statistics. The number of stocks included in the sample ranges from sixty-two in Columbia to 17,310 in the US. Data is available from January 1990 for most countries, but the earliest data pertains to 1991 for China, 1993 for Brazil, Nigeria, and Poland, 1994 for Israel, 1995 for Russia, 2000 for Saudi Arabia, and 2001 for the United

⁵ See, for example, Chui, Titman, and Wei (2010), Hou, Karolyi, and Kho (2011), and Fama and French (2017).

Arab Emirates. GDP per capita as of 2018 for sample countries ranges from \$US 2,036 for India to \$US 82,950 for Switzerland, and the average GDP growth rate for sample countries ranges from 0.76% per year for Italy to 9.61% for China. End-of-period market capitalization for sample stocks ranges from \$US 27 billion for Nigeria to \$US 26.6 trillion for the US. Finally, an indication of the importance of stock markets in each sample country, the ratio of average market capitalization to GDP, ranges from 0.07 for Turkey to 6.08 for Hong Kong.⁶

The Compustat data upon which we rely for non-US stock returns does not include information regarding post-delisting share values or post-delisting payments to shareholders. Following Shumway (1997), we set the final return on non-US stocks with an incomplete return series, as well as stocks indicated to be delisted for reasons of bankruptcy or liquidation, to -30%. For US stocks we incorporate CRSP delisting returns where available, while setting the final return to -30% in the few cases where the delisting return is missing and the CRSP delisting code is 500, 520, 551-573, 580, 574, or 584.

Table 2 provides information regarding the stock exchanges studied in each country. For example, the US sample includes 3,113 New York Stock Exchange stocks, 1,551 American Stock Exchange Stocks, and 12,646 NASDAQ stocks, while the China sample includes 1,506 stocks listed on the Shanghai Stock Exchange and 2,192 listed on the Shenzhen Stock Exchange.⁷ Table 2 also reports the percentage of dollar trading volume that occurs on each exchange within each sample country.⁸

2.2 Measuring stock return performance

Let $R_t = R_{ct} + R_{dt}$ denote the time t return to shareholders, where R_{ct} is the capital gain component of the period t return, and R_{dt} is the dividend component. The buy-and-hold return, inclusive

⁶ Hong Kong (6.09) and Singapore (1.65) are exceptional, with many large firms listed on their exchanges. A number of large Chinese firms in particular are listed in Hong Kong, and five members of the Jardine Group, which is headquartered in Hong Kong, shifted from Hong Kong to Singapore in 1994 (Chan, Hameed, and Lau, 2003). Prior to the change in listing, Jardine comprised about 10% of the total market capitalization in Hong Kong.

⁷ Firms can change exchange listings over time. The counts reported are based on the exchange where the firm initially appears.

⁸ Percentages can sum to less than 100% for a country, since minor exchanges are excluded from the study.

of reinvested dividends, from time 0 to T is simply $BHR_t = (1 + R_1) \times (1 + R_2) \dots \times (1 + R_T) - 1$.

This represents the accumulated value per unit of initial investment to a hypothetical investor who purchased shares at time 0 and did not transact thereafter, except to reinvest dividends in additional shares. We also compute for each stock the geometric mean return inclusive of dividends, which for a sample of T returns is $GM = (1 + BHR)^{1/T} - 1$, as well as the arithmetic mean return inclusive of dividends, which is simply the sum of the T returns divided by T.

While the buy-and-hold return is a simple and intuitive measure, in part because the buy-and-hold strategy is straightforward to implement, it is not representative of the experience of shareholders in aggregate. Since secondary market transactions are always zero-sum across the buyer and the seller, shareholders in aggregate cannot use secondary market trading to reinvest dividends in additional shares of stock. Further, shareholders in aggregate receive the proceeds of any share repurchases and fund any new stock issuances.

We assess outcomes to shareholders in aggregate in two ways. First, we compute the Internal Rate of Return (IRR) to all shareholders. Second, we compute the dollar amount of wealth created for shareholders in aggregate resulting from their investing capital in stocks as opposed to Treasury bills. We do so using the following framework. Let I_t denote the time t value of a firm's common stock, D_t denote the firm's time t dividend payment to shareholders, and F_t denote the firm's net share repurchase at time t (with new equity issuances represented by $F_t < 0$). Dividends are $D_t = I_{t-1} \times R_{dt}$, while the value of common stock evolves according to $I_t = I_{t-1} \times (1 + R_{ct}) - F_t$.

The time series of cash flows experienced by investors in aggregate from time 0 to time T can be expressed as follows.⁹ The initial acquisition of shares results in $CF_0 = -I_0$. Interim cash flows,

⁹ Our calculations necessarily begin with the earliest return and market capitalization data available in the database. We therefore do not capture outcomes to shareholders that occur prior to the stock's initial public offering, or after the IPO but before the date of the first monthly observation in the database. A dramatic example of wealth creation that occurred prior to the first monthly observation contained in our data is Petro China. The New York Times, in an article dated November 5, 2007 and titled, "Petro China shares triple value in record IPO", reports that Petro China "became the world's first company to pass \$1 trillion in market capitalization when it debuted on the Shanghai Stock Exchange" and that "...it trebled in value on its opening day and surpassed U.S. energy behemoth

from $t = 1$ to $T - 1$, comprised of dividends and net share repurchases, are $CF_t = D_t + F_t = I_{t-1}(1 + R_t) - I_t$. The time T cash flow is the sum of the final dividend, any net repurchase, and the liquidation value of the final position, $CF_T = D_T + F_T + I_T = I_{T-1}(1 + R_T)$. The compound return per period to shareholders in aggregate, sometimes (e.g., Dichev, 2007) referred to as the “dollar-weighted” return, is obtained for each stock as the internal rate of return (IRR) of this series of cash flows.¹⁰

To assess the dollar amount of wealth created, let R_{ft} denote the time t return on a one-month Treasury bill, and let $FV_{t,T} = (1 + R_{ft+1}) \times (1 + R_{ft+2}) \times (1 + R_{ft+3}) \times \dots \times (1 + R_{fT})$ denote an interest accumulation factor obtained by compounding forward from time t to time T at prevailing one-month Treasury interest rates.¹¹ We focus on the time T equivalent, i.e., the end-of-sample “future value” of the cash flows to shareholders in aggregate. The time T value of the initial cash flow is $CF_0 \times FV_{0,T} = -I_0(1 + R_{f1})FV_{1,T}$. The time T value of cash flows from $t = 1, \dots, T-1$ can be stated as $CF_t \times FV_{t,T} = I_{t-1}(1 + R_t)FV_{t,T} - I_t(1 + R_{ft+1})FV_{t+1,T}$, while the time T value of the final cash flow is simply the cash flow itself, $I_{T-1}(1 + R_T)$. Summing the accumulated values of cash flows to investors in aggregate across t , we obtain:

$$NFV = \sum_{t=0}^T CF_t \times FV_{t,T} = I_0(R_1 - R_{f1})FV_{1,T} + I_1(R_2 - R_{f2})FV_{2,T} + \dots + I_{T-2}(R_{T-1} - R_{fT-1})FV_{T-1,T} + I_{T-1}(R_T - R_{fT}). \quad (1)$$

Expression (1) is identical to the expression for net wealth creation in Bessembinder (2018), and can be interpreted as the difference between the end-of-sample value of shareholders’ actual investment in common stock and the value that would have obtained had the same capital investment earned Treasury bill returns. The analysis here shows that net wealth creation for each common stock is also, borrowing

Exxon Mobil as the world's most valuable company.... At its intra-day high in Shanghai trading, it was valued at almost \$1.2 trillion.”

¹⁰ We rely on the IRR function from the MATLAB Financial Toolbox. The IRR existed and was unique for all but two firms, where all cash flows to investors were negative. We set the IRR for these firms to -100%.

¹¹ The future value factor is defined based on the one-month Treasury returns because the Treasury rate comprises the opportunity cost of capital in this calculation.

from the terminology of corporate finance, simply the Net Future Value (NFV) of the series of cash flows to investors in aggregate. Since it seems natural to measure aggregate investor experience at the level of firms rather than stock issues, we sum cash flows across stock issues for those firms that issued more than one class of common stock. While we refer to wealth creation, all measures constructed for this paper are based on realized stock market outcomes. To the extent that stock markets are informationally efficient, enhancements to shareholder wealth attributable to corporate operating and financial decisions can be measured based on stock market outcomes.

To implement expression (1), we measure aggregate shareholders' investment, I , for each stock/month based on the stock's prior-month market capitalization. In those cases where the shares of one sample stock are owned by another publicly traded company whose stock is also included in the sample, the sum of market capitalization across stocks exceeds the actual combined investment by shareholders. As a consequence, we may double count wealth creation to some degree. For US stocks, the degree of double counting is minor, as Duchin, Gilbert, Harford, and Hrdlicka (2017) document that S&P 500 firms in aggregate hold equity investments amounting to only 0.30% of the market value of their own equity. However, the degree of double counting could be greater in some foreign markets.¹² This may in particular be the case for the Asian markets, where cross-holdings are prominent among large conglomerates such as Samsung Group in South Korea and CK Hutchison Holdings Group in Hong Kong (see, for example, Claessens, Djankov, Fan, and Lang, 2002).

3 Rates of Return to Global Common Stocks

Table 3 reports on monthly returns to sample common stocks, as well as buy-and-hold returns computed at the annual, decade, and full-sample horizons. For each horizon, we report returns for the full (global) sample, for the global sample excluding US stocks, for developed and developing country

¹² Note that the double counting issue arises only due to the ownership of equity in sample stocks by other companies also included in the sample. Holdings of multiple companies by a given non-corporate shareholder do not lead to double counting. Further, buy-and-hold returns are unaffected by this issue.

subsamples, for the developed market subsample excluding US stocks, and across the North America (US and Canada), Europe, and Asia-Pacific regions. Results for the full January 1990 to December 2018 period are also reported for each of the forty-two individual countries.

3.1 Monthly returns

The sample includes 7.68 million monthly returns on the 61,981 sample stocks, reflecting that stocks are included in the database for an average of 10.3 years. Figure 1 displays the frequency distribution of monthly returns (rounded to 1%, to a maximum of 200%), separately for US and non-US stocks.¹³

Panel A of Table 3 shows that the full sample mean monthly return is 0.96%. The sample mean monthly return varies from 0.67% per month for European stocks to 1.16% per month for North American stocks. In contrast to the positive mean, the median monthly return is -0.01% for the full sample and zero (to four decimal places) for the developed economies and North American subsamples. The median monthly return ranges from -0.27% per month for the Asia Pacific region to 0.02% for the Europe region. The percentage of monthly returns that exceed zero is 49.1% for the full sample, and ranges in subsamples from 48.6% in the Asia Pacific region to 50.1% in Europe.

The facts that (i) the median monthly return is approximately zero even while the mean monthly return is positive and (ii) less than half of sample monthly returns are positive can be attributed to the substantial positive skewness in monthly returns. The standardized skewness coefficient is 8.22 for the full sample, and in subsamples ranges from 6.41 for the Europe region to 8.55 for the full sample excluding the US and 9.22 for the North American region. By comparison, Bessembinder (2018) reports a skewness coefficient of 6.96 for monthly US stock returns during the 1926 to 2016 period. The data therefore indicates somewhat greater skewness in the monthly returns to international stocks in the recent

¹³ A notable feature of the distribution of monthly returns to US stocks is the peak at zero, which is presumably attributable to non-trading and price rounding. For non-US stocks the peak at zero is less notable, which reflects that a zero return in local currency may not equate, even with rounding, to a zero return in US dollars.

January 1990 to December 2018 sample as compared to the 1926 to 2016 sample of US stocks studied by Bessembinder (2018).

The data shows that global common stocks most often do not outperform one-month US Treasury bills at the monthly horizon. For the global sample, 48.4% of monthly common stock returns exceed the US Treasury interest rate. The percentage of stocks that outperform Treasury bills ranges from 47.8% in the Asia-Pacific region to 49.0% in Europe.

That fact that stock returns are positively skewed also manifests itself in the fact that most individual stocks' returns are lower than the mean return computed across stocks. For each month we compute the cross-sectional average stock return, weighted by firm values (market capitalization in dollars) as of the end of the prior month. The right column of Table 3 Panel A reports on the percentage of individual stock returns that exceed the value-weighted average stock return in the same month, which is 46.2% for the full sample. This percentage ranges from 45.2% for the Asia-Pacific region to 47.6% for North America.

3.2 Annual and decade buy-and-hold returns

Panels B and C of Table 3 report on buy-and-hold returns computed over annual and decade horizons, respectively.¹⁴ In those cases where a stock enters or departs the dataset within a calendar year or decade, the return is computed based on the partial year or partial decade when data is available, thereby avoiding survivorship bias.

Figure 2 displays the frequency distribution of annual buy-and-hold returns, rounded to the nearest one percent and to a maximum of 400% (i.e., to a maximum gross return of five times the initial investment). Figure 3 displays the frequency distribution of decade buy-and-hold returns, rounded to the nearest five percent and to a maximum of 900% (i.e., to a maximum gross return of ten times the initial

¹⁴ We define decades as January 1990 to December 1999, January 2000 to December 2009, and January 2010 to December 2018.

investment). The contrast between Figures 2 and 3 is notable. While Figure 2 gives some impression of the positive skewness in annual returns, e.g. by the existence of numerous outcomes that exceed 100%, the most frequently observed annual returns are clustered in the vicinity of zero. In contrast, on Figure 3 the most frequently observed returns for both US and non-US stocks at the decade horizon (rounded to 5%) are -95% and -100%. Observed frequencies of decade horizon returns decline almost monotonically for higher returns, but for a small increase at zero.¹⁵

The data in Table 3 verifies that buy-and-hold returns at the annual and decade horizons are more positively skewed than monthly returns. In particular, the standardized skewness coefficient for decade returns (Panel C) in the global sample is 68.20, compared to 18.55 at the annual horizon (Panel B) and 8.22 at the monthly horizon (Panel A). This result is consistent with the implications of the simulations reported by Bessembinder (2018), which show that compounding induces greater skewness in buy-and-hold returns over longer time intervals.

The mean buy-and-hold return across all global stocks increases with return horizon, from 0.96% at the monthly horizon to 14.11% at the annual horizon and 95.62% the decade horizon. However, reflecting the positive skewness, full-sample median returns are far lower: -0.01% at the monthly horizon, 1.27% at the annual horizon, and -2.16% at the decade horizon. The percentage of stocks in the full sample that generate a buy-and-hold return that exceeds the return to the one-month US Treasury bill over the same period is 48.4% in monthly returns, 49.4% in annual returns, and 45.1% in decade returns. Within the decade-horizon results, the percentage of stocks with returns that outperform Treasury bills ranges from 41.5% for the emerging economy subsample to 47.9% for the North America subsample. The percentage of stocks in the full sample that generate buy-and-hold returns that exceed the value-

¹⁵ An interesting feature of the distribution of decade-horizon returns is that outcomes of -85% or less are more frequent for US stocks, while returns that are negative but greater than -80% are more frequent for non-US stocks. This may reflect that the CRSP data includes actual delisting returns for US stocks, while in the absence of such data we impose a -30% delisting return for non-US stocks that delist for negative reasons, as recommended by Shumway (1997).

weighted average stock return over the matched time period is 46.2% in monthly returns, 44.0% in annual returns, and 37.1% in decade returns.

3.3 Full sample buy-and-hold returns

Panel D of Table 3 reports on buy-and-hold returns to global common stocks, based on the full January 1990 to December 2018 sample period. Figure 4 displays the frequency distribution of full-sample buy-and-hold returns (rounded to the nearest 5%, to a maximum of 900% or a gross return of ten times the initial investment). The mean full-sample buy-and-hold return across all 61,981 sample stocks is 260.4%. Notably, however, the median buy-and-hold return for the full sample is -14.9%, and only 45.6% of sample stocks have a positive full-sample buy-and-hold return. Turning to the question posed in the title of this paper, only 40.5% of global common stocks have a full-sample buy-and-hold return that exceeds the return to one-month US Treasury bills over the matched time horizons. Across subsamples, the percentage of individual stocks with buy-and-hold returns that exceed the time-matched one-month US Treasury bill return ranges from 37.6% for emerging market countries to 43.9% for North American stocks. These results reflect the substantial positive skewness in the distribution of full-sample individual common stock returns.

The results described in the previous paragraph reflect the fact that the distribution of individual common stock buy-and-hold returns is positively skewed, with a standardized skewness coefficient across all stocks of 47.92. That is, the positive mean buy-and-hold return for the full sample of stocks is attributable to large returns to a relatively few stocks, while the majority of stocks generate buy-and-hold returns that fall short of returns to one-month Treasury bills. The skewness of returns also manifests itself in the fact that less than one third (31.1%) of individual common stocks have a full-sample buy-and-hold return that exceeds the value-weighted average stock return over the matched time horizon. That is, most stocks underperform their own cross-sectional (value-weighted) average.

Bessembinder (2018) previously reported positive skewness in the distribution of individual CRSP common stock returns over the 1926 to 2016 period. The results reported here for 17,310 US-traded common stocks in the more recent sample are quite similar, as the cross-sectional mean buy-and-hold return of 355.2% greatly exceeds the cross-sectional median, which is -6.4%. Only 43.7% of US common stocks have in the present study have full-sample buy-and-hold returns that outperform one-month US Treasury Bills, and only 37.8% have returns that exceed the value-weighted average return over matched time horizons.

More informative is the evidence on Table 3 indicating (i) that the positive skewness in the distribution of individual common stock returns is a global, not a US-specific, phenomenon and (ii) that the effects of skewness are actually stronger for non-US than for US stocks. Focusing on the 44,476 non-US stocks in the sample, the mean across stocks of the full-sample buy-and-hold return is 221.6%, while the median is -17.0%. Only 44.4% of non-US stocks have positive buy-and-hold returns over the full sample, and only 39.3% have buy-and-hold returns that exceed returns to one-month US Treasury bills. The standardized skewness coefficient for full sample period buy-and-hold returns to non-US common stocks is 57.20, as compared to 37.08 for US stocks. On balance, the evidence supports the conclusion that the positive skewness in non-US common stocks is more pronounced than in US stocks.

Positive skewness is empirically important for common stocks in both developed and emerging economies. The standardized skewness coefficient of full sample buy-and-hold returns is 70.63 for the non-US developed economy stocks, and 37.41 for the emerging economy stocks. Slightly more than one third (37.6%) of emerging economic stocks have full-sample buy-and-hold returns that exceed returns to one-month US Treasury bills, while less than three out of every seven (40.0%) of non-US developed economy stocks have full sample buy-and-hold returns better than one-month US Treasury bills.

Turning to individual countries, the cross-sectional mean buy-and-hold return for the full January 1990 to December 2018 sample period is positive in all forty-two countries, ranging from 20.4% for Greek stocks to 837.9% for Columbian stocks. In contrast, the cross-sectional median buy-and-hold

return is negative, implying negative outcomes for more than half of the individual stocks, in twenty-five of the forty-two countries. The median buy-and-hold return is notably small in Greece (-74.8%), Poland (-51.7%), Australia (-49.5%), Turkey (-49.2%), Nigeria (-46.8%), Indonesia (-41.6%), Russia (-38.6%), Malaysia (-37.6%), and Germany (-37.2%). The divergences between mean and median buy-and-hold returns reflect positive skewness in the return distribution in every country. The standardized skewness coefficient for full-sample buy-and-hold returns ranges from 2.90 in Saudi Arabia to 37.08 in the US.

The effects of positive skewness in the individual stock return distribution can also be observed in the fact that less than half of individual stocks outperform the value-weighted market over their full lifetimes in forty of the forty-two countries, including 16.2% in Turkey, 16.7% in Greece, 19.0% in Japan, 22.2% in China, 27.4% in Australia, and 27.6% in the United Kingdom. In only two countries, Switzerland (50.1%) and Columbia (58.1%), did over half of the individual stocks outperform the value-weighted average stock return.

While the majority of global common stocks have full-sample buy-and-hold returns that fail to match one-month US Treasury returns, this finding does not extend to each individual country. In seven sample countries more than half of individual common stocks outperformed US Treasuries, including 55.0% of Belgian stocks, 59.3% of Finnish stocks, 50.9% of Dutch stocks, 65.6% of Swiss stocks, 55.3% of Israeli stocks, 54.1% of New Zealand stocks, and 67.7% of Columbian stocks. In contrast, only 22.8% of Greek stocks, 30.7% of Italian stocks, 33.2% of Australian stocks, and 33.8% of Japanese stocks had full-sample buy-and-hold returns that exceed those of US Treasury bills. In Section 5 below we assess more systematically the determinants of cross-country variation in the percentage of common stocks that outperform Treasury bills over their lives.

3.4 Average monthly returns based on the full sample: Arithmetic, geometric, and dollar-weighted

Table 3 reports on buy-and-hold returns computed over intervals ranging from monthly to full sample (1990 to 2018). In contrast, most studies of stock market performance focus on the properties of

returns that are computed over short horizons. We report on Table 4 information regarding the distribution of mean monthly returns to individual companies, when the mean for each firm is computed from the full time series, by three distinct methods. First, we compute for each stock the arithmetic mean of the full time series of monthly returns (inclusive of dividends). Academic studies most frequently focus on arithmetic mean monthly returns, either directly or by use of regression specifications that employ monthly returns as the dependent variable.¹⁶ Second, we compute for each stock the geometric mean monthly return for its full time series. (Compounding the geometric return gives the full sample buy-and-hold return for each stock). Finally, we compute for each stock the Internal Rate of Return (IRR) to shareholders in aggregate for each stock. The IRR is referred to by some authors (e.g., Dichev, 2007; Dichev and Yu, 2011) as the “dollar-weighted return”, because it implicitly considers the size of the investment position at the time of each individual return.

Table 4 reports the cross-sectional mean and median of these measures, as well as the percentages that are positive and the percentages that exceed the geometric mean Treasury-bill return over the matched horizon.¹⁷ Each is computed from the full January 1990 to December 2018 time series of monthly returns.

The arithmetic mean return is positive for 71.4%, and exceeds the matched Treasury bill mean return for 68.3%, of the 61,981 stocks in the global sample. The percentage of stocks for which the arithmetic mean return exceeds the matched Treasury bill return is not less than half in any country, ranging from 50.0% in Russia to 79.3% in Switzerland. However, this result reflects the well-known fact that the arithmetic mean return exceeds the geometric mean return for any given security in any sample with positive volatility. Despite the fact that the majority of stocks have arithmetic mean returns that exceed Treasury bill mean returns it remains the case that only a minority (40.5%, Panel D of Table 3) of

¹⁶ Such regression specifications recover conditional arithmetic mean monthly returns.

¹⁷ The percentages of geometric mean returns that are positive and that exceed the matched Treasury-bill return are identical to corresponding percentages for full-sample buy-and-hold returns as reported on Panel D of Table 3, and are reproduced on Table 4 for convenience.

stocks generated buy-and-hold returns that exceeded the accumulated returns to Treasury bills. Note also that for the global sample, the cross-sectional median of the stock-level arithmetic mean returns is 0.90%, which exceeds the cross-sectional mean of 0.42%. That is, the distribution of arithmetic mean returns does not reveal the positive skewness that characterizes the distribution of buy-and-hold returns.

The cross-sectional median of the stock geometric returns for the global sample is -0.17%, while the cross-sectional mean is -1.24%. The finding that the cross-sectional median of the firm geometric mean returns is greater than the cross-sectional mean extends to all but two (Columbia and Saudi Arabia) of the forty-two countries in the sample. This pattern reflects that the cross-sectional distribution of the stock geometric mean returns is negatively skewed. Further, since the buy-and-hold return for each stock is simply the compounded equivalent of the geometric mean, this pattern also implies that the strong positive skewness observed in the distribution of buy-and-hold returns is attributable to the compounding of returns over time.

The dollar-weighted return, computed as the IRR of the cash flows to or from stock investors in aggregate, can for any given stock be greater or less than the geometric mean buy-and-hold return. The IRR will be greater than the geometric mean buy-and-hold return if distributions to shareholders tend to occur prior to low returns or new investments by shareholders tend to occur prior to high returns. The IRR will be less than the geometric mean buy-and-hold return if distributions to shareholders tend to occur prior to high returns or new investments by shareholders tend to occur prior to low returns.

Sears Canada, which filed for bankruptcy in June 2017 and delisted in July 2017, provides an example where the IRR to shareholders in aggregate indicates better performance than the return to a hypothetical buy-and-hold investor. The geometric mean return, assuming reinvestment of dividends in additional shares, on Sears Canada shares during our sample period was -4.51% per year, equating to a buy-and-hold return over the January 1990 to July 2017 period of -71.86%. However, Sears Canada provided significant payments to its shareholders prior to the steepest declines in its share price, including a regular quarterly dividend up until 2005, and substantial share repurchases between 2005 and 2013.

The internal rate of return to Sears Canada's shareholders in aggregate from January 1990 to July 2017 was 7.85% per year. By comparison the geometric mean US Treasury bill return over the same period was 2.78% per year, and by our calculations Sears Canada generated \$US 1.77 billion in shareholder wealth during our sample period.

Vodafone Group, Plc provides an example of the opposite outcome, where the IRR to shareholders in aggregate reflects worse performance as compared to the return to a buy-and-hold investor. In February 2000 Vodafone acquired Mannesmann, a German industrial conglomerate, for £112 billion, the largest corporate acquisition in history as of that date.¹⁸ The acquisition approximately doubled Vodafone's market capitalization. Vodafone's share price subsequently languished, dropping from \$US 5.61 in February 2000 to \$US 1.95 at the end of our sample in December 2018. The geometric mean return to Vodafone stock during our full January 1990 to December 2018 sample was 9.21% per year, equating to a full sample buy-and-hold return (inclusive of reinvested dividends) of 1,179%. In contrast, the internal rate of return to shareholders in aggregate was just 0.75% per year. The divergence is substantially attributable to the poor performance of Vodafone shares subsequent to the increase in the magnitude of shareholder investment in February 2000. By comparison, the geometric mean return to one-month US Treasury bills during the sample period was 2.72% per year, and by our calculations Vodafone's stock market performance reduced shareholder wealth by \$US 53.1 billion during the 1990 to 2018 period.

In the global sample of 61,981 stocks, 44.1% have a positive IRR. By comparison, 45.5% have a positive geometric mean return. Similarly, 39.1% of global sample stocks have IRRs that exceed the IRR earned by Treasury bills over the matched horizon, while 40.5% have a geometric mean return that exceeds the Treasury-bill IRR. The implication is that accounting for the amount and timing of cash flows (including dividends, share repurchases, and new equity issuances) to shareholders in aggregate

¹⁸ <https://en.wikipedia.org/wiki/Vodafone>.

reinforces rather than mitigates the conclusion that most global common stocks underperform one-month US Treasury bills in the long run. The finding that the IRR is lower than the geometric return for more than half of stocks holds not only for the global sample, but for the global non-US, developed economy, non-US developed economy, emerging economy, North America, Europe, and Asian Pacific subsamples, as well as for thirty-four of the forty-two individual countries in the study.

4 Stock Market Wealth Creation

We measure stock market wealth creation by implementing expression (1) for each of the 61,100 companies in the sample, using all available data during the January 1990 to December 2018 sample period. As noted, expression (1) can be viewed as computing the Net Future Value (as of December 2018) of the cash flows (including end-of-sample market capitalization) to shareholders in aggregate. Alternatively, it can be viewed as quantifying the increase in end-of-period wealth to shareholders because they earned the stock's actual returns rather than one-month Treasury bill returns on their invested capital.¹⁹

By our computations, the 61,100 sample companies collectively created \$US 44.74 trillion in shareholder wealth between January 1990 and December 2018. (Table 6). The sample includes 23,905 firms (39.12% of total) with positive wealth creation, and 37,195 (60.88% of total) with negative wealth creation. Focusing only on those firms for which wealth creation was positive, the sum is \$US 66.57 trillion in wealth creation (Table 6). This total was offset by \$US 21.83 trillion in wealth destruction by the remaining sample firms. We will refer to the sum of wealth creation across firms with positive outcomes as “gross wealth” created and to the sum across all firms as “net wealth” created.

Table 5A reports on wealth creation for the fifty firms that created the most wealth during the sample period. The table also reports the first month and the last month that the firm appears in the

¹⁹ Recall that cash flows are summed across share classes for those companies that issued multiple classes of common stock, so that wealth creation figures pertain to companies rather than stock issues.

database, and the annualized internal (or dollar-weighted) rate of return to shareholders in aggregate.²⁰

The firm ranked first in terms of wealth creation during the January 1990 to December 2018 period is Apple, with wealth creation of \$US 1.006 trillion. The rest of the top five firms are Microsoft (\$US 955 billion in wealth creation), Amazon (\$US 697 billion), Alphabet (\$US 529 billion), and Exxon Mobile (\$US 516 billion). Note that Amazon and Alphabet entered the sample in 1997 and 2004, respectively, while the other three firms in the top five were present since the beginning of the sample in January 1990.

Thirty-five of the top fifty wealth creating firms listed on Table 5A are American. The non-US firms include Tencent (ranked 9th, \$US 377 billion wealth created), Nestle (11th, \$US 354 billion), Samsung Electronics (15th, \$US 285 billion), Roche (18th, \$US 276 billion), Novartis (22nd, \$US 250 billion), China Mobile (28th, \$US 220 billion), Taiwan Semiconductor (29th, \$US 217 billion), China Construction Bank (33rd, \$US 211 billion), Unilever (36th, \$US 206 billion), Industrial and Commercial Bank of China (40th, \$US 193 billion), Toyota Motor (43rd, \$US 178 billion), Total SA (44th, \$US 174 billion), HSBC (45th, \$US 167 billion), LVMH Moët Hennessy Louis Vuitton (47th, \$US 159 billion), and L'Oreal SA (48th, \$US 156 billion).

The annualized internal rate of return (IRR) to Tencent shareholders was 50.10%, which was the highest among the top fifty wealth creating firms listed on Table 5A. Other firms that generated shareholder IRRs that exceeded 20% per year include Mastercard (32.18%), Amazon (29.35%), Facebook (25.59%), Visa (22.85%), Apple (21.00%), and United Health Group (21.28%).

As noted, Apple created \$US 1.006 trillion in stock market wealth during the January 1990 to December 2018 sample. Thus, Apple alone accounted for 2.25% of the \$US 44.74 trillion in net global wealth created by the 61,100 sample firms. Apple also accounted for 1.51% of the \$US 66.57 trillion in gross wealth created by those firms that created rather than destroyed stock market wealth during the sample. Table 5A also reports the percentage of global net (across all firms) and gross (across firms with

²⁰ The annualized IRR is obtained from the monthly IRR as $(1 + \text{monthly IRR})^{12} - 1$.

positive outcomes) wealth creation during the January 1990 to December 2018 sample period accounted for by the indicated firm and those listed above it. It can be observed that the top five firms, (Apple, Microsoft, Alphabet, Amazon and Exxon Mobile), which comprise 0.008% of the 61,100 firms in the sample, accounted for 8.27% of global net wealth creation and 5.56% of global gross wealth creation. The top 20 firms (0.033% of the firms in the sample) accounted for 19.37% of global net wealth creation and 13.02% of global gross wealth creation. The top 50 firms (0.082% of the firms in the sample) accounted for 32.93% of global net wealth creation and 22.13% of global gross wealth creation.

Figure 5 displays the cumulative percentages of gross and net wealth creation when firms are ranked from highest to lowest wealth creation, for all 61,100 firms in the sample. The net wealth creation curve ends at 100% by construction, and reaches a maximum of 148.8%, which reflects that gross wealth creation (summed across only firms with positive wealth creation) was 48.8% larger than net wealth creation (which includes the effects of wealth destruction at the majority of firms.) The gross wealth creation curve reaches a maximum of 100% by construction, and ends at 67.2%, which reflects that net wealth creation is 32.8% less than gross wealth creation.

Figure 6 displays the same data as Figure 5, but only for the 1,000 firms with the greatest wealth creation. The net wealth creation curve reaches 25% at 31 firms (0.051% of the total), 50% at 121 firms (0.198% of the total), 75% at 329 firms (0.538% of the total), and 100% at 811 firms (1.327% of the total). That is, the top-performing 1.3% of firms in the sample created net wealth of \$US 44.74 trillion, equivalent to the wealth creation of the entire sample of global firms, while the remaining 98.7% of firms collectively matched the returns to one-month US Treasury bills. By comparison, Bessembinder (2018) reports that 4.1% of stocks contained in the CRSP (US) database account for all net dollar wealth creation during the 1926 to 2016 sample period.

In addition to the 811 firms that created wealth equivalent to the full sample, another 23,094 firms (37.8% of the total) generated positive wealth for their shareholders. However, the wealth creation of these 23,094 firms just offset the wealth destruction of the remaining 37,195 (60.9% of firms), such that

the 60,289 firms (98.7% of total) outside of the top 811 best performers collectively generated returns on invested capital that just matched one-month Treasury bills.

The finding that just 1.3% of firms generated wealth (measured in dollars) equivalent to total global stock market wealth creation can be attributed to several factors, including dispersion in firm sizes and in the length of time that firms are included in the sample, and purely random outcomes. It also reflects the practical importance of positive skewness in the distribution of long horizon stock returns.

Appendix Tables A1 to A42 report on the top 20 firms in terms of full sample wealth creation for each of the individual countries included in this study. For each firm, the Appendix Tables report wealth creation in dollars, the percentage of global gross wealth creation accounted for by the firm, as well as the percentage of national gross wealth creation accounted for by the firm. The Appendix tables reveal that the single top-performing firm typically explains a substantial portion of gross wealth creation in each country. Prominent examples include Anheuser-Busch Inbev (31.2% of gross wealth creation in Belgium), Novo Nordisk (31.8% in Denmark), Nokia (13.8% in Finland), Samsung Electronics (28.8% in South Korea), Taiwan Semiconductor (27.4% in Taiwan), Vale SA (16.1% in Brazil), Royal Dutch Petroleum (16.1% in the Netherlands), Equinor ASA (20.5% in Norway), Nestle (21.9% in Switzerland), Saudi Basic Industries Corp (35.6% in Saudi Arabia), Tencent Holdings (13.3% in Hong Kong), Total SA (7.7% in France), SAP SE (6.9% in Germany), Eni SpA (18.6% in Italy), Toyota Motor (8.0% in Japan), Jardine Matheson Holdings (10.2% in Singapore), Industrial and Commercial Bank of China (6.6% in China), Tata Consultancy Services (6.5% in India), Walmart de Mexico SA (14.4% in Mexico), Sberbank of Russia (19.3% in Russia), Naspers (16.6% in South Africa), Royal Bank of Canada (6.9% in Canada), and HSBC Holdings (4.5% in the United Kingdom). By comparison, Apple, despite having the largest wealth creation of any individual firm in the sample, accounts for a more moderate 3.0% of gross wealth creation among US firms.

Table 5B reports on the twenty firms with the greatest wealth destruction (most negative wealth creation) in the global sample. Petro China was responsible for largest wealth destruction by our

calculations, \$US 470 billion. The rest of the bottom ten are all Chinese or Japanese firms, including six Japanese Banks (Industrial Bank of Japan, Bank Tokyo-Mitsubishi, Fuji Bank, Dai-Ichi Kangyo Bank, Sakura Bank, and Sanwa Bank). The worst performing firm from other countries was WorldCom Inc., with \$US 92 billion in wealth destruction.

Table 6 provides summary data regarding the concentration of wealth creation for the global sample, as well as for individual countries and regions. The table reports on the percentage of net wealth creation (summed across all firms) and gross wealth creation (summed across firms with positive wealth creation) accounted for by the best-performing 0.25%, 0.5%, 1.0%, and 5.0% of firms in each sub-sample.

The data on Table 6 shows that wealth creation is even *more* concentrated among non-US firms than among US firms. The top-performing 0.25% of US firms accounted for a very substantial 43.1% of net wealth creation for all US firms. By comparison, the top-performing 0.25% of non-US firms accounted for over two thirds, 69.2%, of net wealth creation in the non-US sample. The top-performing one percent of US firms accounted for 71.4% of US net wealth creation, while, strikingly, the top-performing one percent of non-US firms accounted for 121.4% of net wealth creation among all non-US firms in the sample. Stated alternatively, the \$US 16.4 trillion in aggregate net wealth created by the 44,034 non-US firms in the January 1990 to December 2018 sample is fully explained by less than one percent of the non-US firms in the sample.

The concentration of gross wealth creation is, in contrast, more similar across US and non-US firms. For example, the top-performing one percent of non-US firms accounted for 59.9% of gross wealth creation in the non-US sample, while the top performing one percent of US firms accounted for 60.1 % of gross wealth creation in the US sample. The fact that the concentration of net wealth creation is notably greater for non-US countries while the concentration of gross wealth creation is similar across US and non-US firms can be attributed to greater wealth destruction by underperforming non-US firms. Gross wealth creation (attributable to companies with positive wealth creation) is nearly equal across US

(\$US 34.0 trillion) and non-US (\$US 32.3 trillion) companies in the sample. In contrast, wealth destruction (attributable to companies with negative wealth creation) was \$US 16.4 trillion for non-US companies, compared to \$US 5.4 trillion for US companies. The result is that net wealth creation is substantially less and is more concentrated for non-US as compared to US companies.

Stock market wealth creation, as well as the degree to which wealth creation is concentrated, varies considerably across countries. Net wealth creation at the national level (obtained by summing firm-level wealth creation across all firms in a country) is *negative* during the sample period for five countries. The most notable is Japan, where wealth creation aggregated across all 4,524 sample firms was -\$US 3.83 trillion.²¹ In addition, net wealth creation across all sample firms was negative in China (-\$US 324 billion), Greece (-\$US 67 billion), Nigeria (-\$US 23 billion), and Turkey (-\$US 22 billion).²² For these countries the percentage of net wealth creation would be negative for all firms creating positive wealth. Hence, we focus on gross wealth creation, obtained by summing firm-level wealth creation across those firms with positive wealth creation outcomes.

Focusing on the top-performing 1% of firms in each country, the least concentration is observed in New Zealand, where the best performing firms accounted for 21.7% of gross wealth creation. In contrast, the percentage of gross wealth creation accounted for by the top-performing 1% of firms exceeded sixty percent in the US, Greece, Australia, Hong Kong, South Korea, Taiwan, India, Indonesia, and Poland. Focusing on the top-performing 5% of firms in each country, the percentage of gross wealth creation explained ranges from 49.6% in Columbia to 94.1% in Poland. Wealth creation is slightly more concentrated in the Asia-Pacific region (one percent of firms account for 63.7% of gross wealth creation)

²¹ Of course, the Japanese stock market performed very well in the years preceding 1990 (the Nikkei Index reached its all-time high on December 29, 1989), so this result would differ over a longer sample period.

²² The outcome that measured wealth creation for Chinese stocks is negative is attributable in part to a large degree of underpricing of IPOs in China. For example, Chan, Wang, and Wei (2004) document underpricing, from the IPO offer price to open market trading prices, averaging 178% in Chinese A share IPOs. Since our study focuses on the experience of shareholders who transact in the public markets, we do not capture the gains to insiders or those who are allocated shares at the IPO price. The outcome for China contributes to a debate as to whether the long-run performance of China's stock market matches its economic long-run performance. See, for instance, Allen, Qian, Shan, and Zhu (2018) and Carpenter, Lu, and Whitelaw (2018) for the debate.

as compared to North America and Europe, where the top one percent of firms account for 59.9% and 55.7%, respectively, of gross wealth creation.

5. Assessing Cross-Country Variation in Underperformance Rates and Concentration of Wealth Creation

The results reported in the preceding sections show that, similar to the results reported by Bessembinder (2018) for US stocks, long-run returns to the majority of global common stocks are less than matched-horizon returns to one-month US Treasury bills, and the net wealth creation revealed by stock market prices is attributable to a relatively few stocks. However, the degree to which these results hold varies across countries. For example, the percentage of stocks with long-run returns that exceed those of US Treasury bills varies from 22.8% in Greece and 33.2% in Australia to 65.6% in Switzerland and 67.7% in Columbia. We next assess the empirical determinants of cross-country variation in outcomes.

The key findings in this paper are attributable to the empirical fact that the distribution of long horizon return outcomes is positively skewed across stocks. Bessembinder (2018) provides simulation evidence and Farago and Hjalmarsson (2019) show theoretically that long horizon returns will be positively skewed, even if short horizon return are distributed symmetrically and returns are independent across time, and that the skewness in long horizon resulting from compounding is greater if the volatility of short horizon returns is higher. We therefore include in our cross-sectional analysis the average, across stocks within each country, of the standard deviation of the time series of monthly returns to each stock.

It is intuitive that, other things equal, positive skewness in the distribution of short horizon returns will lead to greater skewness in long horizon outcomes. We therefore also include in our cross-sectional analysis the average, across stocks within each country, of the skewness of the time series of monthly returns to each stock. We also include the cross-sectional average of the time series mean return to the individual stocks in each country. While it is somewhat self-evident that a higher mean return across

stocks will be associated with a greater rate of outperformance relative to Treasury bills, this inclusion allows for the assessment of robustness of outcomes regarding volatility and skewness.

In addition, we control for the potential effects of macroeconomic performance by including in the cross-sectional analysis 2018 GDP per capita in US dollars as well as the annual growth rate from beginning to end of sample in real GDP. Further, we follow Chui, Titman, and Wei (2010), who report that the measures of individualism provided by Hofstede (2001) have explanatory power across countries for the degree of momentum in stock returns, and we propose that investor risk taking behavior may also be associated with our outcome variables. We obtain the Global Preference Survey (GPS) risk taking preference from Falk, Becker, Dohmen, Enke, Huffman, and Sunde (2018).

We consider the possibility that individualism is associated with overconfidence, and that countries with more overconfident and risk-taking individuals would be willing to invest to a greater extent in uncertain projects with the potential for high payoffs. If so, we expect individualism and risk taking to be associated with fewer stocks outperforming Treasury bills, and greater concentration of wealth creation. The Hofstede individualism measure is available for thirty-eight sample countries, while the GPS risk-taking measure is available for thirty-three sample countries. We use an indicator variable set equal to one in the multiple regressions in those cases where a Hofstede variable or the GPS risk-taking measure is missing.

On Table 7 we report the results of cross-sectional regressions estimated across the forty-one sample countries (“homeless ADRs” are excluded). We focus on explaining the proportion of stocks in each country whose long-term returns exceed matched-horizon returns to US Treasury bills (Panel A) and the proportion of total gross wealth created in each country by the 0.5% best-performing firms (Panel B). Since we seek in each case to explain a proportion that is necessarily bounded by zero and one, the dependent variable in each case is the logistic transformation of the original variable (X), $\ln(X/(1 - X))$.

In column (1) of Panels A and B we report results obtained when the only explanatory variables are the average standard deviation and average skewness of monthly returns. The resulting coefficient estimates strongly support the implications of Bessembinder (2018) and Farago and Hjalmarsson (2019) that the volatility of short horizon returns is a determinant of the degree of positive skewness in long horizon returns. In particular, the average standard deviation of monthly returns is negatively associated (t -statistic = -3.36) with the proportion of stocks that outperform US Treasury bills in the long run, and is positively associated (t -statistic = 3.12) with the proportion of wealth creation attributed to the top performing stocks in each country. In contrast, the average skewness of monthly individual stock returns does not have significant explanatory power (t -statistics are -0.84 and -0.29) for either variable.

In column (2) of Panels A and B we report results obtained when the cross-sectional regression also includes the average across stocks of the time-series mean monthly return to each stock in the country. As would be anticipated, a higher average stock return in a given country is associated (column 2 of Panel A) with more stocks outperforming the US Treasury bill benchmark (t -statistic = 4.60). The mean stock return in a country is not a significant predictor (t -statistic = -1.29) of the degree of concentration in wealth creation. More informative, inclusion of the average stock return in the regression does not alter the result that the average standard deviation of stock returns has significant explanatory power for both dependent variables. While the average skewness of stock returns remains insignificant in explaining the concentration of wealth creation, the inclusion of the mean stock return results in a significant coefficient (t -statistic = -2.46) on average skewness when explaining the proportion of stocks that outperform the US Treasury bill benchmark. This result supports the implication that the skewness of short horizons returns can also contribute to the skewness of long horizon returns.

Column (3) of Panels A and B on Table 7 reports estimates obtained when we use the two macroeconomic variables as explanatory variables. We find that national GDP per capita has significant explanatory power (t -statistic = 2.06) for the proportion of stocks in a country with cumulative returns

that exceed US Treasury bills. We also find that the rate of growth in real GDP has significant (t -statistic = 2.34) explanatory power for the degree of concentration in wealth creation.

Column (4) of Panels A and B on Table 7 reports results obtained when the Hofstede (2001) individualism variable and the risk-taking measure are included, along with an indicator variable for developed economies. In contrast to our conjectures, none of these variables have significant explanatory power for either the proportion of stocks that outperform the US Treasury bill benchmark or the concentration of wealth creation.

Finally, to assess robustness, we report in column (5) of Panels A and B results obtained when all explanatory variables are simultaneously included in the regressions. The results confirm that the average standard deviation of monthly returns continues to have significant explanatory power for both dependent variables, and that the average monthly return continues to have significant explanatory power for the proportion of stocks that outperform the US Treasury bill benchmark. Other variables, including the average skewness of monthly returns, measures of macroeconomic performance, and the cultural variables are insignificant, excepting GDP per capita which has significant (t -statistic = 2.07) explanatory power for the concentration of wealth creation. That is, the results confirm that, aside from the somewhat self-evident result that higher mean returns in a given country are associated with greater rates of outperformance for stocks in that country, the main determinant of the percentage of stocks with long-run returns that exceed US Treasury bills and the degree to which wealth creation is concentrated is the volatility of short horizon returns.

6. Conclusions

Prior work focusing on stock return skewness has mainly studied US markets. Here, we examine compound return data for nearly 62,000 global common stocks traded on public markets over the 1990 to 2018 period. We find that only 40.5% of global common stocks, including 43.7% of US stocks and 39.3% of non-US stocks, have full-sample buy-and-hold return that exceeds the accumulated return to

one-month US Treasury bills over matched time horizons. We also consider “dollar-weighted” returns, which consider that investors in aggregate do not reinvest dividends but do fund net equity issuances. We find that the dollar-weighted return exceeds the Treasury-bill return over the matched time horizon for an even lower proportion of stocks, 39.1% globally and 37.5% for non-US stocks.

We calculate that the stocks traded in public global markets created over \$US 44.7 trillion in shareholder wealth during the twenty-nine-year sample period we study. Focusing only on firms with positive value creation, we calculate gross stock market wealth creation of \$US 66.57 trillion. However, wealth creation is highly concentrated. We find that the five firms (0.008% of the total) with the largest wealth creation during the January 1990 to December 2018 period accounted for 8.27% of global net wealth creation and 5.56% of global gross wealth creation. The best performing 811 firms (1.33% of total) accounted for *all* of the net global wealth creation, and 67.20% of gross global wealth creation. The global sample therefore provides even stronger evidence that stock market wealth creation is concentrated in a few top-performing stocks as compared to Bessembinder’s (2018) study US stock returns. Outside the US, the 441 top-performing firms (1% of total) accounted for 59.94% of gross and 121.37% of net wealth creation by non-US firms. That is, less than one percent of non-US firms in the sample accounted for all of the net stock market wealth creation outside the US during the January 1990 to December 2018 sample period.

We assess empirically the determinants of cross-country variation in the proportion of stocks that outperform US Treasury bills and in the degree to which wealth creation is concentrated in a relative few stocks. We find that higher stock return volatility is associated with a lower proportion of stocks outperforming Treasury bills and greater concentration of wealth creation, results which are consistent with the simulations reported by Bessembinder (2018) and the theoretical predictions of Farago and Hjalmarsson (2019).

The results reported here are important from a number of perspectives. Most empirical analyses of stock markets consider unconditional or (by use of regression specifications) conditional arithmetic

means of returns measured over short (e.g., monthly) horizons. However, the investment horizons of individuals or fund managers (particularly pension funds) can stretch to decades. The results therefore raise questions regarding the evaluation of portfolio and investment manager performance. Two frequently used measures include versions of Jensen's alpha and the Sharpe ratio, typically implemented using monthly returns. The former is a conditional arithmetic mean, while the latter is the ratio of the arithmetic mean return to the variance. As such, neither allows for the fact that for any given time series of returns the arithmetic mean return overstates the outcome to a buy-and-hold investor (since for any given sample it exceeds the geometric mean, which when compounded, gives the buy-and-hold return) or the empirical fact that stock returns are positively skewed, especially at longer horizons.

The result that the wealth created by stock market investing is largely attributable to positive outcomes to a relatively few stocks is relevant to the debate regarding active stock selection vs. the passive holding of broadly diversified stock indices. For those investors without a comparative advantage in identifying the few stocks that will create the most wealth (or in selecting a manager with the ability to do so) and without a substantial preference for positive skewness, the results reinforce the desirability of passive indexing. On the other hand, for the (presumably few) investors with the appropriate comparative advantage, the results highlight the degree to which successful stock selection can enhance wealth. Finally, the findings raise the question of whether existing industrial organization models of firm entry, growth, merger, competition, and failure can account for the observed degree of concentration in wealth creation, or whether new models are required.

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Figure 1: Percent of Stock/Months with Indicated Return

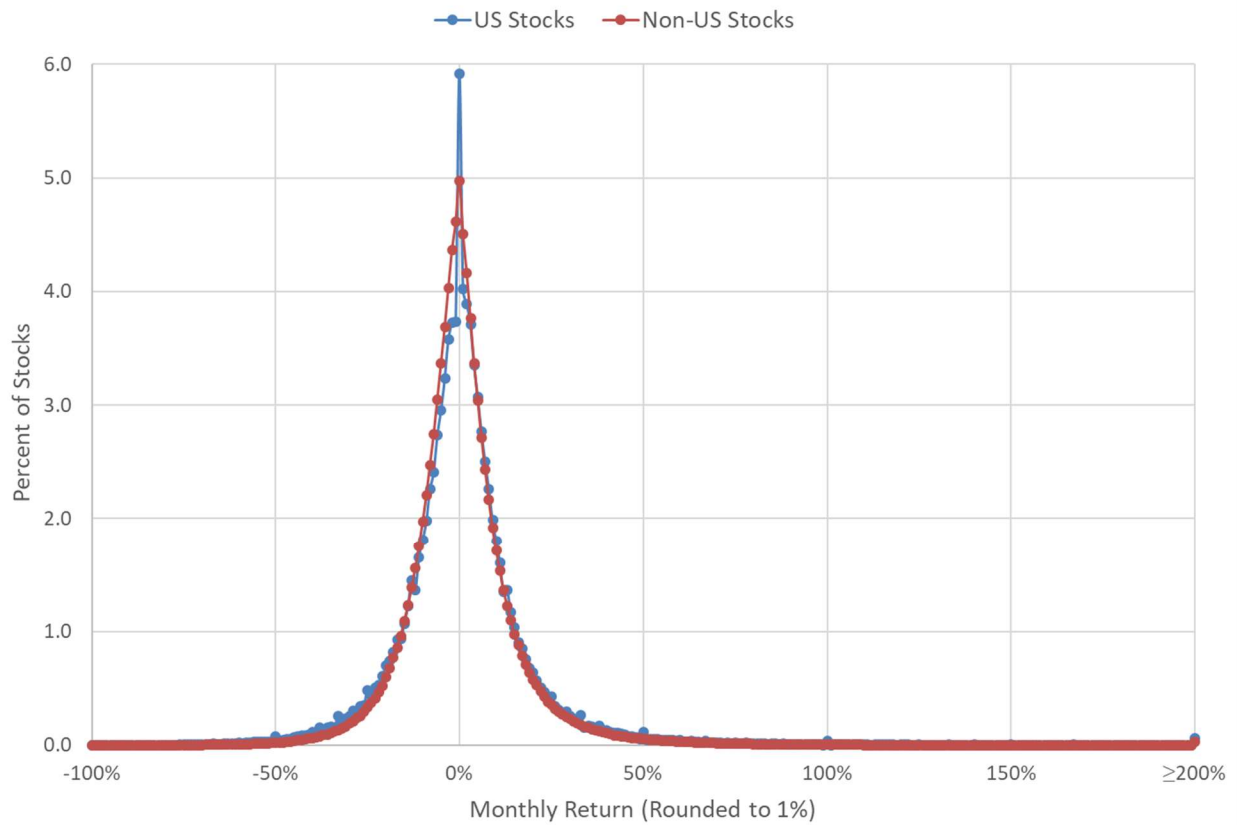


Figure 2: Percent of Stock-Years with Indicated Buy-and-Hold Return

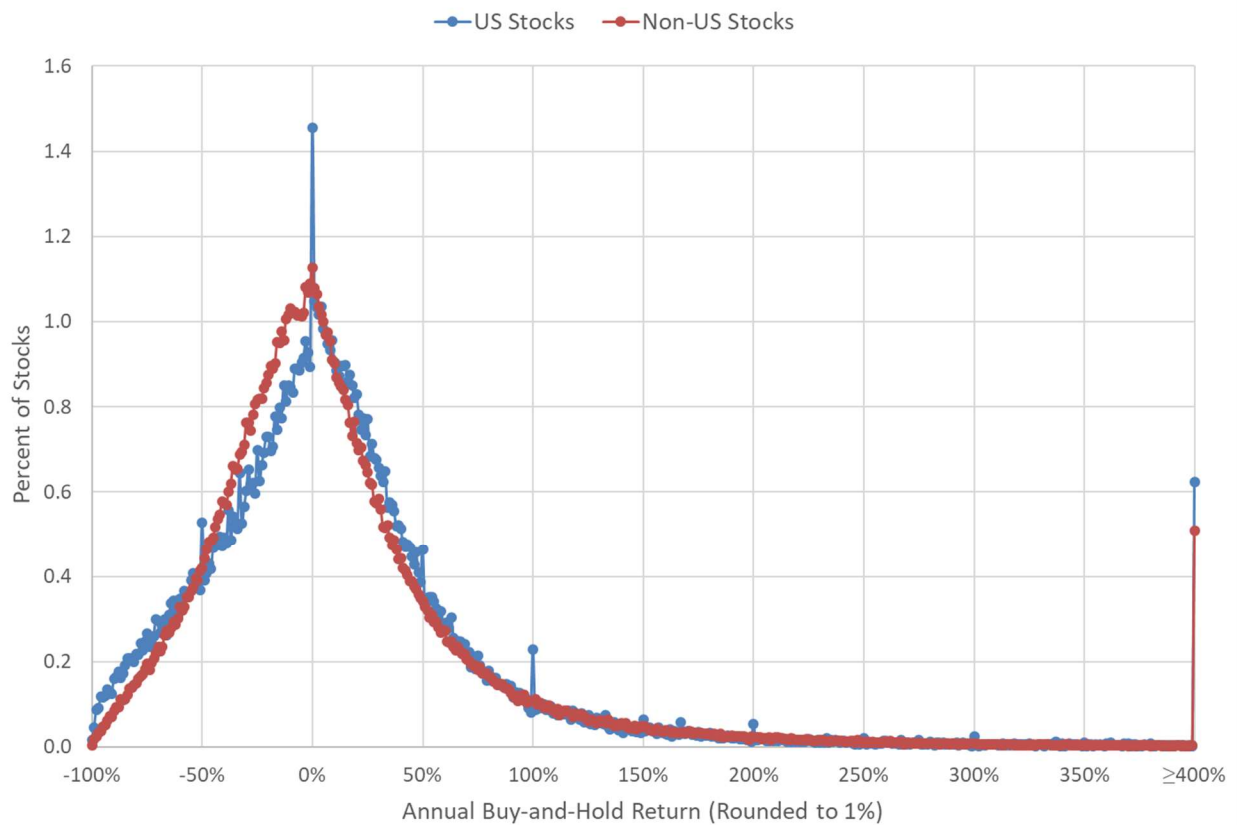


Figure 3: Percent of Stock-Decades with Indicated Buy-and-Hold Return

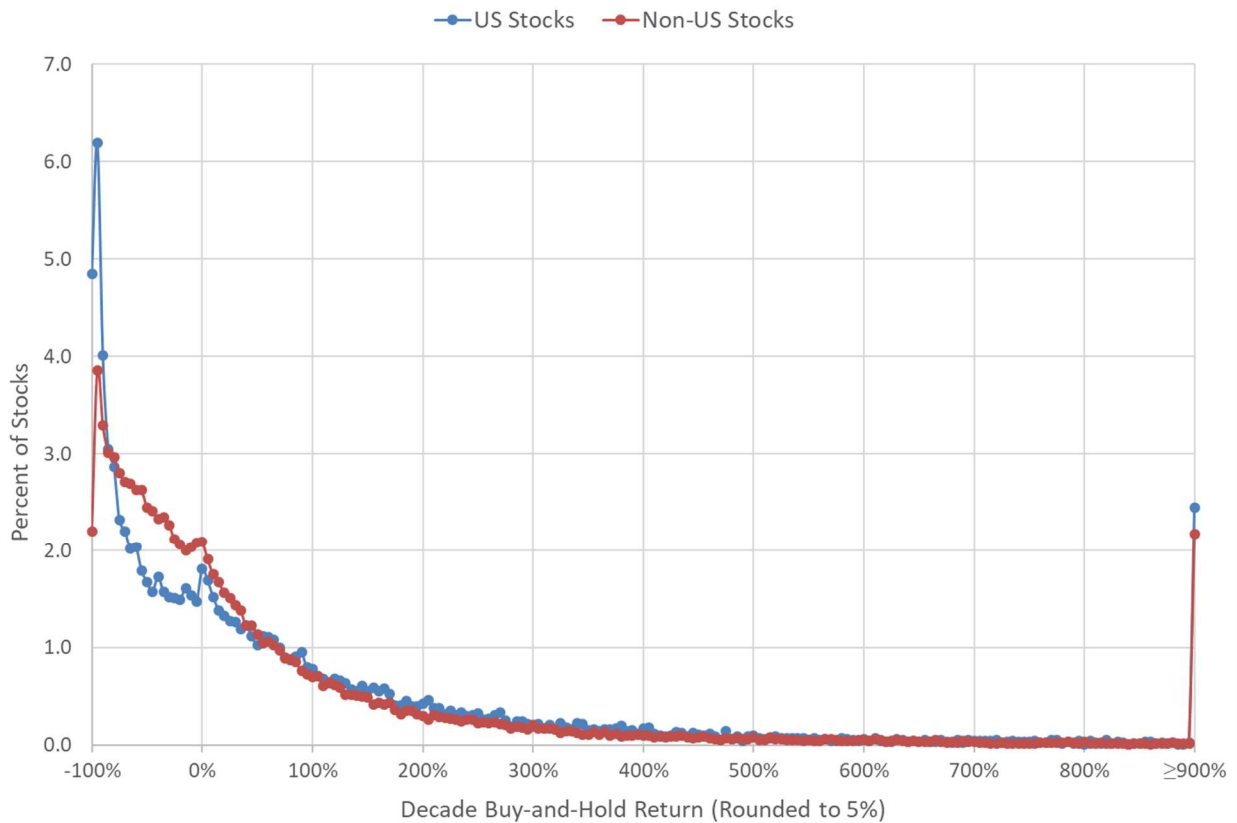
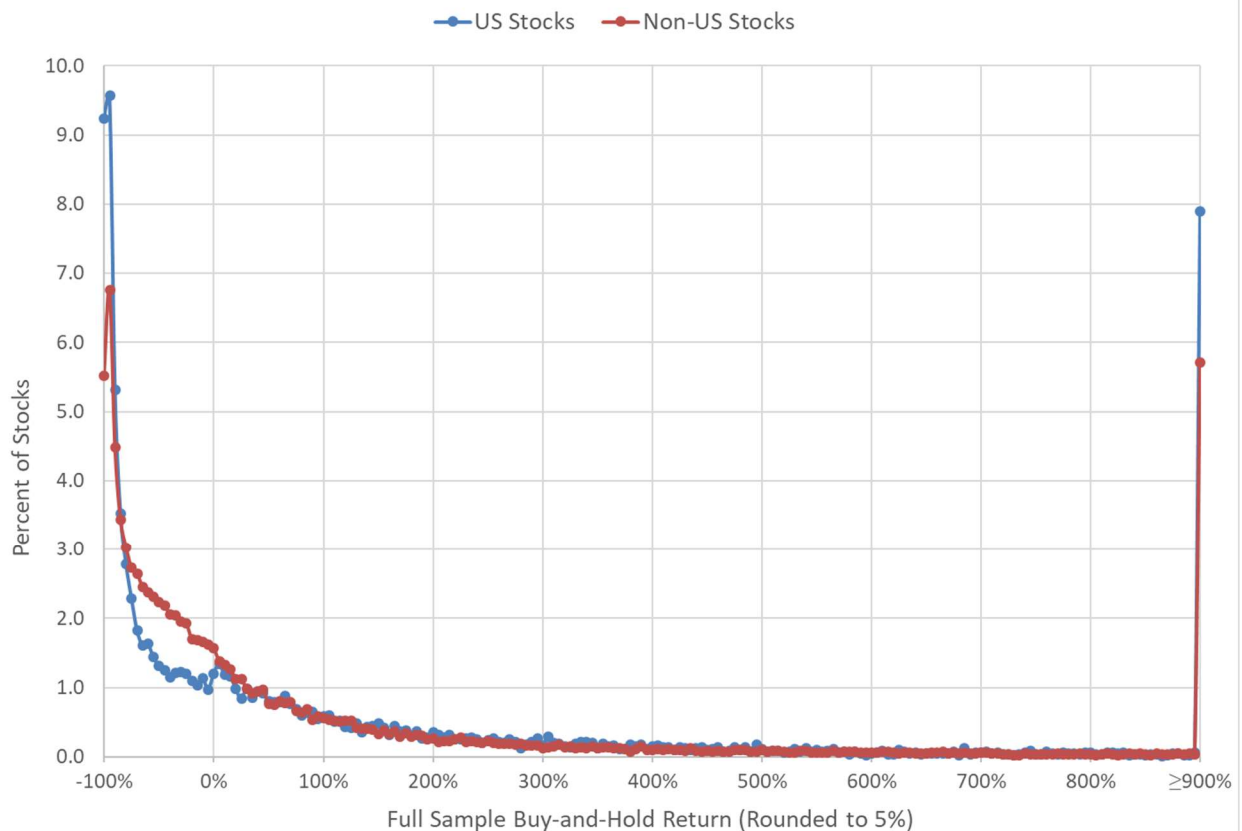


Figure 4: Percent of Stocks with Indicated Full Sample Period Buy-and-Hold Return



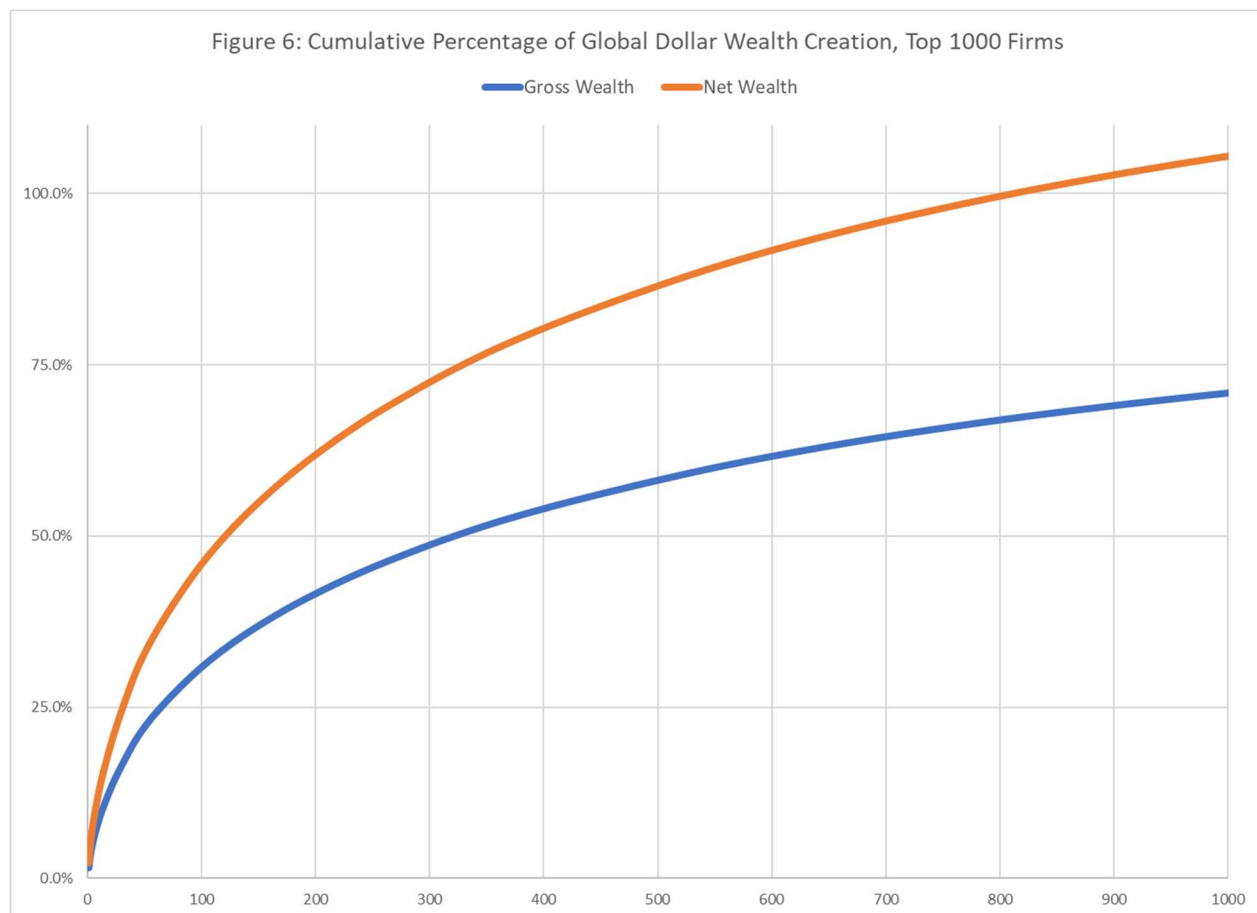
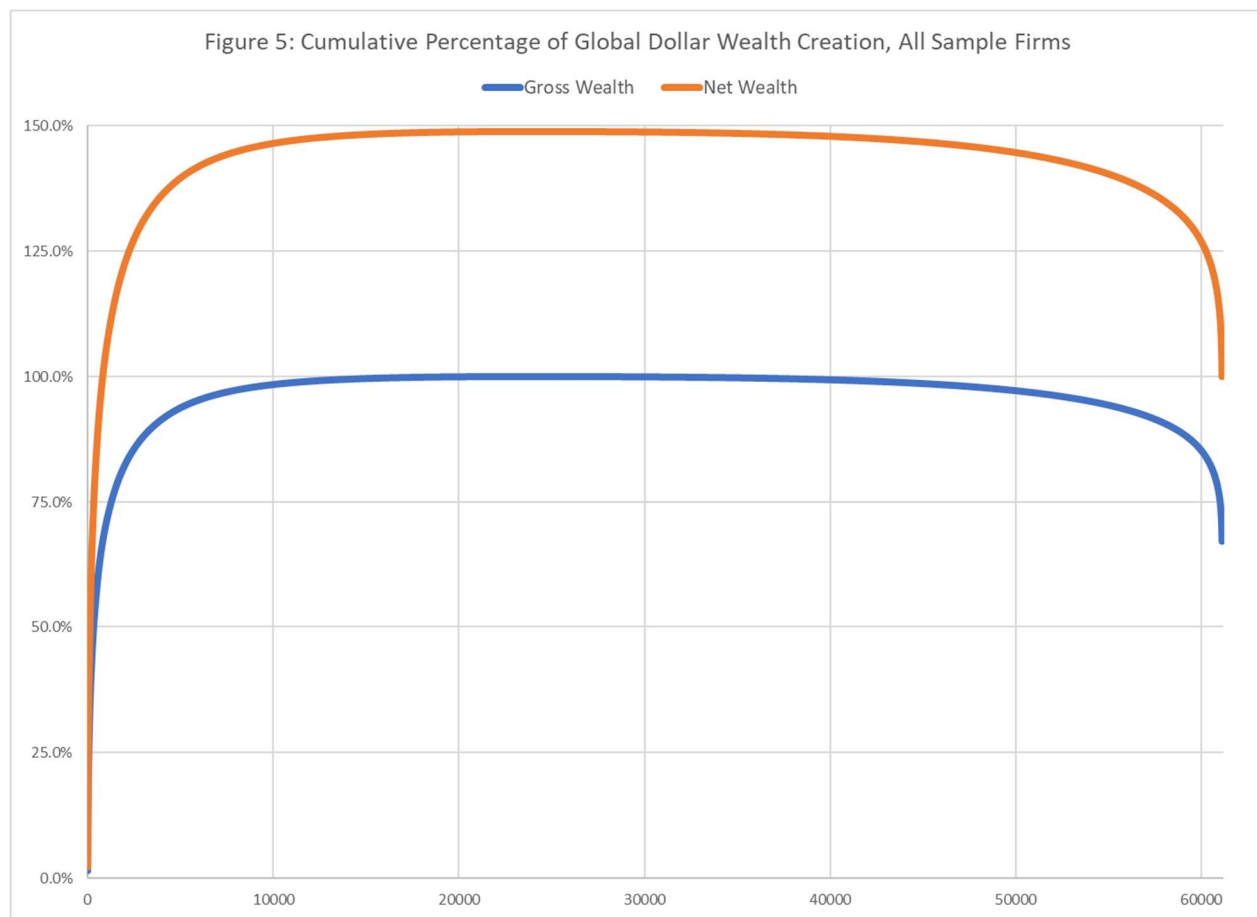


Table 1: Summary Statistics by Country

Country	Number of Stocks	First Month	Last Month	GDP per Capita (\$US , 2018)	Avg. GDP Growth Rate (%)	GDP Creation (\$US billions)	Total Market Cap (\$US billions, 2018)	Avg. Market Cap (\$US billions)	Avg. Market Cap to GDP
United States	17,310	199001	201812	62,606	2.48	14,852	26,597	13,831	1.12
Canada	1,995	199001	201812	46,261	2.23	1,144	1,460	888	0.79
Austria	161	199001	201812	51,509	2.02	324	118	71	0.23
Belgium	289	199001	201812	46,724	1.78	372	338	213	0.58
Denmark	348	199001	201812	60,692	1.67	238	349	161	0.65
Finland	263	199001	201812	49,845	1.71	156	232	153	0.80
France	1,632	199001	201812	42,878	1.61	1,749	2,160	1,325	0.65
Germany	1,490	199001	201812	48,264	1.69	2,748	1,898	1,168	0.41
Greece	408	199001	201812	20,408	0.93	140	32	61	0.31
Italy	590	199001	201812	34,260	0.76	1,134	579	492	0.30
Netherlands	324	199001	201812	53,106	2.15	653	533	404	0.64
Norway	495	199001	201812	81,695	2.34	332	233	134	0.46
Portugal	119	199001	201812	23,186	1.66	179	65	57	0.33
Spain	360	199001	201812	30,697	2.19	1,014	650	469	0.47
Sweden	991	199001	201812	53,873	2.14	338	631	356	0.92
Switzerland	401	199001	201812	82,950	1.67	502	1,324	735	1.64
United Kingdom	4,050	199001	201812	42,558	1.97	1,823	2,884	2,345	1.11
Australia	2,820	199001	201812	56,352	3.05	1,111	1,179	692	0.87
Hong Kong	2,260	199001	201812	48,517	3.77	294	3,272	1,226	6.08
Israel	597	199411	201812	41,644	4.13	296	148	99	0.50
Japan	4,524	199001	201812	39,306	1.13	1,917	5,203	3,715	0.79
New Zealand	266	199001	201812	41,267	2.86	159	83	30	0.28
Singapore	1,021	199001	201812	64,041	5.97	330	458	271	1.64
South Korea	2,892	199001	201812	31,346	5.11	1,376	1,289	555	0.65
Taiwan	2,321	199001	201812	24,971	4.65	437	1,048	508	1.36
Argentina	114	199001	201812	11,627	2.92	430	32	36	0.10
Brazil	341	199308	201812	8,968	2.53	1,486	548	292	0.22
China	3,698	199101	201812	9,608	9.61	13,009	5,975	2,117	0.48
Colombia	62	199001	201812	6,684	3.54	278	92	63	0.32
India	3,731	199001	201812	2,036	6.67	2,417	2,049	670	0.63
Indonesia	602	199001	201812	3,871	5.18	900	359	125	0.27
Malaysia	1,315	199001	201812	10,942	5.81	313	388	236	1.34
Mexico	241	199001	201812	9,807	2.67	977	329	204	0.25
Nigeria	157	199311	201812	2,049	5.53	345	27	28	0.10
Poland	962	199306	201812	15,431	4.23	497	162	95	0.28
Russia	268	199508	201812	11,327	2.80	1,334	511	265	0.24
Saudi Arabia	191	200001	201812	23,566	3.64	621	492	323	0.66
South Africa	846	199001	201812	6,377	2.30	269	420	274	1.17
Thailand	874	199001	201812	7,187	4.51	413	458	166	0.69
Turkey	339	199002	201812	9,346	4.72	619	39	36	0.07
United Arab Emirates	118	200105	201812	40,711	3.96	321	221	127	0.45

* Source of GDP: International Monetary Fund (IMF) World Economic Outlook Database, April 2019

Table 2: Summary Statistics by Exchange

Country	Exchange	N	Average % of Trading Volume
United States	New York Stock Exchange	3,113	58.62%
United States	Amex	1,551	2.77%
United States	NASDAQ	12,646	38.60%
Homeless Firms (US ADRs)	New York Stock Exchange	88	76.87%
Homeless Firms (US ADRs)	Amex	5	0.02%
Homeless Firms (US ADRs)	NASDAQ	102	23.12%
Canada	Toronto Stock Exchange	1,995	98.26%
Austria	Wiener Boerse AG	161	98.29%
Belgium	NYSE Euronext Brussels	289	99.48%
Denmark	OMX Nordic Exchange Copenhagen AS	348	100.00%
Finland	NASDAQ OMX Helsinki Ltd	263	100.00%
France	NYSE Euronext Paris	1,632	99.58%
Germany	Deutsche Boerse AG	1,054	13.28%
Germany	XETRA	436	85.25%
Greece	Athens Exchange SA Cash Market	408	100.00%
Italy	Borsa Italiana Electronic Share Market	590	99.97%
Netherlands	NYSE Euronext Amsterdam	324	99.99%
Norway	Oslo Bors ASA	495	100.00%
Portugal	NYSE Euronext Lisbon	119	100.00%
Spain	Bolsa De Madrid	360	99.98%
Sweden	NASDAQ OMX Nordic	991	99.85%
Switzerland	Swiss Exchange	333	95.86%
Switzerland	Zurich	68	4.10%
United Kingdom	London Stock Exchange	4,050	99.78%
Australia	ASX All Markets	2,820	100.00%
Hong Kong	Hong Kong Exchanges and Clearing Ltd	2,260	100.00%
Israel	Tel Aviv Stock Exchange	597	99.99%
Japan	Tokyo Stock Exchange	4,524	99.20%
New Zealand	New Zealand Exchange Ltd	211	81.81%
New Zealand	Wellington	55	18.19%
Singapore	Singapore Exchange	1,021	99.94%
South Korea	Korea Exchange KOSDAQ	1,571	18.67%
South Korea	Korea Exchange Stock Market	1,321	81.33%
Taiwan	Taipei Exchange	1,283	15.70%
Taiwan	Taiwan Stock Exchange	1,038	84.30%
Argentina	Bolsa De Comercio De Buenos Aires	114	100.00%
Brazil	BM and F Bovespa SA Bolsa De Valores Mercadorias E Futuros	341	99.33%
China	Shanghai Stock Exchange	1,506	58.56%
China	Shenzhen Stock Exchange	2,192	41.44%
Colombia	Bolsa De Valores De Colombia	62	100.00%
India	BSE Ltd	2,456	10.10%
India	National Stock Exchange of India	1,275	89.90%
Indonesia	Indonesia Stock Exchange	602	100.00%
Malaysia	Bursa Malaysia	1,315	100.00%
Mexico	Bolsa Mexicana De Valores Mexican Stock Exchange	241	100.00%
Nigeria	Nigerian Stock Exchange	157	100.00%
Poland	Warsaw Stock Exchange	962	100.00%
Russia	MICEX Stock Exchange	268	92.39%
Saudi Arabia	Saudi Stock Exchange	191	100.00%
South Africa	Johannesburg Stock Exchange	846	100.00%
Thailand	Stock Exchange of Thailand	874	99.99%
Turkey	Istanbul Stock Exchange	339	100.00%
United Arab Emirates	Abu Dhabi Securities Exchange	66	35.88%
United Arab Emirates	Dubai Financial Market	52	61.66%

Table 3: Accumulated Buy-and-Hold Returns, with Dividends Reinvested in Stock, 1990-2018

Panel A: Monthly Horizon								
Sample	N	Mean	Median	SD	Skewness	% > 0	% > T-bill	% > VW Market
Global	7,676,602	0.0096	-0.0001	0.177	8.218	49.1%	48.4%	46.2%
Global (Excl. US)	5,776,801	0.0091	-0.0016	0.170	8.545	49.1%	48.3%	45.7%
By Development								
Developed	6,064,253	0.0094	0.0000	0.177	8.190	49.2%	48.5%	46.4%
Developed (Excl. US)	4,164,452	0.0085	-0.0014	0.167	8.634	49.2%	48.3%	45.8%
Emerging	1,612,349	0.0104	-0.0020	0.177	8.323	48.8%	48.1%	45.5%
By Region								
North America	2,152,492	0.0116	0.0000	0.199	9.224	49.5%	48.9%	47.6%
Europe	1,444,398	0.0067	0.0002	0.155	6.412	50.1%	49.0%	46.5%
Asia Pacific	2,467,363	0.0091	-0.0027	0.168	7.126	48.6%	47.8%	45.2%
Panel B: Annual Horizon								
Sample	N	Mean	Median	SD	Skewness	% > 0	% > T-bill	% > VW Market
Global	688,936	0.1411	0.0127	0.890	18.545	51.4%	49.4%	44.0%
Global (Excl. US)	518,056	0.1380	0.0048	0.869	16.825	50.5%	48.7%	43.1%
By Development								
Developed	543,577	0.1340	0.0199	0.884	20.003	52.2%	50.0%	44.6%
Developed (Excl. US)	372,697	0.1264	0.0125	0.851	18.348	51.4%	49.3%	43.6%
Emerging	145,359	0.1676	-0.0192	0.913	13.610	48.2%	47.1%	41.9%
By Region								
North America	193,457	0.1598	0.0386	0.999	22.270	54.0%	51.5%	46.8%
Europe	132,628	0.1045	0.0255	0.713	13.789	53.1%	50.5%	45.2%
Asia Pacific	217,492	0.1290	0.0013	0.868	17.871	50.1%	48.3%	42.3%
Panel C: Decade Horizon								
Sample	N	Mean	Median	SD	Skewness	% > 0	% > T-bill	% > VW Market
Global	108,521	0.9562	-0.0216	7.465	68.200	49.1%	45.1%	37.1%
Global (Excl. US)	79,551	0.8714	-0.0468	5.217	32.679	48.1%	44.1%	35.5%
By Development								
Developed	85,790	0.9524	0.0079	7.860	71.744	50.3%	46.0%	38.2%
Developed (Excl. US)	56,820	0.8318	-0.0111	4.994	39.394	49.5%	45.2%	36.6%
Emerging	22,731	0.9704	-0.1273	5.735	21.051	44.5%	41.5%	32.8%
By Region								
North America	32,721	1.2262	0.0627	11.064	61.248	52.2%	47.9%	41.7%
Europe	21,479	0.7607	0.0085	3.482	19.187	50.4%	45.4%	37.3%
Asia Pacific	31,590	0.7991	-0.0434	5.712	43.270	48.3%	44.5%	35.3%
Panel D: Full Sample Horizon								
Sample	N	Mean	Median	SD	Skewness	% > 0	% > T-bill	% > VW Market
Global	61,981	2.6042	-0.1489	21.976	47.922	45.6%	40.5%	31.1%
Global (Excl. US)	44,476	2.2155	-0.1702	18.521	57.204	44.4%	39.3%	28.6%
By Development								
Developed	48,122	2.6499	-0.1145	22.386	50.132	46.8%	41.3%	32.6%
Developed (Excl. US)	30,617	2.1115	-0.1335	17.558	70.627	45.9%	40.0%	29.6%
Emerging	13,859	2.4453	-0.2280	20.489	37.414	41.3%	37.6%	26.2%
By Region								
North America	19,500	3.6368	-0.0598	27.863	36.524	48.7%	43.9%	37.8%
Europe	11,921	2.0603	-0.0926	9.086	10.336	47.2%	41.4%	32.3%
Asia Pacific	16,701	1.9184	-0.1722	21.835	66.393	44.4%	38.2%	26.6%

By Country								
Developed								
United States	17,310	3.5516	-0.0642	27.381	37.075	48.6%	43.7%	37.8%
Homeless Firms (US ADRs)	195	7.1397	-0.3269	92.872	13.833	34.9%	33.3%	26.7%
Canada	1,995	4.0339	0.0424	15.587	9.539	50.6%	46.4%	39.1%
Austria	161	1.6119	0.0524	4.146	3.573	50.9%	44.7%	31.1%
Belgium	289	3.0017	0.4707	8.765	7.358	61.9%	55.0%	45.7%
Denmark	348	3.2488	0.1664	12.084	8.025	54.3%	48.6%	37.4%
Finland	263	4.0420	0.4887	10.917	5.687	61.6%	59.3%	47.1%
France	1,632	2.1147	0.0636	7.304	6.087	52.0%	46.0%	35.9%
Germany	1,490	1.6036	-0.3720	10.070	15.191	40.2%	35.4%	26.1%
Greece	408	0.2041	-0.7478	2.565	4.152	25.7%	22.8%	16.7%
Italy	590	0.8064	-0.1900	4.437	9.008	41.5%	30.7%	24.9%
Netherlands	324	3.1514	0.3459	8.613	5.531	61.1%	50.9%	39.5%
Norway	495	1.9426	0.0067	10.881	13.756	50.5%	43.8%	37.2%
Portugal	119	0.9600	0.0369	3.322	4.341	52.9%	42.9%	30.3%
Spain	360	2.5174	0.0776	10.417	8.982	53.6%	46.1%	41.7%
Sweden	991	3.1726	0.0481	12.366	6.767	51.9%	49.0%	40.5%
Switzerland	401	4.9793	0.8825	12.764	6.595	70.3%	65.6%	50.1%
United Kingdom	4,050	1.6536	-0.2839	8.248	10.543	42.2%	36.6%	27.6%
Australia	2,820	3.7261	-0.4947	49.833	32.594	36.2%	33.2%	27.4%
Hong Kong	2,260	2.3744	-0.1717	13.049	16.156	43.3%	40.4%	30.0%
Israel	597	2.2205	0.3468	5.645	3.817	57.8%	55.3%	47.1%
Japan	4,524	0.9953	-0.0985	5.293	16.742	46.4%	33.8%	19.0%
New Zealand	266	4.8998	0.4518	18.759	7.247	60.5%	54.1%	46.6%
Singapore	1,021	1.7893	-0.1601	7.185	10.734	45.1%	40.1%	28.6%
South Korea	2,892	1.3396	-0.2699	6.759	8.644	41.9%	37.6%	26.1%
Taiwan	2,321	1.4362	-0.0340	7.641	16.760	48.8%	44.4%	29.6%
Emerging								
Argentina	114	4.4356	0.1035	19.801	8.827	52.6%	45.6%	30.7%
Brazil	341	2.1552	-0.1550	8.685	5.858	44.6%	41.9%	32.6%
China	3,698	0.9498	-0.0797	5.113	13.341	46.1%	41.2%	22.2%
Colombia	62	8.3791	1.2140	21.034	4.241	67.7%	67.7%	58.1%
India	3,731	4.1504	-0.2671	29.059	29.612	41.4%	39.8%	33.0%
Indonesia	602	3.2771	-0.4164	18.054	7.724	35.5%	32.7%	25.1%
Malaysia	1,315	1.7005	-0.3758	11.535	15.235	38.8%	31.9%	19.2%
Mexico	241	3.4171	0.1362	9.787	5.120	51.5%	48.1%	36.5%
Nigeria	157	1.6429	-0.4675	11.365	8.697	30.6%	29.3%	31.2%
Poland	962	0.4463	-0.5166	4.281	9.136	28.9%	26.9%	17.5%
Russia	268	0.8825	-0.3855	10.670	14.260	30.6%	30.2%	20.9%
Saudi Arabia	191	1.0755	-0.1949	3.059	2.903	45.0%	43.5%	25.7%
South Africa	846	4.6433	-0.2938	46.789	18.800	37.9%	32.6%	27.4%
Thailand	874	3.2830	-0.0955	11.744	6.786	46.9%	42.1%	30.4%
Turkey	339	0.2483	-0.4919	3.596	11.996	28.9%	23.3%	16.2%
United Arab Emirates	118	0.7460	-0.2138	2.753	2.928	39.8%	37.3%	22.9%

Table 4: Mean Monthly Returns, 1990-2018

	Arithmetic Mean Return					Geometric Mean Return					Dollar-Weighted Return (IRR)				
Sample	Mean	Median	Skew.	% > 0	% > T-bill	Mean	Median	Skew.	% > 0	% > T-bill	Mean	Median	Skew.	% > 0	% > T-bill
Global	0.0042	0.0090	-0.613	71.4%	68.3%	-0.0124	-0.0017	-2.660	45.5%	40.5%	-0.0164	-0.0022	-3.658	44.1%	39.1%
Global (Excl. US)	0.0039	0.0081	-0.800	71.2%	68.1%	-0.0107	-0.0018	-2.543	44.4%	39.3%	-0.0137	-0.0024	-3.527	42.6%	37.5%
By Development															
Developed	0.0042	0.0091	-0.606	72.1%	68.6%	-0.0129	-0.0013	-2.662	46.8%	41.3%	-0.0177	-0.0016	-3.582	45.9%	40.5%
Developed (Excl. US)	0.0038	0.0079	-0.865	72.3%	68.5%	-0.0108	-0.0013	-2.631	45.8%	40.0%	-0.0145	-0.0017	-3.602	44.7%	38.9%
Emerging	0.0043	0.0085	-0.629	68.9%	67.2%	-0.0104	-0.0028	-2.169	41.3%	37.6%	-0.0120	-0.0040	-2.502	38.0%	34.5%
By Region															
North America	0.0056	0.0116	-0.445	72.4%	69.5%	-0.0157	-0.0008	-2.496	48.6%	43.9%	-0.0223	-0.0012	-3.226	48.0%	43.4%
Europe	0.0007	0.0072	-0.840	67.8%	63.5%	-0.0128	-0.0012	-2.526	47.1%	41.4%	-0.0149	-0.0011	-2.901	47.2%	40.9%
Asian Pacific	0.0051	0.0079	-0.789	74.8%	71.3%	-0.0097	-0.0015	-2.554	44.4%	38.2%	-0.0142	-0.0020	-4.300	42.4%	36.7%
By Country															
Developed															
United States	0.0051	0.0115	-0.387	71.8%	68.8%	-0.0166	-0.0008	-2.457	48.6%	43.7%	-0.0233	-0.0012	-3.233	48.1%	43.4%
Homeless Firms (US ADRs)	0.0026	0.0042	-0.873	63.6%	60.5%	-0.0138	-0.0081	-1.219	34.9%	33.3%	-0.0147	-0.0095	-1.162	33.3%	32.8%
Canada	0.0107	0.0127	-1.032	78.7%	75.7%	-0.0083	0.0006	-2.793	50.6%	46.4%	-0.0145	-0.0009	-2.239	48.7%	44.4%
Austria	0.0025	0.0065	-2.514	70.2%	63.4%	-0.0079	0.0003	-3.216	50.9%	44.7%	-0.0084	0.0000	-4.040	50.3%	42.2%
Belgium	0.0047	0.0092	-2.393	76.8%	74.4%	-0.0038	0.0038	-2.580	61.9%	55.0%	-0.0048	0.0042	-2.810	62.6%	55.0%
Denmark	0.0025	0.0076	-1.959	73.3%	69.0%	-0.0074	0.0017	-2.276	54.3%	48.6%	-0.0092	0.0020	-2.298	55.7%	49.4%
Finland	0.0079	0.0106	-1.793	79.8%	75.3%	-0.0004	0.0038	-2.424	61.2%	59.3%	-0.0017	0.0036	-2.544	62.7%	58.2%
France	0.0052	0.0084	1.005	72.5%	68.5%	-0.0080	0.0007	-1.709	52.0%	46.0%	-0.0091	0.0007	-3.474	51.4%	45.0%
Germany	-0.0073	0.0054	-2.144	62.1%	58.7%	-0.0250	-0.0047	-2.380	40.2%	35.4%	-0.0272	-0.0043	-1.426	40.6%	35.4%
Greece	0.0035	0.0059	-0.466	67.6%	63.5%	-0.0127	-0.0096	-1.266	25.7%	22.8%	-0.0164	-0.0096	-2.082	25.0%	22.3%
Italy	-0.0001	0.0037	-1.916	61.4%	54.6%	-0.0079	-0.0028	-2.224	41.5%	30.7%	-0.0084	-0.0023	-1.766	41.5%	32.7%
Netherlands	0.0035	0.0088	-2.407	76.5%	71.3%	-0.0056	0.0028	-2.855	61.1%	50.9%	-0.0089	0.0027	-3.021	62.3%	50.3%
Norway	0.0043	0.0093	-1.929	70.9%	67.1%	-0.0098	0.0001	-2.544	50.5%	43.8%	-0.0118	0.0002	-2.834	50.1%	45.9%
Portugal	0.0064	0.0074	-1.106	73.9%	65.5%	-0.0017	0.0005	-2.138	52.9%	42.9%	-0.0031	0.0009	-1.737	54.6%	42.0%
Spain	0.0057	0.0066	1.650	71.7%	65.8%	-0.0012	0.0008	0.581	53.6%	46.1%	-0.0025	0.0010	-1.541	54.7%	47.2%
Sweden	0.0025	0.0097	-2.041	69.2%	67.5%	-0.0117	0.0010	-3.048	51.9%	49.0%	-0.0134	0.0018	-4.132	53.5%	50.5%
Switzerland	0.0065	0.0091	-2.347	84.0%	79.3%	-0.0005	0.0053	-3.070	70.3%	65.6%	-0.0016	0.0049	-3.339	70.1%	63.8%
United Kingdom	-0.0015	0.0066	0.246	63.3%	58.7%	-0.0169	-0.0045	-1.913	42.2%	36.6%	-0.0199	-0.0046	-2.780	41.9%	34.9%
Australia	0.0034	0.0088	-0.871	65.5%	62.9%	-0.0171	-0.0095	-1.402	36.2%	33.2%	-0.0242	-0.0135	-1.777	34.0%	31.5%
Hong Kong	0.0071	0.0100	0.903	71.5%	69.7%	-0.0100	-0.0026	-1.707	43.3%	40.4%	-0.0129	-0.0044	-1.523	40.8%	38.1%
Israel	0.0039	0.0110	-2.917	73.7%	72.2%	-0.0073	0.0037	-3.161	57.8%	55.3%	-0.0085	0.0028	-4.805	57.3%	54.6%

Japan	0.0062	0.0060	-2.530	83.6%	75.6%	-0.0032	-0.0005	-4.278	46.4%	33.8%	-0.0052	-0.0009	-11.782	43.5%	31.9%
New Zealand	0.0058	0.0100	-3.170	71.8%	68.8%	-0.0040	0.0050	-4.849	60.5%	54.1%	-0.0050	0.0047	-4.439	60.2%	55.3%
Singapore	0.0079	0.0099	-0.559	75.6%	73.7%	-0.0057	-0.0018	-1.220	45.1%	40.1%	-0.0087	-0.0028	-1.833	43.7%	39.1%
South Korea	0.0036	0.0110	-0.650	71.9%	70.6%	-0.0173	-0.0032	-2.135	41.9%	37.6%	-0.0285	-0.0034	-2.988	40.8%	36.9%
Taiwan	0.0040	0.0083	-2.895	75.8%	74.2%	-0.0067	-0.0003	-3.712	48.8%	44.4%	-0.0078	-0.0006	-4.037	47.8%	43.3%
Emerging															
Argentina	0.0149	0.0133	0.886	81.6%	78.9%	-0.0038	0.0007	-3.664	52.6%	45.6%	-0.0053	-0.0002	-3.266	48.2%	39.5%
Brazil	0.0118	0.0100	1.055	72.7%	70.7%	-0.0059	-0.0025	-1.154	44.6%	41.9%	-0.0051	-0.0016	-1.168	44.6%	41.9%
China	0.0056	0.0097	-0.262	77.6%	77.0%	-0.0059	-0.0007	-1.756	46.1%	41.2%	-0.0079	-0.0025	-1.551	37.1%	33.1%
Colombia	0.0152	0.0116	0.262	75.8%	75.8%	0.0071	0.0069	-0.374	67.7%	67.7%	0.0078	0.0060	1.209	67.7%	66.1%
India	0.0031	0.0095	-1.278	64.6%	63.7%	-0.0132	-0.0049	-1.727	41.4%	39.8%	-0.0144	-0.0055	-1.746	40.4%	38.7%
Indonesia	0.0019	0.0054	-0.352	57.1%	55.1%	-0.0194	-0.0098	-2.098	35.5%	32.7%	-0.0222	-0.0089	-2.178	34.6%	31.1%
Malaysia	0.0059	0.0079	-1.120	77.9%	74.5%	-0.0071	-0.0034	-1.732	38.8%	31.9%	-0.0083	-0.0033	-1.883	37.0%	31.5%
Mexico	0.0065	0.0091	-1.744	69.7%	68.5%	-0.0050	0.0019	-3.069	51.5%	48.1%	-0.0031	0.0017	-0.931	53.5%	49.0%
Nigeria	0.0014	0.0019	1.927	53.5%	52.2%	-0.0146	-0.0116	0.133	30.6%	29.3%	-0.0179	-0.0137	-0.139	26.8%	25.5%
Poland	-0.0025	0.0023	-0.900	54.3%	52.6%	-0.0194	-0.0104	-2.324	28.9%	26.9%	-0.0217	-0.0112	-2.114	27.1%	24.7%
Russia	0.0033	0.0003	2.695	50.4%	50.0%	-0.0100	-0.0084	0.065	30.6%	30.2%	-0.0097	-0.0084	0.112	31.0%	30.2%
Saudi Arabia	0.0075	0.0066	2.427	72.8%	71.2%	-0.0019	-0.0026	1.792	45.0%	43.5%	-0.0017	-0.0023	1.786	43.5%	41.4%
South Africa	0.0029	0.0069	-0.250	64.7%	58.4%	-0.0150	-0.0059	-2.086	37.8%	32.6%	-0.0170	-0.0058	-2.663	37.2%	31.0%
Thailand	0.0051	0.0108	-1.138	71.5%	69.2%	-0.0089	-0.0012	-2.040	46.9%	42.1%	-0.0114	-0.0017	-4.837	44.9%	38.6%
Turkey	0.0075	0.0064	-0.103	60.8%	58.4%	-0.0130	-0.0107	-0.669	28.9%	23.3%	-0.0128	-0.0104	-0.546	29.2%	24.5%
United Arab Emirates	0.0043	0.0050	-0.251	67.8%	64.4%	-0.0032	-0.0022	-0.517	39.8%	37.3%	-0.0040	-0.0023	-1.334	35.6%	33.1%

Table 5A: Full Sample Wealth Creation, Top 50 Global Firms, 1990-2018

Firm Name	Country	PERMCO/GVKEY*	Wealth Created (\$US millions)	Accumulated % of Global Gross Wealth Creation	Accumulated % of Global Net Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
APPLE INC	U.S.	7	1,006,035	1.51%	2.25%	21.00%	199002	201812
MICROSOFT CORP	U.S.	8048	954,787	2.95%	4.38%	17.77%	199002	201812
AMAZON COM INC	U.S.	15473	696,738	3.99%	5.94%	29.35%	199706	201812
ALPHABET INC	U.S.	45483	528,536	4.79%	7.12%	17.62%	200409	201812
EXXON MOBIL CORP	U.S.	20678	515,827	5.56%	8.27%	11.26%	199002	201812
BERKSHIRE HATHAWAY INC DEL	U.S.	540	438,959	6.22%	9.26%	12.12%	199002	201812
JOHNSON & JOHNSON	U.S.	21018	437,430	6.88%	10.23%	13.87%	199002	201812
WALMART INC	U.S.	21880	407,376	7.49%	11.14%	13.13%	199002	201812
TENCENT HOLDINGS LTD	Hong Kong	270615*	377,356	8.06%	11.99%	50.10%	200407	201812
ALTRIA GROUP INC	U.S.	21398	360,711	8.60%	12.79%	17.12%	199002	201812
NESTLE SA/AG	Switzerland	016603*	354,068	9.13%	13.59%	12.88%	199002	201812
PROCTER & GAMBLE CO	U.S.	21446	315,778	9.60%	14.29%	12.59%	199002	201812
INTEL CORP	U.S.	2367	312,027	10.07%	14.99%	16.23%	199002	201812
JPMORGAN CHASE & CO	U.S.	20436	298,095	10.52%	15.65%	9.16%	199002	201812
SAMSUNG ELECTRONICS CO LTD	South Korea	104604*	284,884	10.95%	16.29%	18.61%	199002	201812
HOME DEPOT INC	U.S.	5085	282,676	11.37%	16.92%	16.17%	199002	201812
COCA COLA CO	U.S.	20468	281,365	11.80%	17.55%	12.99%	199002	201812
ROCHE HOLDING AG	Switzerland	025648*	276,330	12.21%	18.17%	13.80%	199002	201812
CHEVRON CORP NEW	U.S.	20440	270,235	12.62%	18.77%	11.05%	199002	201812
MERCK & CO INC NEW	U.S.	21188	266,496	13.02%	19.37%	11.96%	199002	201812
UNITEDHEALTH GROUP INC	U.S.	7267	264,762	13.42%	19.96%	21.28%	199002	201812
NOVARTIS AG	Switzerland	101310*	249,576	13.79%	20.52%	9.96%	199002	201812
ORACLE CORP	U.S.	8045	245,690	14.16%	21.07%	19.43%	199002	201812
PFIZER INC	U.S.	21394	231,589	14.51%	21.59%	7.04%	199002	201812
VISA INC	U.S.	52983	231,202	14.85%	22.10%	22.85%	200804	201812
WELLS FARGO & CO NEW	U.S.	21305	230,975	15.20%	22.62%	10.74%	199002	201812
INTERNATIONAL BUSINESS MACHS COR	U.S.	20990	225,554	15.54%	23.12%	9.70%	199002	201812
CHINA MOBILE LTD	Hong Kong	065662*	219,742	15.87%	23.61%	10.22%	199711	201812
TAIWAN SEMICONDUCTOR MFG CO	Taiwan	201395*	217,222	16.20%	24.10%	16.12%	199502	201812
BOEING CO	U.S.	20315	215,011	16.52%	24.58%	10.72%	199002	201812
FACEBOOK INC	U.S.	54084	214,652	16.84%	25.06%	25.59%	201206	201812
GENERAL ELECTRIC CO	U.S.	20792	213,156	17.16%	25.54%	9.76%	199002	201812

CHINA CONSTR BANK CORP	China & Hong Kong	274364*	210,889	17.48%	26.01%	12.68%	200512	201812
PEPSICO INC	U.S.	21384	209,516	17.79%	26.48%	12.45%	199002	201812
CISCO SYSTEMS INC	U.S.	10486	208,676	18.11%	26.94%	9.60%	199003	201812
UNILEVER NV & UNILEVER PLC	Netherlands & U.K.	010845* & 010846*	205,942	18.42%	27.40%	11.61%	199002	201812
ABBOTT LABORATORIES	U.S.	20017	205,571	18.73%	27.86%	14.16%	199002	201812
MCDONALDS CORP	U.S.	21177	205,367	19.03%	28.32%	13.00%	199002	201812
MASTERCARD INC	U.S.	50700	202,779	19.34%	28.78%	32.18%	200606	201812
INDUSTRIAL & COMM BANKCHINA	China & Hong Kong	279378*	193,090	19.63%	29.21%	8.13%	200612	201812
DISNEY WALT CO	U.S.	20587	179,925	19.90%	29.61%	9.49%	199002	201812
AMGEN INC	U.S.	216	179,113	20.17%	30.01%	18.43%	199002	201812
TOYOTA MOTOR CORP	Japan	019661*	178,467	20.44%	30.41%	6.95%	199002	201812
TOTAL SA	France	024625*	173,932	20.70%	30.80%	9.95%	199002	201812
HSBC HLDGS PLC	U.K.	015509*	166,739	20.95%	31.17%	9.33%	199002	201812
VERIZON COMMUNICATIONS INC	U.S.	20288	164,482	21.19%	31.54%	6.77%	199002	201812
LVMH MOET HENNESSY LOUIS VUITTON	France	014447*	159,051	21.43%	31.89%	10.76%	199002	201812
L'OREAL SA	France	100581*	156,479	21.67%	32.24%	14.81%	199002	201812
3M CO	U.S.	21205	153,250	21.90%	32.59%	11.58%	199002	201812
COMCAST CORP NEW	U.S.	43613	152,602	22.13%	32.93%	10.59%	200212	201812

Table 5B: Full Sample Wealth Destruction, Bottom 20 Global Firms (1990-2018)

Firm Name	Country	PERMCO/GVKEY*	Wealth Created (\$US millions)	% of Global Gross Wealth Destruction	Accumulated % of Global Gross Wealth Destruction	Annualized Dollar Weighted Return	First Month	Last Month
PETROCHINA CO LTD	China & Hong Kong	133870*	-470,053	2.15%	2.15%	-9.70%	200006	201812
INDUSTRIAL BANK OF JAPAN LTD	Japan	015685*	-172,957	0.79%	2.95%	-13.36%	199002	200009
NIPPON TELEGRAPH & TELEPHONE	Japan	007908*	-163,989	0.75%	3.70%	0.08%	199002	201812
SUMITOMO MITSUI FINANCIAL GR	Japan	010137*	-152,908	0.70%	4.40%	-3.41%	199002	201812
BANK TOKYO-MITSUBISHI	Japan	015627*	-125,306	0.57%	4.97%	-8.91%	199002	200103
CHINA SHENHUA ENERGY CO LTD	China & Hong Kong	273153*	-117,850	0.54%	5.51%	-8.41%	200608	201812
FUJI BANK LTD	Japan	015556*	-109,675	0.50%	6.01%	-8.55%	199002	200009
DAI-ICHI KANGYO BANK LTD	Japan	015550*	-98,586	0.45%	6.46%	-8.04%	199002	200009
SAKURA BANK LTD	Japan	015624*	-94,908	0.43%	6.90%	-9.63%	199002	200103
SANWA BANK LTD	Japan	006775*	-94,634	0.43%	7.33%	-9.86%	199002	200103
WORLDCOM INC GA NEW	U.S.	61	-92,076	0.42%	7.75%	-50.47%	199002	200205
TOKYO ELECTRIC POWER CO HOLD	Japan	100688*	-89,891	0.41%	8.17%	-5.06%	199002	201812
VIAVI SOLUTIONS INC	U.S.	12583	-83,374	0.38%	8.55%	-15.99%	199312	201812
LUCENT TECHNOLOGIES INC	U.S.	31614	-82,061	0.38%	8.92%	-19.45%	199605	200611
NOMURA HOLDINGS INC	Japan	015613*	-79,571	0.36%	9.29%	-3.80%	199002	201812
UNICREDIT SPA	Italy	015549*	-73,572	0.34%	9.63%	-12.09%	199002	201812
ROYAL BANK OF SCOTLAND GROUP	U.K.	015634*	-72,443	0.33%	9.96%	-11.69%	199002	201812
GAZPROM PJSC	Russia	206454*	-65,712	0.30%	10.26%	-7.18%	201002	201812
WACHOVIA CORP 2ND NEW	U.S.	1869	-63,746	0.29%	10.55%	-25.39%	199002	200812
TOKAI BANK LTD	Japan	015658*	-63,317	0.29%	10.84%	-12.73%	199002	200103

Table 6: Concentration of Gross and Net Wealth Creation (1990-2018)

	Total Firms			Top 0.25% of Firms			Top 0.5% of Firms			Top 1% of Firms			Top 5% of Firms		
Sample	# Firms	Gross Wealth (\$US Bil.)	Net Wealth (\$US Bil.)	# Firms	% of Gross Wealth	% of Net Wealth	# Firms	% of Gross Wealth	% of Net Wealth	# Firms	% of Gross Wealth	% of Net Wealth	# Firms	% of Gross Wealth	% of Net Wealth
Global	61,100	66,570	44,739	153	37.23%	55.39%	306	49.08%	73.03%	611	61.98%	92.23%	3,055	88.31%	131.40%
Global (Excl. US)	44,034	32,308	15,955	111	34.18%	69.20%	221	46.32%	93.79%	441	59.94%	121.37%	2,202	87.70%	177.58%
By Development															
Developed	47,483	59,202	41,919	119	37.04%	52.32%	238	48.79%	68.91%	475	61.69%	87.12%	2,375	88.19%	124.55%
Developed (Excl. US)	30,417	24,939	13,135	77	34.66%	65.82%	153	46.73%	88.73%	305	60.28%	114.45%	1,521	87.86%	166.82%
Emerging	13,722	7,415	2,820	35	29.82%	78.40%	69	42.75%	112.41%	138	56.40%	148.30%	687	86.35%	227.06%
By Region															
North America	19,022	36,149	30,224	48	35.71%	42.71%	96	47.07%	56.30%	191	59.86%	71.59%	952	86.77%	103.78%
Europe	11,767	14,006	10,772	30	29.62%	38.51%	59	41.63%	54.13%	118	55.74%	72.47%	589	85.48%	111.15%
Asian Pacific	16,697	9,047	923	42	38.94%	381.60%	84	51.16%	501.32%	167	63.69%	624.12%	835	89.03%	872.39%
By Country															
Developed															
United States	16,872	33,992	28,607	43	36.26%	43.09%	85	47.28%	56.18%	169	60.05%	71.36%	844	86.83%	103.17%
Homeless Firms (US ADRs)	195	271	176	1	41.65%	63.89%	1	41.65%	63.89%	2	55.59%	85.27%	10	84.08%	128.97%
Canada	1,955	1,886	1,440	5	26.24%	34.36%	10	37.02%	48.48%	20	51.83%	67.88%	98	83.65%	109.54%
Austria	161	84	27	1	17.85%	54.95%	1	17.85%	54.95%	2	26.74%	82.33%	9	65.33%	201.10%
Belgium	289	359	286	1	31.20%	39.17%	2	38.76%	48.66%	3	42.93%	53.89%	15	72.84%	91.43%
Denmark	329	421	392	1	31.82%	34.16%	2	37.17%	39.90%	4	47.23%	50.70%	17	78.95%	84.75%
Finland	237	305	264	1	13.81%	15.96%	2	26.23%	30.33%	3	38.20%	44.17%	12	74.07%	85.65%
France	1,631	2,257	1,808	5	31.45%	39.26%	9	43.46%	54.26%	17	59.09%	73.77%	82	87.18%	108.84%
Germany	1,466	1,708	1,180	4	23.96%	34.67%	8	37.41%	54.12%	15	53.96%	78.07%	74	86.23%	124.75%
Greece	408	51	-67	2	45.00%	NA	3	51.72%	NA	5	60.18%	NA	21	85.90%	NA
Italy	590	653	217	2	25.15%	75.56%	3	30.65%	92.08%	6	45.10%	135.47%	30	79.87%	239.93%
Netherlands	323	888	791	1	16.05%	18.01%	2	29.43%	33.03%	4	49.64%	55.71%	17	86.77%	97.38%
Norway	481	316	241	2	35.43%	46.58%	3	48.41%	63.64%	5	57.34%	75.38%	25	81.49%	107.13%
Portugal	119	54	5	1	15.27%	152.97%	1	15.27%	152.97%	2	30.02%	300.77%	6	66.22%	663.42%
Spain	359	741	565	1	12.02%	15.77%	2	22.79%	29.92%	4	39.43%	51.76%	18	77.21%	101.34%
Sweden	930	835	780	3	18.38%	19.67%	5	28.32%	30.30%	10	45.73%	48.93%	47	81.21%	86.89%
Switzerland	398	1,615	1,530	1	21.93%	23.14%	2	39.04%	41.20%	4	57.78%	60.98%	20	81.37%	85.87%
United Kingdom	4,048	3,720	2,751	11	33.37%	45.12%	21	47.19%	63.81%	41	59.33%	80.23%	203	87.49%	118.30%
Australia	2,820	1,483	1,257	8	42.43%	50.04%	15	53.43%	63.01%	29	64.23%	75.75%	141	90.19%	106.37%
Hong Kong	2,256	2,831	2,330	6	37.43%	45.49%	12	48.55%	58.99%	23	61.71%	74.99%	113	88.54%	107.59%
Israel	597	144	127	2	16.50%	18.74%	3	22.67%	25.76%	6	35.77%	40.64%	30	67.82%	77.05%
Japan	4,524	2,235	-3,829	12	30.01%	NA	23	43.15%	NA	46	59.76%	NA	227	88.62%	NA
New Zealand	266	87	74	1	8.95%	10.60%	2	15.62%	18.50%	3	21.71%	25.72%	14	55.52%	65.76%
Singapore	1,021	485	376	3	27.12%	34.96%	6	45.09%	58.12%	11	56.60%	72.97%	52	86.09%	110.99%

South Korea	2,892	989	528	8	44.91%	84.09%	15	55.22%	103.41%	29	67.53%	126.45%	145	89.98%	168.49%
Taiwan	2,321	793	60	6	45.43%	602.73%	12	56.18%	745.41%	24	67.36%	893.74%	117	86.92%	1153.33%
Emerging															
Argentina	112	33	7	1	35.04%	167.02%	1	35.04%	167.02%	2	55.58%	264.92%	6	71.76%	342.04%
Brazil	341	430	259	1	16.10%	26.77%	2	27.70%	46.07%	4	41.42%	68.88%	18	73.64%	122.45%
China	3,610	2,240	-324	10	33.13%	NA	19	44.88%	NA	37	56.66%	NA	181	81.80%	NA
Colombia	62	98	90	1	27.86%	30.22%	1	27.86%	30.22%	1	27.86%	30.22%	4	49.57%	53.77%
India	3,731	1,620	1,214	10	35.68%	47.60%	19	48.56%	64.78%	38	62.33%	83.14%	187	90.15%	120.25%
Indonesia	602	312	185	2	27.54%	46.33%	4	47.69%	80.21%	7	65.72%	110.55%	31	89.79%	151.04%
Malaysia	1,315	338	150	4	23.58%	53.32%	7	32.04%	72.44%	14	47.86%	108.21%	66	84.70%	191.50%
Mexico	199	385	288	1	14.44%	19.28%	1	14.44%	19.28%	2	27.40%	36.59%	10	68.89%	91.99%
Nigeria	157	15	-23	1	25.91%	NA	1	25.91%	NA	2	51.78%	NA	8	84.06%	NA
Poland	962	141	74	3	33.87%	64.39%	5	44.26%	84.13%	10	63.35%	120.42%	49	94.14%	178.95%
Russia	268	313	81	1	19.33%	74.55%	2	35.92%	138.58%	3	48.05%	185.37%	14	91.04%	351.17%
Saudi Arabia	191	393	303	1	35.59%	46.09%	1	35.59%	46.09%	2	46.29%	59.96%	10	75.25%	97.47%
South Africa	845	488	230	3	28.94%	61.45%	5	38.69%	82.13%	9	49.94%	106.03%	43	87.46%	185.67%
Thailand	874	410	259	3	26.47%	41.91%	5	36.08%	57.12%	9	50.23%	79.52%	44	85.21%	134.90%
Turkey	335	48	-22	1	19.62%	NA	2	34.83%	NA	4	57.84%	NA	17	91.53%	NA
United Arab Emirates	118	151	48	1	35.80%	112.28%	1	35.80%	112.28%	2	52.36%	164.19%	6	84.64%	265.44%

Table 7: Cross-Country Analysis

	Panel A					Panel B				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
	Buy-and-Hold Returns > T-bill (%)					Gross Wealth Creation by the Top 0.5% (%)				
<i>Standard Deviation</i>	-7.586*** (-3.36)	-6.136*** (-3.32)			-7.751*** (-3.34)	9.071*** (3.12)	8.428*** (2.88)			8.774** (2.52)
<i>Skewness</i>	-0.207 (-0.84)	-0.517** (-2.46)			-0.257 (-0.96)	-0.093 (-0.29)	0.045 (0.13)			-0.017 (-0.04)
<i>Mean</i>		55.803*** (4.60)			58.123*** (4.47)		-24.739 (-1.29)			-29.454 (-1.51)
<i>Real GDP Growth %</i>			0.005 (0.11)		0.061 (1.54)			0.122** (2.34)		0.031 (0.53)
<i>GDP Per Capita (\$US thousands)</i>			0.010** (2.06)		0.006 (0.97)			0.011* (1.85)		0.018** (2.07)
<i>Individualism</i>				0.052 (0.13)	0.150 (0.39)				-0.362 (-0.76)	-0.874 (-1.53)
<i>Risk Taking</i>				0.056 (0.18)	0.180 (0.81)				0.492 (1.29)	0.303 (0.91)
<i>I_{Missing Individualism}</i>				-0.081 (-0.22)	-0.298 (-0.97)				-0.468 (-1.04)	-0.585 (-1.28)
<i>I_{Missing Risk Taking}</i>				0.076 (0.40)	-0.101 (-0.74)				0.101 (0.43)	0.114 (0.56)
<i>I_{Developed}</i>				0.205 (1.08)	0.077 (0.43)				0.119 (0.51)	-0.107 (-0.40)
<i>Number of Stocks</i>					0.003 (0.15)					0.040 (1.33)
<i>Constant</i>	1.067*** (3.25)	0.889*** (3.32)	-0.589** (-2.52)	-0.495** (-2.56)	0.456 (0.98)	-1.914*** (-4.53)	-1.835*** (-4.33)	-1.234*** (-4.41)	-0.467* (-1.99)	-1.885** (-2.71)
Adj. R ²	0.30	0.54	0.09	-0.04	0.54	0.19	0.21	0.09	-0.06	0.29
N	41	41	41	41	41	41	41	41	41	41

Data Appendix: Sources, sample election, and filters

In this appendix, we describe our data sources, sample selection, as well as the error filters and corrections that we rely on. While many authors study returns to equity portfolios or country-level market outcomes, we wish to study a broad cross-section of individual stocks. Data errors, resulting from either the inclusion of an incorrect observation or the filtering of a correct observation, are potentially more damaging in our study of the distribution of compound returns to individual stocks as compared, for example, to a study of mean returns to value-weighted stock portfolios. Prior studies typically exclude or winsorize unusually large returns. While we also exclude data for certain stocks and time periods where it is probable that large errors are prevalent, in other cases we alter the raw data (e.g., shifting a decimal) on occasions where it seems likely that errors can be corrected or mitigated. While our algorithms are undoubtedly imperfect, the goal is to mitigate potentially influential data errors, while retaining as much data as possible, including outlier observations that do not represent errors.

Our focus is on outcomes to investors in publicly traded stocks. We therefore exclude preferred stocks and real estate investment trusts. Also, to avoid double counting of investment results, we exclude mutual funds, hedge funds and exchange-traded funds. Further, for stocks traded on multiple exchanges we avoid double counting by focusing only on outcomes in a single “home” country.

1. Data sources and return definitions

For US firms data is retrieved from the CRSP monthly stock files. For Canadian firms, data is obtained from Compustat/North America. For all other countries, we retrieve data from Compustat/Global. We include common stocks listed on the major stock exchange(s) in each country. Local currency outcomes are converted into US dollars using end-of-month exchange rates from Compustat/Global and Compustat/North America. We obtain the 30-day US Treasury bill data from Professor Ken French’s data library.

We rely on several Compustat data libraries, including the Security Daily library (secd) and the Security Monthly library (secm) for Canadian stocks and ADRs, and for other countries the Security Daily library (gsecd). We also obtain data from the Compustat - Fundamentals Quarterly library (fundq). The CRSP/Compustat Merged (CCM) library is used to link CRSP stocks to Compustat companies and securities. We rely also on reference data, including Country (r_country), Exchange Trading Codes (r_ex_codes), and Global Industry Codes (r_giccd). We obtain *security* descriptor data from the Data Group (security), including the Unique Issue ID (IID), Security Description (DSCI), Stock Exchange (EXCHG), Stock Exchange Country Code (EXCNTRY), and Issue Type (TPCI). We obtain *company* descriptor data from the Data Group (company), including Global Company Key (GVKEY), Primary Issue Tag - Rest of World (PRIROW), Primary Issue Tag - Canada (PRICAN), Primary Issue Tag - United States of America (PRIUSA), Business Description (BUSDESC), Company Name (CONM), Company Legal Name (CONML), ISO Country Code - Headquarters (LOC), GICS Groups (GGROUP), GICS Industries (GIND), and GICS Sub-Industries (GSUBIND).

For US stocks, CRSP reports monthly rates of return and end-of-month share prices measured in US dollars. For international stocks, returns must be computed from daily data series included in the Compustat databases. We rely on the following Compustat variables:

- PRCCD -- Price - Close - Daily
- CSHOC -- Shares Outstanding
- AJEXDI -- Adjustment Factor (Issue)-Cumulative by Ex-Date
- TRFD -- Daily Total Return Factor
- CURCDD -- ISO Currency Code - Daily
- QUNIT -- Price Quotation Unit
- ADRRC -- ADR Ratio - Daily

We compute the time t share price in US dollars as $PRCFX(t) = PRCCD(t)/FX(t)/QUNIT(t)$, where $FX(t)$ is the currency exchange rate per US dollar provided by Compustat from the Data Group

(*exrt_mth*). Market capitalization in US dollars at time t is $SZ(t) = PRCFX(t) \times CSHOC(t)$. (For ADR stocks, CSHOC is modified as CSHOC/ADRR, to obtain total shares outstanding.) We extract monthly data from the daily data by focusing on the latest date with a non-zero daily closing price. To retain a stock/month in the sample, we require at least five daily observations with positive closing prices. The month t return in dollars, inclusive of dividends is computed as $RET(t) = [PRCFX(t) / AJEXDI(t) \times TRFD(t)] / [PRCFX(tlag) / AJEXDI(tlag) \times TRFD(tlag)] - 1$. Typically, $tlag$ refers to month $t-1$, but in some cases where data is missing the return is computed over a longer horizon.

2. Countries included in the study

We obtain data for stocks in the forty largest countries, based on average GDP during the 1990-2018 period as reported by the International Monetary Fund (IMF). In addition, we include Singapore and New Zealand because of their status as prominent developed economies. We exclude Iran even though it is among the forty largest economies, because stock return data is available for only ten years. We also include those stocks that are traded as American Deposit Receipts (ADRs), but do not have during the period of ADR trading publicly listed primary common stock data in Compustat. Including these “homeless ADRs” as a distinct “country”, our final sample includes stocks for forty-two countries.

We assign each country as a developed or an emerging market based on IMF classifications. The developed markets include Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, South Korea, Spain, Sweden, Switzerland, Taiwan, the U.K., and the U.S. The emerging markets included in the study are Argentina, Brazil, China, Columbia, India, Indonesia, Malaysia, Mexico, Nigeria, Poland, Russia, Saudi Arabia, South Africa, Thailand, Turkey, and the United Arab Emirates. We further group the developed markets into three regions: North America (Canada and the U.S.), Europe (Austria, Belgium, Denmark, Finland, France, Germany, Greece, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the U.K.), and Asia Pacific (Australia, Hong Kong, Israel, Japan, New Zealand,

Singapore, South Korea, and Taiwan). The remaining countries are not grouped into a region because theirs are typically smaller markets, scattered across Latin America, Asia, and the Middle East.

3. Selecting common stocks

We identify US common stocks based on the CRSP sharecode variable, SHRCD = 10, 11, and 12. The identification of common stocks outside the US is more complicated. We initially select securities with Compustat Issue Type TPCI = '0' as well as ADR stocks with TPCI = 'F'. We exclude securities that contain the “%” symbol in the DSCI field (as they are likely preferred stocks with a fixed dividend). We also exclude stocks where the EXCHG field contains “Broker”, “Fund Manager”, “Fund Managers”, “OTC”, “OTC Bulletin Board”, “Other-OTC”, “Subsidiary/Private”, “Unlisted Evaluated Equity”, “Unlisted Securities Market”, “Non-traded Company or Security”, or “Stock Connect”.

4. Main stock exchange filters

Although a country may have multiple stock exchanges, some can be very small. We exclude minor exchanges based on dollar trading volume. Since volume data can contain errors, the assessment of the exchanges to be excluded relies on some filters. We first compute dollar trading volume on a stock/month basis as $DVOL = CSHTR \times PRCFX$, where $CSHTR$ is share trading volume and $PRCFX$ is the stock price per share in US dollars. We truncate share turnover ($CSHTR/shares\ outstanding$) at 100, and for each country, winsorize monthly dollar trading volume at the top one percent. We exclude from our study stocks listed on exchanges that account for less than two percent of total dollar trading volume among exchanges in that country, as well as stocks listed on exchanges that never have more than thirty listed stocks.²³ Having implemented these filters, our sample contains 67,843 stocks and 8,481,540 stock-month observations for the January 1990 to December 2018 period.

5. Screening investment funds and trusts

²³ The latter condition removes only two exchanges, the Moscow Stock Exchange and Nasdaq Dubai.

Visual examination reveals that the set of non-US securities with Issue Type TPCI = ‘0’ contains a substantial number of investment funds and trusts, including mutual funds, hedge funds, and exchange-traded funds. To avoid double counting and to focus on returns and wealth creation from publicly traded common stocks, we seek to exclude these securities. We do so as follows:²⁴

- a. First, we identify securities that are likely to be Real Estate Investment Trusts, based on GSUBIND = 40401010 or GIND = 404020.
- b. Second, we identify securities that are likely to be funds or trusts, based on their company name variables. In particular, we focus on securities for which CONM or CONML contains “_fund”, “_trust”, “venture_capital_trust”, “_vct”, or “_reit”.
- c. Third, we consider the information provided by the business description variable BUSDESC. To avoid false positives in the following step, we remove the whole words that represent company name (i.e., CONM or CONML) as well as the following words and phrases (when they occur) from the BUSDESC variable: ‘_fund_advisor’, ‘_fund_manage’, ‘_fund_benchmark’, ‘_fund_raising’, ‘_fund_administration’, ‘_fund_transfer’, ‘_fund_service’, ‘_fund_product’, ‘_fund_sponsor’, ‘_fund_plan_sponsor’, ‘_fund_corp’, ‘_fund_company’, ‘_fund_group’, ‘_trust_advisor’, ‘_trust_bank’, ‘_trust_manager’, ‘_trust_sponsor’, or ‘_reit_manager’. We then identify securities that may be funds or trusts based on BUSDESC, if any of the following are true:
 - (i) BUSDESC contains any word patterns in the sequences of {str1,str2,str3} or {str1,str2,str4,str3}, where str1 = {‘_is_a_’, ‘_was_a_’, ‘_as_a_’, ‘_is_an_’, ‘_was_an_’, ‘_as_an_’}, str2 = {‘_fund_’, ‘_fund’, ‘_etf’, ‘_reit’, ‘property_trust’, ‘investment_trust’, ‘venture_capital_trust’, ‘_vct’}, str3 = {‘.’} and str4 = {‘firm’, ‘company’, ‘companies’, ‘_by_’, ‘_for_’, ‘_of_’, ‘_to_’, ‘_through_’}.

²⁴ Here, we use the underline character () to indicate a blank space.

- (ii) BUSDESC contains {'_fund_invests', '_fund_invests', '_fund_engages', '_fund_engages', '_fund_operates', '_fund_operates'} or {'_trust_invests', '_trust_invests', '_trust_engages', '_trust_engages', '_trust_operates', '_trust_operates'}.
- (iii) BUSDESC contains {'_reit', '_an_investment_trust', or '_real_estate_investment_trust'} in the beginning 100 characters and there is none of {'_by_', '_for_', '_of_', '_to_', '_through_'} prior to the indicated string found.

Having completed steps (a) to (c) above, we exclude those securities that are identified as potential mutual funds or trusts, except for those that can be identified as banks (an example being Northern Trust Corporation). We identify securities as banks, and do not exclude them, if they meet condition (b) and the industry code GGROUP equals 4010, if the company name fields CONM or CONML contains 'bank' at the beginning or end, or if the company name fields contain "trust_&_banking" or "securities_co".

6. Decimal errors

Visual examination indicates that data errors for non-US stocks are not infrequent, especially for emerging markets and for smaller capitalization issues. One form of data error is an erroneous shift of the decimal, either to the left or to the right, for one or more periods. We seek to repair temporary decimal shift errors in six key variables that are used to compute market capitalization and returns for non-US stocks. These are *PRC*, the stock price per share; *AJEX*, a cumulative adjustment factor; *TRF*, a total return factor; *CSHO*, common shares outstanding; *QUNIT* the price quotation unit; and, for ADRs, *ADRR*, the ADR-to-regular stock ratio.

Let X be a variable of interest and define $RatX(t; s_1; s_2)$ as the ratio of $X(t + s_1)/X(t + s_2)$, where t identifies the current month.

- a. Repairing a decimal error that pertains to only a single date (immediate reversal). An example would be the sequence of stock prices 8.56, 69.50, and 7.32 in $t = -1, 0$, and $+1$. The natural

repair would be to use $6.95 = 69.50/10$ to replace 69.50 at $t=0$. More broadly, we proceed as follows. If $RatX(t; 0; -1) > 5 \times 10^{N-1}$ and $RatX(t; 0; 1) > 5 \times 10^{N-1}$, we replace $X(t)$ by $Xa(t) = X(t)/10^N$, where N is the maximal positive integer such that $5 \times 10^{N-1} < \min\{RatX(t; 0; -1), RatX(t; 0; 1)\}$. In other words, $X(t)$ is scaled by 10 if both $RatX(t; 0; -1)$ and $RatX(t; 0; 1)$ are in the interval $[5, 50)$ and scaled by 100 in the interval $[50, 500)$, etc. Analogously, if $RatX(t; 0; -1) < 1/(5 \times 10^{N-1})$ and $RatX(t; 0; 1) < 1/(5 \times 10^{N-1})$, we replace $X(t)$ by $Xa(t) = X(t) \times 10^N$, where N is the maximal positive integer such that $1/(5 \times 10^{N-1}) > \max\{RatX(t; 0; -1), RatX(t; 0; 1)\}$. That is, $X(t)$ is multiplied by 10 if both $RatX(t; 0; -1)$ and $RatX(t; 0; 1)$ are in the interval $[1/50, 1/5)$ and multiplied by 100 in the interval $[1/500, 1/50)$ and so on so forth.

- b. Repairing decimal errors that persist up to three periods. An example is the Brazilian stock MMX MINERACAO E METALICOS SA, with CSHO (shares outstanding) values of 305.12 from August 2009 to January 2010, 35.242 from February 2010 to April 2010, and 472.973 from May 2010 to December 2010. In this case, we would like to replace the value 35.242 with 352.42 during the three periods that it occurs. Formally, we search for $nlag$ from 1 to 3 and $n fwd$ from 1 to 3. If $RatX(t; 0; -nlag) > 5 \times 10^{N-1}$ and $RatX(t; 0; n fwd) > 5 \times 10^{N-1}$ together with the conditions that $RatX(t; -n1; -nlag) > 5 \times 10^{N-1}$, $Rat(t; n2; n fwd) > 5 \times 10^{N-1}$, and $|Rat(t; 0; n2) - 1| < 30\%$ for all $n1 = 1, \dots, nlag-1$ and for all $n2 = 1, \dots, n fwd-1$, we then replace $X(t)$ by $Xa(t) = X(t)/10^N$, where N is the maximal positive integer such that $5 \times 10^{N-1} < \min\{RatX(t; 0; -nlag), RatX(t; 0; n fwd)\}$. On the other hand, if $RatX(t, 0, -nlag) < 1/(5 \times 10^{N-1})$ and $RatX(t, 0, n fwd) < 1/(5 \times 10^{N-1})$ together with the conditions that $RatX(t; -n1; -nlag) < 1/(5 \times 10^{N-1})$, $RatX(t; n2; n fwd) < 1/(5 \times 10^{N-1})$, and $|RatX(t; 0; n2) - 1| < 30\%$ for all $n1$ and all $n2$ defined above, we then replace $X(t) = X(t) \times 10^N$, where N is the maximal positive integer such that $1/(5 \times 10^{N-1}) > \max\{RatX(t; 0; -nlag), RatX(t; 0; n fwd)\}$.

- c. We first apply the 3-period correction procedure to {'TRF', 'QUNIT', 'ADRR'} independently. For share price and shares outstanding we apply the 3-period decimal error correction procedure to the adjusted share price (ex-dividend) $adjPRC = PRCFX/AJEX$ and the adjusted shares outstanding $adjCSHO = CSHOC \times AJEX$.

7. Merging Compustat North America data for common stock selection

The variable CSHOC (shares outstanding at the security level) is not available prior to April 1998 in the Compustat North America data, which we rely on for Canadian stocks and ADRs. As a consequence, we must rely for the pre-1998 period on data regarding shares outstanding at the firm level. We proceed as follows:

- a. For Canadian stocks and US ADR stocks in Compustat North America, we use data from Compustat secd data library if available and otherwise use the data from the Compustat secm data library. If secd data is available for less than five daily observations, we drop the month.
- b. If CSHOC is missing, we delete observations prior to April 1998 if aggregate stock-level *CSHOC* summed over stock issues has an unexplained jump (up or down) of 10% or more at the first available fiscal quarter end. Otherwise, we use the proportional firm-level shares outstanding provided by Compustat - Fundamentals Quarterly library (fundq) to replace *CSHOC*. For ADR stocks, we drop observations if ADRR is missing.
- c. We select unique stock issues using GVKEY, IID (unique issue ID), and DSCI. For companies with stock listed in both Canada and the US, we focus on the stock with the longest period of listing. If DSCI for the primary issue contains the keyword of "CL" or "SER" (e.g., CL A), which indicates multiple share classes, we continue to search for other IIDs (e.g., CL B, CL C, etc.) with the same GVKEY. Specifically, for each company j , we select the fiscal quarter end $DATE(j, q)$ such that its stock-level shares outstanding $CSHOC(js, qm)$ are available for the first time at $qm = DATE(j, q)$, where $js = j1, j2, \dots, jN$ for N different unique share classes. If $|sum(CSHOC(js, qm))/CSHOC(j, q) - 1| < 10\%$, we then fill the missing value for

$CSHOC(js, m)$ in month m with $w(js) \times CSHOQ(j, q^*) \times AJEX(js, q^*)/AJEX(js, m)$, where the weight $w(js) = CSHOC(js, qm)/\text{sum}\{CSHOC(js, qm); js = j1, j2, \dots, jN\}$ and q^* is the quarter that exceeds month m but no more than one year.

8. Filters based on trading volume, share price, market capitalization, and initial errors

Our repair of decimal errors described in Section 6 does not address all types of errors that are observable in the data. Visual examination indicates that the remaining errors tend to be most frequent in low trading volume, low share price, and low market capitalization stocks, and also in the earliest months that a stock is contained in the data. We further filter securities according to the following sequence.

- a. *Trading volume filter*: We compute the average number of daily observations with positive trading volume for each stock and month. We then eliminate from the study the two percent of stocks with the lowest average. This filter reduces the sample to 67,164 stocks and 8,466,048 stock-month observations.
- b. *AJEX and QUNIT filtering*: We drop stocks where the adjustment factor *AJEX* ever takes the value of zero. We also drop stocks with changes in *QUNIT* (the price quotation unit) in the absence of a change in the currency code *CURCD*, and where the contemporaneous change in *adjPRCFX* (the adjusted share price in US dollars) is higher than 50% or below -50%. This filter reduces the sample to 67,145 stocks and 8,464,493 stock-month observations.
- c. *Low share price and low market capitalization filtering*: We delete the remaining history for any stock if its share price drops below \$US 0.01 or its market capitalization drops below \$US 1 million. All observations are deleted for stocks where the initial observation indicates a share price below \$US 0.01 or market capitalization below \$US 1 million. After this filter, the sample contains 65,680 stocks and 7,855,410 stock-month observations.

- d. *Filtering data gaps and measurement of returns*: In those cases where the data contain gaps of more than 11 months, we drop the first month after the data resumes.²⁵ In cases where there is a gap of eleven or fewer months we measure the return by $RET(t) = adjPRCFX(t)/adjPRCFX(tlag) \times TRF(t)/TRF(tlag) - 1$, where TRF is the Compustat adjustment factor for cash equivalent distributions. If TRF is missing, it is filled with the value within the past one year if available, or the value within the future one year if available. If no TRF observation is available with one year, the return is set to be missing. With these filters, the sample is reduced to 65,242 stocks and 7,785,937 stock-month observations.
- e. *Adjusted shares outstanding filtering*: We observe some large changes in the number of adjusted (for stock splits) shares outstanding, in the absence of commensurate changes in market capitalization. With the exception of stocks listed in China, where reverse mergers are common (see Lee, Qu, and Shen, 2017), we deem monthly changes in adjusted shares outstanding by a factor of more than 5x to be indicative of likely data errors. Letting $adjCSHO$ denote adjusted shares outstanding and SZ denote market capitalization, we proceed as follows:
- (1) Define an observation as a jump if
 - $adjCSHO(t)/adjCSHO(tlag) \geq 5$ and $SZ(t)/SZ(tlag) \geq 2.5$ (for China: $adjCSHO(t)/adjCSHO(tlag) \geq 50$ and $SZ(t)/SZ(tlag) \geq 25$) or
 - $adjCSHO(t)/adjCSHO(tlag) \leq 0.2$ and $SZ(t)/SZ(tlag) \leq 0.4$.
 - (2) If any jump appears during the lesser of the first 24 months of data for the firm or during the first 20% of the observation for that firm, we delete all observations up to and including the jump.
 - (3) For observations not deleted by criterion (2) above, we proceed as follows.

²⁵ When we compare each stock's accumulated returns to benchmarks defined by the Treasury bill rates or value-weighted market returns we omit the same months from the benchmark returns.

- For paired jumps (i.e., for jumps that are reversed by an offsetting jump), we adjust $adjCSHO$ inside the interval smoothly. Specifically, we use the smaller value between two ends, $\min\{adjCSHO(t-1), adjCSHO(s)\}$, to replace the values of $adjCSHO$ inside the interval $[t, s-1]$.
- For unpaired jumps (where there is no reversal), we adjust $adjCSHO$ to be smaller for observations before or after the jumps. For up-jumps, $adjCSHO$ starting at t until the end of sample is multiplied by $adjCSHO(t-1)/adjCSHO(t)$. For down-jumps, $adjCSHO$ since the beginning and ending at t is multiplied by $adjCSHO(t)/adjCSHO(t-1)$.

We then recompute the market capitalization SZ based on the updated $adjCSHO$ data. After these corrections, the sample continues to consist of 65,242 stocks, but is reduced to 7,783,347 stock month observations.

f. *Filtering market capitalization*: We identify cases where the market capitalization variable (SZ) jumps in a manner not supported by returns or shares outstanding. We proceed as follows:

(1) Define an observation as a jump if

- $SZ(t)/SZ(tlag) - 1 > 9$ and $RET < 2$ or
- $SZ(t)/SZ(tlag) - 1 < -0.9$ and $RET > -0.5$.

(2) If any jump appears during the lesser of the first 24 months of data for the firm or during the first 20% of the observation for that firm, we delete all observations up to and including the jump.

(3) For observations not deleted by the above criterion, we adjust the size variable. In the case of paired jumps (where the change is reversed), we define SZ within the interval smoothly.

Specifically, we use the smaller value between two ends, $\min\{SZ(t-1) \times (1 + RETX(t), SZ(s)/(1 + RETX(s))\}$, to replace the values of SZ within the interval $[t, s-1]$. In the case of unpaired jumps $[t, s]$ (when the change is not reversed), we adjust SZ downward before or after the jumps. For up-jumps, SZ starting at t until the end is multiplied by

$SZ(t-1)/SZ(t) \times (1 + RETX(t))$. For down-jumps, SZ since beginning and ending at t is multiplied by $SZ(t)/SZ(t-1)/(1 + RETX(t))$.

After implementing these filters, the sample continues to contain 65,242 stocks, while the number of stock-month observations is reduced to 7,783,290.

- g. *Filtering returns*: We correct large returns that are not accompanied by commensurate changes in market capitalization. In particular, we delete observations where
 (i) $RET > 0.8$ and $SZ(t)/SZ(tlag) - 1 < 0.5$ or (ii) $RET < -0.8$ and $SZ(t)/SZ(tlag) - 1 > -0.5$. After this filter, the sample continues to contain 65,242 stocks, but stock-month observations are reduced to 7,782,964.
- h. *Filtering initial share price and market cap*: We observe a number of erroneous share price ($adjPRCFX$) and market capitalization observations that were not captured by earlier filters, particularly in the first few observations after a stock enters the database. We delete observations that are among the first three for the stock if (i) $adjPRCFX(t)/adjPRCFX(tlag) > 10$ or $SZ(t)/SZ(tlag) > 10$ or (ii) $adjPRCFX(t)/adjPRCFX(tlag) < 1/10$ or $SZ(t)/SZ(tlag) < 1/10$. After implementing this filter the sample contains 65,241 stocks and 7,782,602 stock-month observations.
- i. *Treatment of inactive returns, delisting returns, and short-lived stocks*: We observe a number of non-US stocks for which the security status (SECSTAT) variable indicates that it is active, but for which return data is unavailable (as the identical price is reported in each month) for twelve or more months before the end of the sample. We view these stocks to be effectively delisted and delete observations with identical share prices and following Shumway (1997), we set the final return on these stocks to -30%. For non-US stocks that are indicated to have been delisted, and where the delisting reason is bankruptcy (DLRSN = 02) or liquidation (DLRSN = 03), we also set the final return to -30%. For US stocks we incorporate CRSP delisting returns, and also follow Shumway (1997). In particular, if the delisting return is available, the return is adjusted as $(1 +$

$return) \times (1 + delisted\ return) - 1$. If the delisting return is missing and the delisting code is 500, 520, 551-573, 580, 574, or 584, we set the return to -30%. Finally, we delete stocks with less than six monthly observations in total. After these filters, the sample contains 61,981 stocks and 7,676,602 stock-month observations.

9. Primary issues and multiple share class selection:

Some sample firms issue multiple classes of common stock, and some stocks are traded in more than one country. When studying overall outcomes to a firm's shareholders it is natural to focus on results that are aggregated across share classes. Also, to avoid double counting, we assign each firm to just one country. We proceed as follows.

We identify 17,728 CRSP stocks and 45,344 Compustat stocks. We are able to link 17,232 of the CRSP stocks to the Compustat GVKEY variable. Of these, 713 CRSP stocks were also identified as non-US Compustat stocks. Examples include UNILEVER NV, ROYAL BANK CANADA MONTREAL, MEDTRONIC PLC, and AON PLC. Among these issues, (as well as issues listed in multiple non-US countries), we retain in the study only the issue with the longest listing period. If there are two primary issues with the same listing period, we retain the stock in the headquarter country (LOC). Among the 713 CRSP firms also present in the non-US Compustat data, we ultimately assign 295 as US firms and 418 as non-US firms. We retain (homeless) ADR stocks in the sample only if the database contains no other stock with the same GVKEY during the listing time of the ADR stock.

To identify multiple securities issued by the same US (CRSP) firm, we focus on stocks with differing PERMNO but common PERMCO. For non-US stocks we retain the primary issue with IID (unique issue ID) = PRICAN in Canada or IID = PRIROW in the rest of the world, and we identify multiple class of stock from the same issuer based on the keywords of "CL" or "SER" contained in the Security Description (DSCI).

In a few cases a company's primary issue may not be in the form of common stock (examples include Volkswagen's "preference shares" and Roche's "bearer shares"), even though the primary issues retain the key economic characteristics of common stock. Therefore, for (only) those companies whose primary issues are not common stocks, we retain in the sample *special* stocks (TPCI = "Q"), and *preferred* stocks (TPCI = '1').

In addition, we note that the sample contains seven dual-listed companies, where economically identical stocks trade with different company identifiers. We follow Jong, Rosenthal, and Van Dijk (2009) and Bedi, Richards, and Tennant (2003) to manually combine these. The companies are:

- a. UNILEVER NV & UNILEVER PLC; Netherlands & United Kingdom.
- b. RIO TINTO GROUP (AUS) & RIO TINTO GROUP (GBR); Australia & United Kingdom.
- c. BHP GROUP LTD & BILLITON PLC; Australia & United Kingdom.
- d. RELX NV & RELX PLC; Netherlands & United Kingdom.
- e. CARNIVAL CORP & CARNIVAL CORP/PLC (GBR); United Kingdom & United States.
- f. MONDI PLC/LTD (GBR) & MONDI PLC/LTD (ZAF); South Africa & United Kingdom.
- g. INVESTEC LTD & INVESTEC PLC; South Africa & United Kingdom.

After these adjustments, the final sample contains 61,100 firms, 61,981 stocks and 7,676,602 stock-month observations from January 1990 to December 2018. Among the 7.677 million stock-months in the final sample, 1.6% contain either a return or a market capitalization observation that has been modified by our algorithms as described here.

Table A1: Wealth Creation by Country, Top 20 United States Firms

Firm Name	PERMCO	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
APPLE INC	7	1,006,035	1.51%	2.96%	21.00%	199002	201812
MICROSOFT CORP	8048	954,787	1.43%	2.81%	17.77%	199002	201812
AMAZON COM INC	15473	696,738	1.05%	2.05%	29.35%	199706	201812
ALPHABET INC	45483	528,536	0.79%	1.55%	17.62%	200409	201812
EXXON MOBIL CORP	20678	515,827	0.77%	1.52%	11.26%	199002	201812
BERKSHIRE HATHAWAY INC DEL	540	438,959	0.66%	1.29%	12.12%	199002	201812
JOHNSON & JOHNSON	21018	437,430	0.66%	1.29%	13.87%	199002	201812
WALMART INC	21880	407,376	0.61%	1.20%	13.13%	199002	201812
ALTRIA GROUP INC	21398	360,711	0.54%	1.06%	17.12%	199002	201812
PROCTER & GAMBLE CO	21446	315,778	0.47%	0.93%	12.59%	199002	201812
INTEL CORP	2367	312,027	0.47%	0.92%	16.23%	199002	201812
JPMORGAN CHASE & CO	20436	298,095	0.45%	0.88%	9.16%	199002	201812
HOME DEPOT INC	5085	282,676	0.42%	0.83%	16.17%	199002	201812
COCA COLA CO	20468	281,365	0.42%	0.83%	12.99%	199002	201812
CHEVRON CORP NEW	20440	270,235	0.41%	0.79%	11.05%	199002	201812
MERCK & CO INC NEW	21188	266,496	0.40%	0.78%	11.96%	199002	201812
UNITEDHEALTH GROUP INC	7267	264,762	0.40%	0.78%	21.28%	199002	201812
ORACLE CORP	8045	245,690	0.37%	0.72%	19.43%	199002	201812
PFIZER INC	21394	231,589	0.35%	0.68%	7.04%	199002	201812
VISA INC	52983	231,202	0.35%	0.68%	22.85%	200804	201812

Table A2: Wealth Creation by Country, Top 20 Homeless Firms (US ADRs)

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ALIBABA GROUP HLDG	020530	112,676	0.17%	41.65%	10.56%	201410	201812
BAIDU INC	164532	37,708	0.06%	13.94%	22.88%	200509	201812
NETEASE INC	137435	32,347	0.05%	11.96%	46.20%	200202	201812
NEW ORIENTAL ED & TECH	175442	7,646	0.01%	2.83%	19.86%	200610	201812
SUZANO PAPEL E CELULOSE SA	200512	7,593	0.01%	2.81%	17.47%	200202	201812
HUAZHU GROUP LIMITED - ADR	184453	6,860	0.01%	2.54%	27.55%	201005	201812
SOC QUIMICA Y MINERA DE CHI	028883	6,052	0.01%	2.24%	26.47%	200202	201812
TAL EDUCATION GROUP	185733	5,760	0.01%	2.13%	38.86%	201011	201812
JAMES HARDIE INDUSTRIES PLC	100477	5,658	0.01%	2.09%	12.19%	200202	201812
AUTOHOME INC -ADR	019295	5,148	0.01%	1.90%	27.54%	201401	201812
WEIBO CORP	020004	3,029	0.00%	1.12%	21.71%	201405	201812
VIPSHOP HOLDINGS LTD -ADR	170761	2,963	0.00%	1.10%	72.23%	201204	201812
MOMO INC -ADR	022164	2,740	0.00%	1.01%	26.65%	201501	201812
51JOB INC -ADR	160855	2,729	0.00%	1.01%	11.53%	200411	201812
AMARIN CORP	028027	2,268	0.00%	0.84%	10.91%	200202	201812
ADVANCED ACCELERATR APP -ADR	022266	2,183	0.00%	0.81%	65.14%	201512	201802
PINAFOR HOLDINGS BV	017225	2,036	0.00%	0.75%	10.17%	200202	201009
EMBOTELLADORA ANDINA SA	030436	1,741	0.00%	0.64%	14.62%	200202	201812
TERNIUM SA -ADR	165927	1,716	0.00%	0.63%	3.54%	200603	201812
SHANDA INTERACTIVE-ADR	266320	1,594	0.00%	0.59%	13.15%	200406	201202

Table A3: Wealth Creation by Country, Top 20 Canada Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ROYAL BANK OF CANADA	015633	130,557	0.20%	6.92%	14.34%	199002	201812
TORONTO DOMINION BANK	015706	105,357	0.16%	5.59%	13.21%	199002	201812
BCE INC	002137	100,594	0.15%	5.33%	18.80%	199002	201812
BANK OF NOVA SCOTIA	015582	80,590	0.12%	4.27%	15.87%	199002	201812
CANADIAN NATIONAL RAILWAY CO	002696	77,763	0.12%	4.12%	23.50%	199512	201812
BANK OF MONTREAL	015580	57,325	0.09%	3.04%	14.82%	199002	201812
CANADIAN IMPERIAL BANK	015581	44,855	0.07%	2.38%	12.40%	199002	201812
SUNCOR ENERGY INC	015070	34,342	0.05%	1.82%	10.60%	199304	201812
THOMSON-REUTERS CORP	015704	33,979	0.05%	1.80%	8.18%	199002	201812
SHELL CANADA LTD -CL A	009652	32,971	0.05%	1.75%	15.64%	199002	200704
GREAT-WEST LIFECO INC	016431	31,126	0.05%	1.65%	15.51%	199002	201812
CANADIAN PACIFIC RAILWAY LTD	002698	30,888	0.05%	1.64%	9.14%	199002	201812
ENCANA CORP	011781	30,877	0.05%	1.64%	13.28%	199002	201812
CANADIAN NATURAL RESOURCES	015055	30,458	0.05%	1.61%	16.09%	199002	201812
IMPERIAL OIL LTD	005903	27,546	0.04%	1.46%	8.27%	199002	201812
ALCAN INC	001243	27,168	0.04%	1.44%	12.23%	199002	200711
NUTRIEN LTD	016582	25,529	0.04%	1.35%	16.83%	199002	201812
ENBRIDGE INC	006135	25,485	0.04%	1.35%	12.21%	199002	201812
POWER FINANCIAL CORP	015607	25,209	0.04%	1.34%	14.47%	199002	201812
TRANSCANADA CORP	010671	25,034	0.04%	1.33%	8.43%	199002	201812

Table A4: Wealth Creation by Country, Top 20 Austria Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
OMV AG	102798	14,963	0.02%	17.85%	9.01%	199002	201812
VERBUND AG	102797	7,457	0.01%	8.90%	10.72%	199002	201812
VOESTALPINE AG	226156	6,352	0.01%	7.58%	11.21%	199511	201812
ANDRITZ AG	246759	6,085	0.01%	7.26%	26.39%	200108	201812
ERSTE GROUP BK AG	214659	5,375	0.01%	6.41%	4.33%	199801	201812
BOEHLER-UDDEHOLM	203429	4,733	0.01%	5.65%	17.43%	199510	200808
TELEKOM AUSTRIA AG	141239	4,047	0.01%	4.83%	7.02%	200012	201812
FLUGHAFEN WIEN AG	222139	3,035	0.00%	3.62%	8.26%	199302	201812
MAYR-MELNHOF KARTON AG	210823	2,718	0.00%	3.24%	8.64%	199407	201812
LENZING AG	101434	2,213	0.00%	2.64%	7.23%	199002	201812
OBERBANK AG	015730	2,213	0.00%	2.64%	7.33%	199002	201812
EVN-ENERGIE-VERSORG NIEDEROS	103133	2,119	0.00%	2.53%	6.51%	199007	201812
BUWOG AG	317487	2,068	0.00%	2.47%	18.32%	201406	201812
OESTERREICH POST AG	277897	2,033	0.00%	2.42%	8.44%	200607	201812
BRAU-UNION AG	103070	1,396	0.00%	1.67%	13.98%	199112	200405
SCHOELLER BLECKMANN OILF EQP	214325	1,081	0.00%	1.29%	16.60%	200305	201812
CONSTANTIA PACKAGING AG	222148	1,070	0.00%	1.28%	14.71%	199506	201009
KTM AG	271495	1,019	0.00%	1.22%	12.82%	200501	201606
OESTERREICHISCHE BRAU-BETEIL	101485	1,014	0.00%	1.21%	12.13%	199002	200403
PALFINGER AG	230638	1,002	0.00%	1.20%	12.99%	199910	201812

Table A5: Wealth Creation by Country, Top 20 Belgium Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ANHEUSER-BUSCH INBEV	241637	112,146	0.17%	31.20%	12.10%	200101	201812
KBC GROUP NV	015703	27,161	0.04%	7.56%	9.72%	199002	201812
COLRUYT SA	101465	14,970	0.02%	4.17%	19.19%	199002	201812
GROUPE BRUXELLES LAMBERT	021408	13,789	0.02%	3.84%	7.95%	199008	201812
UCB SA-NV	100751	13,685	0.02%	3.81%	9.94%	199002	201812
SOLVAY SA	101394	11,115	0.02%	3.09%	8.42%	199002	201812
ALMANIJ NV	023899	10,452	0.02%	2.91%	10.19%	199107	200502
PROXIMUS SA	200384	9,269	0.01%	2.58%	7.63%	200404	201812
DELHAIZE GROUP - ETS DLHZ FR	100781	8,906	0.01%	2.48%	7.46%	199002	201607
CIE NATL A PORTEFEUILLE	204024	8,157	0.01%	2.27%	13.33%	199008	201105
SOFINA SA	024056	7,358	0.01%	2.05%	11.21%	199002	201812
TELENET GROUP HOLDING N.V.	270034	7,198	0.01%	2.00%	18.52%	200511	201812
GENERALE DE BANQUE	015592	6,564	0.01%	1.83%	17.75%	199002	199812
PETROFINA	101289	5,576	0.01%	1.55%	8.38%	199002	200012
POWERFIN	100754	5,430	0.01%	1.51%	16.42%	199002	200202
ACKERMANS & VAN HAAREN NV/SA	200034	5,226	0.01%	1.45%	11.15%	199008	201812
UMICORE SA	100773	5,043	0.01%	1.40%	5.13%	199003	201812
ROYALE BELGE GROUP	015635	4,431	0.01%	1.23%	13.36%	199002	199902
GROUP BRUXELLES LAMBERT -OLD	021412	4,175	0.01%	1.16%	12.28%	199002	200104
BANQUE BRUX LAMBERT	015573	4,131	0.01%	1.15%	17.71%	199002	199806

Table A6: Wealth Creation by Country, Top 20 Denmark Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
NOVO NORDISK A/S	008020	133,861	0.20%	31.82%	20.27%	199002	201812
A.P. MOLLER - MAERSK A/S	103292	22,517	0.03%	5.35%	6.55%	199002	201812
COLOPLAST A/S	101538	21,618	0.03%	5.14%	17.87%	199309	201812
DANSKE BANK AS	015552	20,711	0.03%	4.92%	9.12%	199002	201812
TDC A/S	030112	14,095	0.02%	3.35%	9.47%	199406	201805
VESTAS WIND SYSTEMS A/S	225094	13,714	0.02%	3.26%	13.30%	199812	201812
ORSTED A/S	232646	13,637	0.02%	3.24%	30.75%	201607	201812
DSV A/S	213029	13,121	0.02%	3.12%	22.12%	199701	201812
NOVOZYMES A/S	241336	12,731	0.02%	3.03%	17.72%	200012	201812
CHR.HANSEN HOLDINGS AS	294764	10,810	0.02%	2.57%	24.73%	201007	201812
KOBENHAVNS LUFTHAVNE	221130	8,580	0.01%	2.04%	15.86%	199409	201812
WILLIAM DEMANT	204544	8,402	0.01%	2.00%	17.36%	199512	201812
TRYG AS	274282	8,361	0.01%	1.99%	15.50%	200511	201812
H LUNDBECK A/S	232106	8,201	0.01%	1.95%	10.92%	199912	201812
GENMAB AS	245207	7,695	0.01%	1.83%	13.64%	200106	201812
D/S 1912 AB	103289	7,534	0.01%	1.79%	10.51%	199002	200306
TOPDANMARK AS	024826	6,554	0.01%	1.56%	11.32%	199002	201812
CARLSBERG A/S	101130	6,136	0.01%	1.46%	4.35%	199002	201812
GN STORE NORD A/S	101539	5,757	0.01%	1.37%	10.12%	199002	201812
AMBU A/S	221020	5,001	0.01%	1.19%	21.36%	199409	201812

Table A7: Wealth Creation by Country, Top 20 Finland Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
NOKIA CORP	023671	42,155	0.06%	13.81%	10.41%	199108	201812
FORTUM OYJ	225597	37,924	0.06%	12.42%	21.16%	199902	201812
SAMPO PLC	015773	36,566	0.05%	11.98%	14.57%	199002	201812
KONE OYJ	101023	29,338	0.04%	9.61%	18.48%	199002	201812
NESTE OYJ	272746	16,578	0.02%	5.43%	12.19%	200505	201812
UPM-KYMMENE CORP	101718	15,987	0.02%	5.24%	10.02%	199606	201812
WARTSILA OYJ ABP	101557	13,139	0.02%	4.30%	13.19%	199002	201812
STORA ENSO OYJ	101020	9,652	0.01%	3.16%	7.96%	199607	201812
METSO OYJ	102345	6,905	0.01%	2.26%	13.09%	199902	201812
ORION CORP	211453	6,324	0.01%	2.07%	9.97%	199512	201812
KESKO OYJ	101971	6,090	0.01%	1.99%	8.46%	199002	201812
NOKIAN TYRES OYJ	211452	5,502	0.01%	1.80%	19.90%	199512	201812
POHJOLA BANK PLC	021447	5,266	0.01%	1.72%	19.65%	199309	201409
AMER SPORTS CORP	101737	4,650	0.01%	1.52%	8.88%	199002	201812
POHJOLA GROUP	015608	4,540	0.01%	1.49%	14.37%	199002	200606
ELISA CORP	234087	4,080	0.01%	1.34%	4.79%	200002	201812
NESTE OY	220527	3,587	0.01%	1.17%	43.11%	199601	199808
FISKARS OY	101739	3,028	0.00%	0.99%	16.55%	199002	201812
HELSINGIN PUHELIN OYJ	216591	3,015	0.00%	0.99%	51.67%	199804	200006
HUHTAMAKI OYJ	101714	2,842	0.00%	0.93%	6.90%	199002	201812

Table A8: Wealth Creation by Country, Top 20 France Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
TOTAL SA	024625	173,932	0.26%	7.71%	9.95%	199002	201812
LVMH MOET HENNESSY LOUIS V	014447	159,051	0.24%	7.05%	10.76%	199002	201812
L'OREAL SA	100581	156,479	0.24%	6.93%	14.81%	199002	201812
ELF AQUITAINE SA	019364	119,695	0.18%	5.30%	14.31%	199002	201004
SANOFI	101204	100,714	0.15%	4.46%	8.08%	199002	201812
CHRISTIAN DIOR SE	201260	77,340	0.12%	3.43%	14.72%	199202	201812
KERING	222379	68,398	0.10%	3.03%	13.93%	199002	201812
HERMES INTERNATIONAL	203053	63,539	0.10%	2.81%	19.73%	199401	201812
AIRBUS SE	220833	61,907	0.09%	2.74%	8.87%	199907	201812
L'AIR LIQUIDE SA	101202	54,463	0.08%	2.41%	10.39%	199002	201812
SAFRAN SA	101305	48,181	0.07%	2.13%	15.29%	199002	201812
VINCI SA	102296	47,948	0.07%	2.12%	11.75%	199002	201812
DANONE	017452	45,028	0.07%	1.99%	8.51%	199002	201812
PERNOD RICARD SA	101396	41,705	0.06%	1.85%	11.22%	199002	201812
AVENTIS SA	013467	41,584	0.06%	1.84%	11.58%	199303	200509
BNP PARIBAS	015532	40,292	0.06%	1.79%	5.84%	199311	201812
AXA SA	063120	33,503	0.05%	1.48%	5.11%	199002	201812
DASSAULT SYSTEMS SA	063169	27,415	0.04%	1.21%	12.73%	199608	201812
ESSILORLUXOTTICA	101248	26,252	0.04%	1.16%	13.07%	199002	201812
ASSURANCE GEN DE FRANCE	024052	24,455	0.04%	1.08%	11.36%	199102	200706

Table A9: Wealth Creation by Country, Top 20 Germany Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
SAP SE	103487	118,382	0.18%	6.93%	12.59%	199002	201812
BASF SE	017436	103,640	0.16%	6.07%	12.25%	199002	201812
SIEMENS AG	019349	99,326	0.15%	5.82%	7.58%	199002	201812
VODAFONE AG	100181	87,884	0.13%	5.15%	20.60%	199002	200208
VOLKSWAGEN AG	100737	63,337	0.10%	3.71%	7.66%	199002	201812
BAYER MOTOREN WERKE AG	100022	61,775	0.09%	3.62%	11.17%	199002	201812
BAYER AG	100080	59,658	0.09%	3.49%	8.44%	199002	201812
ALLIANZ SE	015724	44,830	0.07%	2.63%	4.51%	199002	201812
HENKEL AG & CO KGAA	101942	44,122	0.07%	2.58%	10.37%	199002	201812
HOECHST AG	100049	43,854	0.07%	2.57%	13.12%	199002	200507
ADIDAS AG	221244	42,800	0.06%	2.51%	13.89%	199512	201812
E.ON SE	100590	41,912	0.06%	2.45%	8.05%	199002	201812
AUDI AG	101120	39,012	0.06%	2.28%	16.60%	199306	201812
MUNICH RE CO	015677	36,903	0.06%	2.16%	6.66%	199002	201812
LINDE AG	100037	34,115	0.05%	2.00%	8.43%	199002	201812
BAYER SCHERING PHARMA AG	101076	29,404	0.04%	1.72%	15.17%	199002	200809
DEUTSCHE BOERSE AG	243774	28,433	0.04%	1.66%	17.11%	200103	201812
BEIERSDORF AG	100083	27,663	0.04%	1.62%	12.47%	199002	201812
CONTINENTAL AG	100609	24,692	0.04%	1.45%	9.21%	199002	201812
DEUTSCHE POST AG	241456	24,502	0.04%	1.43%	6.21%	200012	201812

Table A10: Wealth Creation by Country, Top 20 Greece Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
COSMOTE MOBILE TELECOMMNICTN	140717	13,077	0.02%	25.78%	29.83%	200011	200803
OPAP SA	245672	9,750	0.01%	19.22%	32.56%	200106	201812
MOTOR OIL CORINTH REFINERIES	250580	3,409	0.01%	6.72%	16.40%	200204	201812
JUMBO SA	243198	2,173	0.00%	4.28%	18.89%	200103	201812
ERGOBANK S/A	202058	2,119	0.00%	4.18%	18.10%	199002	200009
TITAN CEMENT CO SA	209048	1,822	0.00%	3.59%	8.51%	199002	201812
GERMANOS SA	239598	1,217	0.00%	2.40%	15.52%	200010	200703
HELLENIC PETROLEUM SA	112296	1,082	0.00%	2.13%	3.79%	199808	201812
KARELIA TOBACCO CO INC SA	241494	967	0.00%	1.91%	15.38%	200012	201812
IONIAN BANK OF GREECE SA	203374	920	0.00%	1.81%	17.07%	199002	200004
BANK OF GREECE	200569	897	0.00%	1.77%	15.28%	199002	201812
NEOCHIMIKI IND AND COMMERCIA	257040	808	0.00%	1.59%	41.15%	200308	200809
XIOSBANK SA	213779	746	0.00%	1.47%	93.68%	199712	200006
METKA INDUSTRIAL CONSTRUCT	210537	700	0.00%	1.38%	12.81%	199412	201706
ALUMINIUM OF GREECE SA	200146	698	0.00%	1.38%	9.95%	199002	200709
FOLLI FOLLIE SA	223389	607	0.00%	1.20%	13.27%	199811	201012
ALPHA-BETA VASSILOPOULOS SA	212802	567	0.00%	1.12%	17.02%	199611	201007
NATIONAL MORTGAGE BNK GREECE	204039	548	0.00%	1.08%	11.57%	199002	199809
ALPHA FINANCE SA	212803	539	0.00%	1.06%	43.66%	199611	200104
ETHNIKI GENERAL INSURANCE CO	202077	483	0.00%	0.95%	10.21%	199002	200712

Table A11: Wealth Creation by Country, Top 20 Italy Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ENI SPA	061616	121,043	0.18%	18.55%	12.62%	199601	201812
TELECOM ITALIA SPA	019540	43,111	0.06%	6.61%	16.56%	199004	200307
TIM-TELECOM ITALIA MOBILE	214259	35,881	0.05%	5.50%	16.02%	199508	200506
TELECOM ITALIA SPA-OLD	101090	33,290	0.05%	5.10%	30.07%	199002	199707
ATLANTIA SPA	102743	30,612	0.05%	4.69%	21.92%	199011	201812
ENEL SPA	201794	30,365	0.05%	4.65%	4.48%	199912	201812
LUXOTTICA GROUP SPA	020196	24,332	0.04%	3.73%	9.88%	200104	201812
SNAM SPA	249457	24,088	0.04%	3.69%	16.00%	200201	201812
SAN PAOLO-IMI SPA	024589	17,913	0.03%	2.74%	8.76%	199205	200612
TENARIS SA	151933	16,859	0.03%	2.58%	20.38%	200302	201812
RAS HOLDINGS SPA	015786	13,803	0.02%	2.12%	10.14%	199002	200610
TERNA SPA	270451	13,226	0.02%	2.03%	14.42%	200408	201812
EXOR SPA	016331	9,810	0.01%	1.50%	19.02%	200904	201812
CAPITALIA SPA	025885	9,669	0.01%	1.48%	7.48%	199111	200709
DAVIDE CAMPARI SPA	246939	9,653	0.01%	1.48%	17.58%	200109	201812
RECORDATI SPA	102429	8,137	0.01%	1.25%	15.29%	199008	201812
ROLO BANCA 1473 SPA	030543	7,935	0.01%	1.22%	21.74%	199312	200206
BANCA CR FIRENZE	240716	7,805	0.01%	1.20%	32.43%	200008	200804
ISTITUTO MOBILIARE ITALIANO	029699	7,374	0.01%	1.13%	21.41%	199403	199810
BANCA FIDEURAM SPA	016397	6,902	0.01%	1.06%	13.37%	199008	200701

Table A12: Wealth Creation by Country, Top 20 Netherlands Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ROYAL DUTCH PETROLEUM NV	009267	142,543	0.21%	16.05%	12.94%	199002	200509
UNILEVER NV	010846	118,852	0.18%	13.38%	11.62%	199002	201812
ABN-AMRO HOLDINGS NV	015504	110,490	0.17%	12.44%	19.55%	199002	200804
ASML HOLDING NV	061214	69,004	0.10%	7.77%	18.49%	199504	201812
HEINEKEN NV	104833	54,230	0.08%	6.11%	14.03%	199002	201812
KONINKLIJKE PHILIPS NV	008546	47,693	0.07%	5.37%	10.07%	199002	201812
ING GROEP NV	015617	44,114	0.07%	4.97%	8.91%	199104	201812
KONINKLIJKE KPN NV	061440	28,273	0.04%	3.18%	11.03%	199407	201812
RELX NV	100075	27,828	0.04%	3.13%	11.28%	199002	201809
AKZO NOBEL NV	015334	25,159	0.04%	2.83%	9.61%	199002	201812
HEINEKEN HOLDING NV	206493	24,708	0.04%	2.78%	12.38%	199306	201812
KONINKLIJKE DSM NV	102454	19,827	0.03%	2.23%	12.24%	199002	201812
WOLTERS KLUWER NV	101361	18,666	0.03%	2.10%	10.56%	199002	201812
KONINKLIJKE AHOLD DELHAIZE	023667	12,878	0.02%	1.45%	4.45%	199002	201812
NUMICO (KONINKLIJKE) NV	100775	11,809	0.02%	1.33%	17.01%	199002	200712
RANDSTAD NV	104761	8,011	0.01%	0.90%	10.27%	199102	201812
NIELSEN HOLDINGS PLC	100873	6,580	0.01%	0.74%	12.52%	199002	200607
NN GROUP N.V.	260840	6,089	0.01%	0.69%	12.98%	201408	201812
VOPAK (KONINKLIJKE) NV	100783	5,970	0.01%	0.67%	9.59%	199002	201812
KONINKLIJKE VENDEX KBB	220375	5,449	0.01%	0.61%	41.03%	199508	200407

Table A13: Wealth Creation by Country, Top 20 Norway Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
EQUINOR ASA	220546	64,876	0.10%	20.50%	9.58%	200107	201812
NORSK HYDRO ASA	012383	47,224	0.07%	14.93%	13.56%	199002	201812
TELENOR ASA	221612	41,073	0.06%	12.98%	14.79%	200101	201812
ORKLA ASA	100913	14,533	0.02%	4.59%	13.84%	199002	201812
YARA INTERNATIONAL ASA	264351	13,703	0.02%	4.33%	19.70%	200404	201812
GJENSIDIGE FORSIKRING BA	296091	7,531	0.01%	2.38%	16.74%	201101	201812
FRONTLINE LTD-OLD	029290	6,833	0.01%	2.16%	36.51%	199802	201511
SCHIBSTED ASA	208224	6,496	0.01%	2.05%	11.69%	199311	201812
SALMAR ASA	284523	5,855	0.01%	1.85%	23.51%	200706	201812
LEROY SEAFOOD GROUP ASA	252666	4,617	0.01%	1.46%	22.33%	200207	201812
AKER ASA	270904	4,582	0.01%	1.45%	19.56%	200410	201812
AKASTOR ASA	270267	4,347	0.01%	1.37%	23.31%	200407	201812
TANDBERG AS	203881	3,833	0.01%	1.21%	39.31%	199011	201004
SUBSEA 7 INC	254138	3,582	0.01%	1.13%	35.44%	200211	201101
TOMRA SYSTEMS A/S	102252	3,341	0.01%	1.06%	12.64%	199002	201812
HAFSLUND ASA	223601	3,204	0.00%	1.01%	8.02%	199002	201707
TGS-NOPEC GEOPHYSICAL CO ASA	216613	2,874	0.00%	0.91%	12.86%	199804	201812
GJENSIDIGE NOR ASA	235896	2,756	0.00%	0.87%	23.21%	199409	200311
CHRISTIANIA BANK-KREDIT	016400	2,684	0.00%	0.85%	23.51%	199409	200101
P/F BAKKAFROST HOLDING	294271	2,409	0.00%	0.76%	30.48%	201005	201812

Table A14: Wealth Creation by Country, Top 20 Portugal Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
EDP ENERGIAS DE PORTUGAL SA	064910	8,173	0.01%	15.27%	4.75%	199707	201812
JERONIMO MARTINS SGPS SA	221102	7,897	0.01%	14.75%	9.49%	199409	201812
GALP ENERGIA SGPS SA	279448	7,436	0.01%	13.89%	7.08%	200611	201812
THE NAVIGATOR COMPANY SA	065439	4,480	0.01%	8.37%	11.19%	199602	201812
PHAROL SGPS SA	031882	4,177	0.01%	7.80%	6.01%	199507	201812
BANCO TOTTA & ACORES SA	200539	3,282	0.00%	6.13%	13.52%	199002	200409
MUNDIAL CONFIANCA SA	221225	2,441	0.00%	4.56%	38.94%	199409	200010
SEMAPA-SOC DE INVEST GESTAO	211952	2,193	0.00%	4.10%	14.21%	199602	201812
SONAE SGPS SA	208164	2,124	0.00%	3.97%	6.29%	199002	201812
BANCO PINTO & SOTTO MAIOR SA	211946	1,871	0.00%	3.50%	22.40%	199602	200011
ALTRI SGPS SA	271928	1,611	0.00%	3.01%	28.42%	200504	201812
CORTICEIRA AMORIM SGPS SA	222135	951	0.00%	1.78%	5.89%	199002	201812
BRISA-AUTO-ESTRADAS PORTUGAL	214839	803	0.00%	1.50%	5.08%	199801	201304
BANCO BPI SA	200671	675	0.00%	1.26%	3.87%	199002	201812
VODAFONE TELECEL COM PESSOAI	108468	503	0.00%	0.94%	7.69%	199701	200305
GESCARTAO SGPS SA	257971	476	0.00%	0.89%	46.43%	200309	200704
MOTA-ENGIL SGPS SA	106276	455	0.00%	0.85%	6.89%	199611	201812
INPARSA INDS E PARTICIPACOES	213493	369	0.00%	0.69%	36.01%	199704	199909
UNICER-UNIAO CERVEJARIA SA	209296	365	0.00%	0.68%	11.27%	199004	200005
SEGUROS TRANQUILIDADE SA	208285	342	0.00%	0.64%	17.86%	199602	200007

Table A15: Wealth Creation by Country, Top 20 Spain Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
IND DE DISENO TEXTIL SA	245663	89,053	0.13%	12.02%	16.09%	200106	201812
TELEFONICA SA	013683	79,842	0.12%	10.77%	10.24%	199002	201812
ENDESA SA	015321	72,974	0.11%	9.85%	14.73%	199002	201812
IBERDROLA SA	100957	50,338	0.08%	6.79%	10.41%	199002	201812
NATURGY ENERGY GROUP SA	220586	30,881	0.05%	4.17%	12.40%	199002	201812
ABERTIS INFRAESTRUCTURAS SA	102629	27,334	0.04%	3.69%	13.47%	199002	201807
AMADEUS IT GROUP SA	294508	26,900	0.04%	3.63%	22.04%	201006	201812
REPSOL SA	015319	26,843	0.04%	3.62%	7.38%	199002	201812
TELEFONICA MOVILES SA	141243	25,515	0.04%	3.44%	10.65%	200012	200607
BANCO SANTANDER SA	014140	22,185	0.03%	2.99%	3.83%	199002	201812
ALTADIS SA	101209	20,762	0.03%	2.80%	18.08%	199002	200802
ACTIVIDADES CONSTR Y SERVICI	222186	17,508	0.03%	2.36%	13.37%	199101	201812
RED ELECTRICA CORP SA	234117	15,911	0.02%	2.15%	18.91%	199908	201812
UNION FENOSA SA	101330	14,725	0.02%	1.99%	18.00%	199002	200908
BBVA	015181	13,648	0.02%	1.84%	4.20%	199002	201812
FERROVIAL SA	271217	13,566	0.02%	1.83%	10.81%	200412	201812
CEPSA-CIA ESPANOLA DE PETROL	100954	12,811	0.02%	1.73%	13.42%	199002	201108
AENA SME S A	319219	11,360	0.02%	1.53%	17.89%	201503	201812
GRIFOLS SA	277812	11,158	0.02%	1.51%	20.64%	200606	201812
ENAGAS SA	252465	9,195	0.01%	1.24%	18.00%	200208	201812

Table A16: Wealth Creation by Country, Top 20 Sweden Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ATLAS COPCO AB	017420	59,429	0.09%	7.12%	16.59%	199002	201812
VOLVO AB	011217	48,537	0.07%	5.81%	13.74%	199002	201812
HENNES & MAURITZ AB	102276	45,509	0.07%	5.45%	25.97%	199003	201812
NORDEA BANK ABP	214881	43,354	0.07%	5.19%	9.60%	199801	201812
INVESTOR AB	015699	39,551	0.06%	4.74%	13.11%	199209	201812
SVENSKA HANDELSBANKEN	015654	35,144	0.05%	4.21%	12.28%	199002	201812
SWEDBANK AB	024578	30,448	0.05%	3.65%	15.00%	199507	201812
SCANIA AB	062557	27,509	0.04%	3.30%	13.18%	199605	201405
SCA-SVENSKA CELLULOSA AB	012368	27,459	0.04%	3.29%	10.18%	199002	201812
SKANDINAVISKA ENSKILDA BANK	015671	24,780	0.04%	2.97%	9.02%	199002	201812
SANDVIK AB	019591	22,223	0.03%	2.66%	11.33%	199208	201812
TELEFONAKTIEBOLAGET LM ERICS	004439	22,199	0.03%	2.66%	6.02%	199002	201812
ASSA ABLOY AB	223492	18,887	0.03%	2.26%	12.99%	199501	201812
HEXAGON AB	103065	15,046	0.02%	1.80%	19.71%	199410	201812
ELECTROLUX AB	014620	14,201	0.02%	1.70%	10.37%	199002	201812
SWEDISH MATCH AB	100956	13,283	0.02%	1.59%	16.29%	199606	201812
ASTRA AB	030938	12,270	0.02%	1.47%	20.50%	199002	199904
ALFA LAVAL AB	251921	10,646	0.02%	1.28%	19.79%	200207	201812
INDUSTRIVARDEN AB	103036	10,282	0.02%	1.23%	10.79%	199002	201812
SKF AB	011749	9,952	0.01%	1.19%	8.76%	199002	201812

Table A17: Wealth Creation by Country, Top 20 Switzerland Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
NESTLE SA/AG	016603	354,068	0.53%	21.93%	12.88%	199002	201812
ROCHE HOLDING AG	025648	276,330	0.42%	17.11%	13.80%	199002	201812
NOVARTIS AG	101310	249,576	0.37%	15.46%	9.96%	199002	201812
SYNGENTA AG	241216	53,077	0.08%	3.29%	18.66%	200012	201801
CIE FINANCIERE RICHEMONT AG	102902	49,640	0.07%	3.07%	13.69%	199002	201812
ACTELION LTD	239717	32,262	0.05%	2.00%	20.22%	200005	201710
CIBA-GEIGY AG	100834	30,056	0.05%	1.86%	21.77%	199002	199612
SWISS RE LTD	063546	27,454	0.04%	1.70%	14.61%	201106	201812
ZURICH VERSICH (ZURICH GRP)	015674	27,303	0.04%	1.69%	23.70%	199002	199903
SCHWEIZERISCHER BANKVEREIN	015788	26,132	0.04%	1.62%	26.33%	199002	199806
ABB LTD	210418	25,062	0.04%	1.55%	4.87%	199002	201812
SGS LTD	101340	22,256	0.03%	1.38%	15.41%	199002	201812
GIVAUDAN SA	237419	22,032	0.03%	1.36%	14.67%	200007	201812
SWISSCOM AG	114927	19,237	0.03%	1.19%	5.55%	199811	201812
GEBERIT AG	228918	18,011	0.03%	1.12%	19.47%	199908	201812
SIKA AG	101350	17,393	0.03%	1.08%	16.10%	199002	201812
PARTNERS GROUP AG	275863	16,887	0.03%	1.05%	23.48%	200605	201812
SYNTHES INC WILMINGTON	228898	15,894	0.02%	0.98%	14.25%	199907	201206
SCHINDLER HOLDING AG	101343	15,790	0.02%	0.98%	15.15%	199002	201812
EMS-CHEMIE HOLDING AG	102283	15,458	0.02%	0.96%	16.37%	199002	201812

Table A18: Wealth Creation by Country, Top 20 United Kingdom Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
HSBC HLDGS PLC	015509	166,739	0.25%	4.48%	9.33%	199002	201812
BP PLC	002410	148,444	0.22%	3.99%	7.20%	199002	201812
ROYAL DUTCH SHELL PLC	012384	127,586	0.19%	3.43%	5.61%	200508	201812
ASTRAZENECA PLC	028272	122,018	0.18%	3.28%	10.52%	199307	201812
BRITISH AMER TOBACCO PLC	001932	120,144	0.18%	3.23%	13.44%	199002	201812
DIAGEO PLC	018636	104,718	0.16%	2.81%	9.97%	199002	201812
SHELL TRANSPORT AND TRADING	009655	94,248	0.14%	2.53%	12.59%	199002	200507
GLAXOSMITHKLINE PLC	005180	93,286	0.14%	2.51%	6.10%	199002	201812
RIO TINTO GROUP (GBR)	019565	88,784	0.13%	2.39%	11.43%	199002	201812
SABMILLER PLC	100472	88,377	0.13%	2.38%	14.68%	199904	201609
UNILEVER PLC	010845	87,091	0.13%	2.34%	11.59%	199002	201812
BG GROUP PLC	014087	68,835	0.10%	1.85%	11.74%	199002	201602
SMITHKLINE BEECHAM (UK) PLC	014261	67,695	0.10%	1.82%	22.32%	199002	200012
RECKITT BENCKISER GROUP PLC	221858	65,622	0.10%	1.76%	12.35%	199002	201812
ORANGE PLC	062489	58,138	0.09%	1.56%	88.12%	199605	200002
PRUDENTIAL PLC	015782	52,499	0.08%	1.41%	10.21%	199002	201812
NATIONAL GRID	211819	47,005	0.07%	1.26%	11.62%	199601	201812
ANGLO AMERICAN PLC	015508	42,543	0.06%	1.14%	10.59%	199204	201812
IMPERIAL BRANDS PLC	212773	41,902	0.06%	1.13%	15.15%	199611	201812
NATL WESTMINSTER BANK	013294	37,105	0.06%	1.00%	17.55%	199002	200004

Table A19: Wealth Creation by Country, Top 20 Australia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
COMMONWEALTH BANK AUSTRALIA	024512	125,736	0.19%	8.48%	16.05%	199111	201812
BHP GROUP LTD	013312	123,766	0.19%	8.35%	11.63%	199002	201812
WESTPAC BANKING	015362	84,518	0.13%	5.70%	12.35%	199002	201812
NATIONAL AUSTRALIA BK	014802	72,538	0.11%	4.89%	13.70%	199002	201812
CSL LTD	223003	68,908	0.10%	4.65%	26.04%	199409	201812
ANZ-AUSTRALIA & NEW ZEALD BK	015889	68,184	0.10%	4.60%	11.72%	199002	201812
RIO TINTO GROUP (AUS)	017535	48,473	0.07%	3.27%	12.60%	199002	201812
WOOLWORTHS GROUP LTD	100894	37,063	0.06%	2.50%	15.48%	199308	201812
WESFARMERS LTD	101601	36,254	0.05%	2.44%	13.35%	199002	201812
MACQUARIE GROUP LTD	212856	29,885	0.04%	2.02%	16.12%	199611	201812
WESTFIELD CORP	015932	28,802	0.04%	1.94%	13.34%	199002	201805
WOODSIDE PETROLEUM LTD	100712	25,158	0.04%	1.70%	12.63%	199002	201812
SYDNEY AIRPORT	252268	14,620	0.02%	0.99%	24.28%	200211	201812
COLES GROUP LTD - OLD	014977	14,250	0.02%	0.96%	12.05%	199002	200711
TRANSURBAN GROUP	212650	14,140	0.02%	0.95%	12.27%	199609	201812
AMCOR LTD	100243	14,063	0.02%	0.95%	9.95%	199002	201812
RINKER GROUP LTD	255798	13,798	0.02%	0.93%	54.13%	200305	200707
CIMIC GROUP LTD	101603	12,967	0.02%	0.87%	21.04%	199002	201812
INSURANCE AUSTRALIA GROUP	239116	12,847	0.02%	0.87%	12.40%	200009	201812
AXA ASIA PACIFIC HLDGS LTD	213112	12,789	0.02%	0.86%	16.23%	199702	201103

Table A20: Wealth Creation by Country, Top 20 Hong Kong Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
TENCENT HOLDINGS LTD	270615	377,356	0.57%	13.33%	50.10%	200407	201812
CHINA MOBILE LTD	065662	219,742	0.33%	7.76%	10.22%	199711	201812
CHINA CONSTR BANK CORP	274364	212,707	0.32%	7.51%	13.77%	200512	201812
CNOOC LTD	142540	92,787	0.14%	3.28%	20.25%	200104	201812
HUTCHISON WHAMPOA LTD	101784	80,999	0.12%	2.86%	16.03%	199002	201505
HANG SENG BANK LTD	015498	76,186	0.11%	2.69%	23.43%	199002	201812
AIA GROUP LTD	295735	71,307	0.11%	2.52%	15.40%	201012	201812
PING AN INSURANCE GROUP	160709	56,339	0.08%	1.99%	19.44%	200407	201812
SUN HUNG KAI PROPERTIES LTD	019421	49,243	0.07%	1.74%	13.09%	199002	201812
BOC HONG KONG HOLDINGS LTD	031648	47,765	0.07%	1.69%	14.14%	200208	201812
INDUSTRIAL & COMM BANKCHINA	279378	46,395	0.07%	1.64%	7.97%	200612	201812
CK HUTCHISON HOLDINGS LTD	101088	43,601	0.07%	1.54%	13.77%	199002	201812
HONG KONG & CHINA GAS CO LTD	018526	42,559	0.06%	1.50%	17.58%	199002	201812
CLP HOLDINGS LTD	100665	40,098	0.06%	1.42%	13.33%	199002	201812
CHINA EVERGRANDE GROUP	293139	39,462	0.06%	1.39%	23.94%	200912	201812
SANDS CHINA LTD	293425	38,969	0.06%	1.38%	23.40%	201001	201812
HONG KONG EXCH & CLEARING	240637	38,731	0.06%	1.37%	22.49%	200011	201812
POWER ASSETS HOLDINGS LTD	016590	31,570	0.05%	1.12%	17.38%	199002	201812
CHINA OVERSEAS LAND & INVEST	105036	29,941	0.04%	1.06%	13.01%	199306	201812
MTR CORP LTD	243011	29,416	0.04%	1.04%	9.52%	200103	201812

Table A21: Wealth Creation by Country, Top 20 Israel Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ICL-ISRAEL CHEMICALS LTD	220211	14,408	0.02%	10.00%	18.68%	199605	201812
BANK HAPOALIM B M	030579	9,354	0.01%	6.49%	12.05%	199503	201812
BANK LEUMI LE-ISRAEL BM	002018	8,899	0.01%	6.18%	10.15%	199602	201812
BEZEQ ISRAEL TELECOMMUNICATN	222070	7,865	0.01%	5.46%	12.35%	199502	201812
FRUTAROM INDUSTRIES	258676	5,879	0.01%	4.08%	26.05%	200311	201810
NICE LTD	061902	5,122	0.01%	3.56%	10.89%	199605	201812
MIZRAHI TEFAHOT BANK LTD	222765	3,945	0.01%	2.74%	15.03%	199811	201812
AZRIELI GROUP LTD	294788	3,685	0.01%	2.56%	11.27%	201007	201812
DELEK GROUP LTD	204409	3,036	0.00%	2.11%	16.01%	199502	201812
THE ISRAEL CORP LTD	220217	2,874	0.00%	2.00%	10.16%	199503	201812
DELEK ENERGI SYSTEM LTD	243107	2,475	0.00%	1.72%	16.15%	200103	201810
OSEM INVESTMENT LTD	204450	2,472	0.00%	1.72%	11.55%	199509	201604
ISRAMCO NEGEV 2 -LP	275644	2,106	0.00%	1.46%	27.16%	200604	201812
SHUFERSAL LTD	204459	1,999	0.00%	1.39%	12.66%	199502	201812
STRAUSS GROUP LTD	023290	1,991	0.00%	1.38%	9.83%	199812	201812
MELISRON LTD	244591	1,926	0.00%	1.34%	24.89%	200105	201812
DELEK AUTOMOTIVE SYSTEMS LTD	224603	1,822	0.00%	1.26%	35.25%	199812	201812
ISRAEL DISCOUNT BANK LTD	006203	1,755	0.00%	1.22%	5.51%	199605	201812
FIRST INTERNATL BANK ISRAEL	204417	1,587	0.00%	1.10%	8.40%	199502	201812
MAKHTESHIM AGAN INDUSTRIES	204440	1,531	0.00%	1.06%	9.48%	199812	201110

Table A22: Wealth Creation by Country, Top 20 Japan Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
TOYOTA MOTOR CORP	019661	178,467	0.27%	7.99%	6.95%	199002	201812
SOFTBANK GROUP CORP	204222	65,750	0.10%	2.94%	10.83%	199502	201812
KEYENCE CORP	103228	58,264	0.09%	2.61%	12.70%	199002	201812
FAST RETAILING CO LTD	212948	57,496	0.09%	2.57%	24.74%	199701	201812
HONDA MOTOR CO LTD	005691	48,755	0.07%	2.18%	7.53%	199002	201812
KDDI CORP	201584	43,607	0.07%	1.95%	6.03%	199311	201812
NTT DOCOMO INC	223911	39,095	0.06%	1.75%	3.81%	199811	201812
CANON INC	002721	38,601	0.06%	1.73%	7.11%	199002	201812
CENTRAL JAPAN RAILWAY CORP	212901	37,501	0.06%	1.68%	9.98%	199711	201812
JAPAN TOBACCO INC	223528	35,418	0.05%	1.58%	5.84%	199501	201812
KAO CORP	018715	34,285	0.05%	1.53%	8.08%	199002	201812
TOKIO MARINE HOLDINGS INC	015659	33,408	0.05%	1.49%	9.65%	200205	201812
NIDEC CORP	221821	32,065	0.05%	1.43%	14.45%	199409	201812
MURATA MANUFACTURING CO LTD	100226	28,716	0.04%	1.29%	8.90%	199002	201812
SHIN-ETSU CHEMICAL CO LTD	100697	27,947	0.04%	1.25%	7.66%	199002	201812
ORIENTAL LAND CO LTD	213193	27,775	0.04%	1.24%	8.38%	199701	201812
NINTENDO CO LTD	102450	26,707	0.04%	1.20%	5.93%	199002	201812
CHUGAI PHARMACEUTICAL CO LTD	100441	26,467	0.04%	1.18%	8.05%	199002	201812
HOYA CORP	100433	26,321	0.04%	1.18%	10.61%	199002	201812
DAIKIN INDUSTRIES LTD	101138	25,736	0.04%	1.15%	8.30%	199002	201812

Table A23: Wealth Creation by Country, Top 20 New Zealand Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
AUCKLAND INTL AIRPORT LTD	222761	7,826	0.01%	8.95%	25.17%	199811	201812
FISHER & PAYKEL HEALTHCARE	101256	5,830	0.01%	6.67%	14.62%	199002	201812
A2 MILK COMPANY LTD (THE)	275289	5,328	0.01%	6.09%	59.32%	200607	201812
MERIDIAN ENERGY LTD	270045	4,117	0.01%	4.71%	25.08%	201312	201812
RYMAN HEALTHCARE LTD	245274	4,049	0.01%	4.63%	29.24%	200106	201812
CONTACT ENERGY LTD	227916	3,611	0.01%	4.13%	11.07%	199906	201812
PORT OF TAURANGA LTD	105118	3,158	0.00%	3.61%	22.38%	199208	201812
AIR NEW ZEALAND LTD	102234	2,406	0.00%	2.75%	6.95%	199002	201812
MAINFREIGHT LTD	212481	2,385	0.00%	2.73%	21.27%	199608	201812
FLETCHER BUILDING LTD	062403	2,145	0.00%	2.45%	7.12%	199605	201812
SKYCITY ENTERTAINMENT GROUP	212167	2,094	0.00%	2.39%	12.29%	199603	201812
MERCURY NZ LTD	270043	2,027	0.00%	2.32%	13.07%	201306	201812
INFRATIL LTD	209828	1,925	0.00%	2.20%	15.84%	199510	201812
EBOS GROUP LTD	201739	1,638	0.00%	1.87%	19.48%	200105	201812
THE NEW ZEALAND REFINING CO	102496	1,594	0.00%	1.82%	47.45%	199003	201812
VECTOR LTD	268629	1,434	0.00%	1.64%	5.70%	200509	201812
INDEPENDENT NEWSPAPERS LTD	101743	1,405	0.00%	1.61%	12.76%	199002	200506
TRADE ME GROUP LTD	311319	1,309	0.00%	1.50%	15.56%	201201	201812
KIWI PROPERTY GROUP LTD	209837	1,183	0.00%	1.35%	10.19%	199510	201812
PORTS OF AUCKLAND LTD	208552	1,130	0.00%	1.29%	43.28%	199312	200507

Table A24: Wealth Creation by Country, Top 20 Singapore Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
JARDINE MATHESON HLDGS LTD	100424	49,249	0.07%	10.15%	11.87%	199502	201812
DBS GROUP HOLDINGS LTD	015743	41,296	0.06%	8.51%	8.90%	199002	201812
JARDINE STRATEGIC HLDGS LTD	016334	41,044	0.06%	8.46%	14.15%	199408	201812
UNITED OVERSEAS BANK LTD	015679	36,408	0.05%	7.50%	12.59%	199002	201812
OVERSEA CHINESE BANKING	015753	33,798	0.05%	6.96%	10.01%	199002	201812
HONGKONG LAND HOLDINGS LTD	210576	17,004	0.03%	3.50%	9.63%	199106	201812
JARDINE CYCLE & CARRIAGE LTD	100296	12,355	0.02%	2.55%	14.36%	199002	201812
ASIA PACIFIC BREWERIES LTD	100126	11,228	0.02%	2.31%	14.35%	199002	201301
THAI BEVERAGE PCL	277881	10,820	0.02%	2.23%	13.23%	200607	201812
SINGAPORE AIRLINES LTD	100713	10,801	0.02%	2.23%	7.27%	199002	201812
GREAT EASTERN HOLDINGS LTD	015565	10,663	0.02%	2.20%	14.11%	199002	201812
SINGAPORE TECHNOLOGIES ENGR	214882	10,360	0.02%	2.13%	12.17%	199802	201812
KEPPEL CORPORATION LTD	100076	10,298	0.02%	2.12%	8.39%	199002	201812
FRASER & NEAVE LTD	100340	8,701	0.01%	1.79%	10.52%	199002	201812
SINGAPORE TELECOMMUNICATIONS	208095	8,248	0.01%	1.70%	3.35%	199403	201812
SINGAPORE EXCHANGE LTD	241904	8,179	0.01%	1.69%	22.34%	200101	201812
SINGAPORE PRESS HOLDINGS LTD	100025	7,643	0.01%	1.58%	12.61%	199002	201812
CAPITALAND LTD	241537	6,986	0.01%	1.44%	6.73%	200012	201812
SATS LTD	242894	4,746	0.01%	0.98%	15.01%	200102	201812
GLOBAL LOGISTIC PROPERTIES	295620	4,712	0.01%	0.97%	6.48%	201011	201801

Table A25: Wealth Creation by Country, Top 20 South Korea Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
SAMSUNG ELECTRONICS CO LTD	104604	284,884	0.43%	28.81%	18.61%	199002	201812
POSCO	104598	29,579	0.04%	2.99%	14.85%	199208	201812
SK TELECOM CO LTD	203480	25,897	0.04%	2.62%	12.34%	199204	201812
HYUNDAI MOTOR CO LTD	104607	24,244	0.04%	2.45%	9.57%	199002	201812
LG CHEMICAL LTD	245036	23,003	0.03%	2.33%	22.44%	200106	201812
NAVER CORP	255244	20,322	0.03%	2.06%	28.98%	200304	201812
SK HYNIX INC	204049	19,756	0.03%	2.00%	5.54%	199707	201812
KEPCO-KOREA ELEC POWER CORP	030874	16,306	0.02%	1.65%	6.55%	199208	201812
KT&G CORP	234376	15,955	0.02%	1.61%	12.44%	200003	201812
LG HOUSEHOLD & HEALTHCARE	245697	15,861	0.02%	1.60%	25.37%	200106	201812
S-OIL CORP	223142	15,568	0.02%	1.57%	14.32%	199411	201812
SAMSUNG BIOLOGICS CO LTD	322757	14,022	0.02%	1.42%	59.21%	201612	201812
SAMSUNG FIRE & MARINE INS	208765	13,740	0.02%	1.39%	16.30%	199004	201812
SHINHAN FINANCIAL GROUP LTD	025714	13,522	0.02%	1.37%	8.74%	200102	201812
HYUNDAI MOBIS	203215	13,322	0.02%	1.35%	13.49%	199204	201812
SK HOLDINGS CO LTD	209610	11,500	0.02%	1.16%	9.25%	199002	201507
SK HOLDINGS CO LTD	293194	10,719	0.02%	1.08%	19.24%	200912	201812
LG CORP	104608	10,252	0.02%	1.04%	7.79%	199002	201812
WOORI FINANCIAL GROUP INC	252819	10,044	0.02%	1.02%	11.04%	200209	201812
HYUNDAI HEAVY INDS CO LTD	240136	9,957	0.01%	1.01%	8.29%	199909	201812

Table A26: Wealth Creation by Country, Top 20 Taiwan Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
TAIWAN SEMICONDUCTOR MFG CO	201395	217,222	0.33%	27.39%	16.12%	199502	201812
HON HAI PRECISION IND CO LTD	203177	45,954	0.07%	5.79%	16.66%	199312	201812
FORMOSA PETROCHEMI	259598	35,476	0.05%	4.47%	12.92%	200402	201812
FORMOSA CHEMICAL & FIBER	202268	21,284	0.03%	2.68%	7.59%	199002	201812
CHUNGHWA TELECOM LTD	241159	20,634	0.03%	2.60%	5.30%	200012	201812
FORMOSA PLASTICS CORP	202271	19,672	0.03%	2.48%	6.80%	199002	201812
NAN YA PLASTICS CORP	204012	19,032	0.03%	2.40%	6.82%	199002	201812
LARGAN PRECISION CO LTD	251937	15,083	0.02%	1.90%	24.75%	200207	201812
FUBON FINANCIAL HOLDING CO	249738	13,680	0.02%	1.73%	8.43%	200102	201812
TAIWAN MOBILE CO LTD	240656	13,191	0.02%	1.66%	9.12%	200011	201812
DELTA ELECTRONICS INC	210201	12,504	0.02%	1.58%	12.21%	199002	201812
UNI-PRESIDENT ENTERPRISE CO	208572	11,787	0.02%	1.49%	7.54%	199002	201812
PRESIDENT CHAIN STORE CORP	231256	11,017	0.02%	1.39%	10.95%	199709	201812
FAR EASTONE TELECOMMUNICTN	252295	10,192	0.02%	1.29%	14.35%	200208	201812
ASE TECHNOLOGY HLDNG CO	213412	9,613	0.01%	1.21%	13.22%	199212	201812
CTBC FINANCIAL HOLDING CO LT	201238	9,316	0.01%	1.17%	6.94%	199312	201812
MEDIA TEK INC	247202	8,900	0.01%	1.12%	9.54%	200109	201812
MEGA FINANCIAL HOLDING CO	213202	6,171	0.01%	0.78%	4.66%	199702	201812
SILICONWARE PRECISION INDS	208068	6,142	0.01%	0.77%	11.10%	199312	201804
CATCHER TECHNOLOGY CO LTD	247618	5,925	0.01%	0.75%	20.22%	200110	201812

Table A27: Wealth Creation by Country, Top 20 Argentina Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
YACIMIENTOS PETE FISCALES SA	028520	11,423	0.02%	35.04%	8.60%	199308	201812
PETROBRAS ARGENTINA SA	220474	6,695	0.01%	20.54%	24.47%	199002	201805
SIDERCA SAIC	208059	2,366	0.00%	7.26%	18.54%	199002	200303
ALUAR ALUMINIO ARGENTINA	200144	1,069	0.00%	3.28%	7.17%	200206	201812
ASTRA CIA ARGENTINA PETROLEO	200301	987	0.00%	3.03%	23.00%	199002	200105
GRUPO FINANCIERO GALICIA SA	137944	852	0.00%	2.61%	3.85%	200012	201812
BANCO MACRO SA	212969	688	0.00%	2.11%	42.66%	199701	200312
TERNIUM ARGENTINA SA	221633	663	0.00%	2.03%	3.54%	199710	201812
GRUPO FINANCIERO VALORES SOC	291533	647	0.00%	1.99%	468.25%	201506	201812
ESTABLECIMIENTO MODELO TERRAB	210211	549	0.00%	1.68%	111.62%	199002	199411
PAMPA ENERGIA SA	212950	529	0.00%	1.62%	3.60%	200607	201812
MOLINOS RIO PLATA	205247	506	0.00%	1.55%	9.12%	199002	201812
BGH SA	212939	466	0.00%	1.43%	7.88%	199710	200007
TRANSPORTADORA DE GAS SUR	030838	465	0.00%	1.43%	4.30%	199410	201812
RENAULT ARGENTINA SA	201296	443	0.00%	1.36%	19.35%	199002	200512
TELEFONICA DE ARGENTINA SA	029178	431	0.00%	1.32%	5.27%	199002	200912
BANCO PATAGONIA SA	279125	419	0.00%	1.29%	5.03%	200709	201812
IMPORT Y EXPORT PATAGONIA	203259	271	0.00%	0.83%	4.83%	199702	201812
LEDESMA SDAD ANONIMA AGRICOL	203579	252	0.00%	0.77%	8.63%	199002	201812
RIGOLLEAU	207107	236	0.00%	0.72%	17.43%	199002	201812

Table A28: Wealth Creation by Country, Top 20 Brazil Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
VALE SA	209382	69,218	0.10%	16.10%	18.81%	199309	201812
ITAU UNIBANCO HLDG SA	030580	49,896	0.07%	11.61%	18.70%	199309	201812
PETROLEO BRASILEIRO SA- PETR	222111	31,134	0.05%	7.24%	10.29%	199309	201812
AMBEV SA	201140	27,844	0.04%	6.48%	8.51%	199309	201812
SOUZA CRUZ SA	208201	18,080	0.03%	4.21%	15.22%	199309	201511
ARCELOR BRASIL SA	200690	17,733	0.03%	4.12%	31.93%	199401	200706
ITAUSA-INV ITAU SA	203401	15,439	0.02%	3.59%	18.06%	199309	201812
PARMALAT BRASIL IND ALIMENTO	213304	13,365	0.02%	3.11%	1089.97%	200908	201002
UNIBANCO UNIAO DE BANCO	061134	11,286	0.02%	2.63%	56.71%	200409	200903
CPFL ENERGIA SA	269005	10,774	0.02%	2.51%	22.84%	200411	201812
CCR SA	255993	8,693	0.01%	2.02%	43.36%	200305	201812
LOJAS RENNER SA	205679	8,169	0.01%	1.90%	28.88%	200508	201812
WEG SA	207375	7,072	0.01%	1.64%	10.12%	200704	201812
NATURA COSMETICOS SA	270509	6,806	0.01%	1.58%	20.26%	200408	201812
MAGAZINE LUIZA SA	297353	6,661	0.01%	1.55%	22.12%	201106	201812
LOCALIZA RENT A CAR SA	267371	5,440	0.01%	1.27%	28.60%	200506	201812
PORTO SEGURO SA	271208	4,811	0.01%	1.12%	17.90%	200412	201812
BANCO SANTANDER BRASIL	285313	4,182	0.01%	0.97%	2.14%	200908	201812
BRASKEM SA	200976	4,074	0.01%	0.95%	11.67%	200105	201812
IRB-BRASIL RESSEGU	325059	3,949	0.01%	0.92%	92.82%	201709	201812

Table A29: Wealth Creation by Country, Top 20 China Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
INDUSTRIAL & COMM BANKCHINA	279378	146,696	0.22%	6.55%	8.18%	200612	201812
KWEICHOW MOUTAI CO LTD	251321	115,773	0.17%	5.17%	34.64%	200205	201812
AGRICULTURAL BANK OF CHINA	269783	85,791	0.13%	3.83%	7.73%	201008	201812
CHINA MERCHANTS BANK CO LTD	254023	73,503	0.11%	3.28%	16.56%	200205	201812
CHINA PETROLEUM & CHEM CORP	140756	66,253	0.10%	2.96%	8.16%	200109	201812
BANK OF CHINA LTD	267461	59,968	0.09%	2.68%	6.08%	200608	201812
PING AN INSURANCE GROUP	160709	58,870	0.09%	2.63%	10.21%	200704	201812
CHINA CNR CORP LTD	293520	47,534	0.07%	2.12%	46.69%	201002	201504
SAIC MOTOR CORP LTD	243804	45,867	0.07%	2.05%	15.45%	200102	201812
CHINA YANGTZE POWER CO	259541	41,778	0.06%	1.87%	14.03%	200312	201812
CHINA VANKE CO LTD	208017	38,807	0.06%	1.73%	20.09%	199312	201812
GREE ELEC APPLLIANCES INC	259980	36,290	0.05%	1.62%	26.21%	200102	201812
HANGZHOU HIK-VISION DIGITAL	294594	30,829	0.05%	1.38%	25.41%	201009	201812
YIBIN WULIANGYE CO LTD	243683	28,336	0.04%	1.27%	14.00%	199806	201812
INDUSTRIAL BANK CO LTD	282483	28,216	0.04%	1.26%	8.92%	200703	201812
JIANGSU HENGRUI MEDICINE CO	271085	27,714	0.04%	1.24%	25.86%	200011	201812
SHANGHAI PUDONG DEV BANK CO	243829	25,849	0.04%	1.15%	7.51%	199912	201812
MIDEA GROUP CO LTD	316100	25,764	0.04%	1.15%	26.15%	201310	201812
CITIC SECURITIES CO LTD	257120	21,319	0.03%	0.95%	13.29%	200302	201812
FOSHAN HAITIAN FLAVOURING	317136	19,364	0.03%	0.86%	28.68%	201403	201812

Table A30: Wealth Creation by Country, Top 20 Colombia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ECOPETROL SA	287882	27,308	0.04%	27.86%	7.88%	200802	201812
BAVARIA SA	200623	7,983	0.01%	8.14%	36.19%	199006	200710
BANCOLOMBIA SA	061067	6,832	0.01%	6.97%	15.94%	199006	201812
BANCO DE BOGOTA	217719	6,468	0.01%	6.60%	17.87%	199003	201812
GRUPO AVAL ACCIONES VALORES	241388	6,013	0.01%	6.13%	18.82%	200205	201812
INTERCONEXION ELECTRICA SA	249448	4,764	0.01%	4.86%	15.73%	200201	201812
CEMENTOS ARGOS SA	201101	4,537	0.01%	4.63%	11.51%	199303	201812
GRUPO DE INVERSIONES SURAMER	203756	4,084	0.01%	4.17%	9.79%	199307	201812
PROMIGAS SA ESP	205898	3,754	0.01%	3.83%	14.71%	199710	201812
GRUPO NUTRESA SA	216320	3,018	0.00%	3.08%	10.44%	199804	201812
GRUPO ARGOS SA	201095	2,968	0.00%	3.03%	7.52%	199302	201812
BANCO DE OCCIDENTE	205914	2,065	0.00%	2.11%	9.93%	199712	201812
ISAGEN S.A. E.S.P.	287564	1,902	0.00%	1.94%	6.45%	200801	201703
CORP FIN COLUMBIANA SA	105163	1,592	0.00%	1.62%	18.26%	200410	201812
CEMENTOS DEL VALLE	237817	1,525	0.00%	1.56%	79.92%	200201	200512
BBVA COLUMBIA SA	200664	1,379	0.00%	1.41%	5.83%	199204	201812
GRUPO ENERGIA BOGOTA SA ESP	286567	1,308	0.00%	1.33%	3.04%	201004	201812
EMPRESA DE ENERGIA PACIFICO	294553	1,244	0.00%	1.27%	8.79%	201006	201812
CELSIA SA ESP	248838	1,058	0.00%	1.08%	20.99%	200111	201812
BANCO SANTANDER COLOMBIA SA	210595	996	0.00%	1.02%	10.30%	199405	201203

Table A31: Wealth Creation by Country, Top 20 India Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
TATA CONSULTANCY SVCS LTD	270885	106,013	0.16%	6.54%	19.87%	200409	201812
RELIANCE INDUSTRIES LTD	207074	93,543	0.14%	5.78%	12.84%	199002	201812
HDFC BANK LTD	144535	71,659	0.11%	4.42%	22.65%	199511	201812
HINDUSTAN UNILEVER LTD	203076	59,261	0.09%	3.66%	15.42%	199002	201812
ITC LTD	203402	55,320	0.08%	3.42%	21.49%	199002	201812
INFOSYS LTD	211009	50,154	0.08%	3.10%	31.08%	199307	201812
HOUSING DEV FINANCE CORP LTD	223055	44,472	0.07%	2.75%	21.42%	199102	201812
OIL & NATURAL GAS CORP LTD	208175	37,393	0.06%	2.31%	10.35%	199511	201812
MARUTI SUZUKI INDIA LTD	201805	32,098	0.05%	1.98%	25.05%	200308	201812
KOTAK MAHINDRA BANK LTD	223062	28,089	0.04%	1.73%	17.81%	199410	201812
STATE BANK OF INDIA	203666	27,825	0.04%	1.72%	13.41%	199102	201812
WIPRO LTD	204867	27,254	0.04%	1.68%	22.35%	199103	201812
ICICI BANK LTD	223148	26,878	0.04%	1.66%	11.05%	199404	201812
LARSEN & TOUBRO LTD	203560	26,650	0.04%	1.65%	11.64%	199002	201812
HINDUSTAN ZINC LTD	206319	23,157	0.03%	1.43%	19.34%	199501	201812
BAJAJ FINANCE LTD	284229	20,188	0.03%	1.25%	39.69%	200705	201812
ASIAN PAINTS LTD	200283	19,757	0.03%	1.22%	19.46%	199002	201812
INDIAN OIL CORP LTD	212777	18,904	0.03%	1.17%	7.95%	199703	201812
AXIS BANK LTD	252278	17,968	0.03%	1.11%	19.35%	200208	201812
BHARTI AIRTEL LTD	251118	16,137	0.02%	1.00%	15.95%	200205	201812

Table A32: Wealth Creation by Country, Top 20 Indonesia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
BANK CENTRAL ASIA TBK (PT)	242865	48,296	0.07%	15.48%	37.85%	200102	201812
HANJAYA MANDALA SAMPOERNA	222066	37,624	0.06%	12.06%	20.66%	199009	201812
PT BANK RAKYAT INDONESIA	157304	35,220	0.05%	11.29%	28.57%	200312	201812
ASTRA INTERNATIONAL TBK (PT)	222309	27,625	0.04%	8.86%	12.83%	199204	201812
BANK MANDIRI (PERSERO) TBK	257421	25,374	0.04%	8.13%	22.41%	200308	201812
TELEKOMUNIKASI INDONESIA	061548	23,364	0.04%	7.49%	7.25%	199512	201812
UNITED TRACTORS	209347	7,528	0.01%	2.41%	13.23%	199002	201812
CHANDRA ASRI PETROCHEMICAL	030531	5,708	0.01%	1.83%	23.37%	200807	201812
INDOFOOD CBP SUKSES MAKMUR	295540	5,580	0.01%	1.79%	12.77%	201011	201812
KALBE FARMA	202830	5,272	0.01%	1.69%	11.89%	199111	201812
PERUSAHAAN GAS NEGARA TBK	260321	5,264	0.01%	1.69%	19.30%	200403	201812
VALE INDONESIA TBK	203269	4,846	0.01%	1.55%	17.82%	199009	201812
PT BUKIT ASAM TBK	256137	4,498	0.01%	1.44%	32.40%	200403	201812
PT SEMEN INDONESIA (PERSERO)	208304	4,488	0.01%	1.44%	7.80%	199109	201812
UNILEVER INDONESIA	209301	4,275	0.01%	1.37%	14.65%	199002	201812
MAYORA INDAH (PT)	205061	4,253	0.01%	1.36%	28.74%	199206	201812
ELANG MAHKOTA TECH TBK	293743	3,160	0.00%	1.01%	30.37%	201002	201812
BAYAN RESOURCES TBK (PT)	289444	3,045	0.00%	0.98%	11.29%	200809	201812
MULTI BINTANG INDONESIA (PT)	205292	2,885	0.00%	0.92%	20.19%	199002	201812
BANK MEGA TBK (PT)	246144	2,669	0.00%	0.86%	26.27%	200107	201812

Table A33: Wealth Creation by Country, Top 20 Malaysia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
PUBLIC BANK BHD	015632	28,855	0.04%	8.53%	15.71%	199002	201812
MALAYAN BANKING BHD	015688	26,381	0.04%	7.80%	11.08%	199002	201812
DIGI.COM BHD	232085	13,547	0.02%	4.00%	19.68%	199912	201812
IOI CORP BHD	103912	10,988	0.02%	3.25%	14.63%	199106	201812
NESTLE (MALAYSIA) BHD	103756	10,085	0.02%	2.98%	13.45%	199007	201812
CIMB GROUP HOLDINGS BERHAD	200932	9,593	0.01%	2.84%	8.65%	199208	201812
MAXIS COMMUNICATIONS BHD	252198	8,933	0.01%	2.64%	29.96%	200208	200706
HONG LEONG BANK BHD	206381	8,663	0.01%	2.56%	8.64%	199508	201812
TELEKOM MALAYSIA BHD	104129	7,876	0.01%	2.33%	7.07%	199101	201812
PETRONAS DAGANGAN	206374	7,802	0.01%	2.31%	12.19%	199508	201812
KUALA LUMPUR KEPONG BHD	018753	7,735	0.01%	2.29%	11.17%	199002	201812
BRITISH AMER TOB (MALAYSIA)	100945	7,514	0.01%	2.22%	16.84%	199002	201812
PETRONAS CHEMICALS GROUP	295951	7,188	0.01%	2.12%	6.07%	201101	201812
HAP SENG CONSOLIDATED BHD	101453	6,747	0.01%	1.99%	15.71%	199002	201812
PPB GROUP BHD	019246	6,432	0.01%	1.90%	9.62%	199002	201812
MISC BERHAD	102138	6,241	0.01%	1.84%	6.77%	199002	201812
GENTING BHD	100050	5,719	0.01%	1.69%	8.80%	199002	201812
HONG LEONG FINANCIAL GP BHD	015921	5,525	0.01%	1.63%	11.61%	199002	201812
PLUS EXPRESSWAYS BHD	252354	4,980	0.01%	1.47%	12.17%	200208	201112
HARTALEGA HOLDINGS BHD	289024	4,886	0.01%	1.44%	46.61%	200807	201812

Table A34: Wealth Creation by Country, Top 20 Mexico Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
WAL MART DE MEXICO SA	105104	55,592	0.08%	14.44%	14.21%	199002	201812
TELMEX-TELEFONOS DE MEXICO	019625	49,871	0.07%	12.96%	17.07%	199007	201212
AMERICA MOVIL SA DE CV	142248	44,697	0.07%	11.61%	15.39%	200103	201812
AMERICA TELECOM SA DE CV	148549	30,903	0.05%	8.03%	67.95%	200206	200701
GRUPO MEXICO SAB DE CV	220509	21,771	0.03%	5.66%	12.73%	199102	201812
GRUPO CARSO SA DE CV	024650	19,107	0.03%	4.96%	11.17%	199207	201812
KIMBERLY-CLARK DE MEXICO SA	210701	11,483	0.02%	2.98%	13.52%	199002	201812
GRUPO FINANCIERO BANORTE SA	211934	11,385	0.02%	2.96%	14.66%	199507	201812
GRUPO ELEKTRA SA DE CV	061761	11,107	0.02%	2.89%	16.48%	199602	201812
EL PUERTO DE LIVERPOOL SA	220496	9,268	0.01%	2.41%	10.74%	199002	201812
GRUPO FINANCIERO INBURSA SA	202553	8,228	0.01%	2.14%	7.16%	199406	201812
INDUSTRIAS PENOLES SAB DE CV	105111	8,080	0.01%	2.10%	11.96%	199002	201812
BANCO SANTANDER MEXICO	243588	7,725	0.01%	2.01%	13.29%	200108	201812
ALFA SAB DE CV	221497	7,165	0.01%	1.86%	9.63%	199106	201812
GRUPO MODELO S.A.B. DE C.V.	204483	5,548	0.01%	1.44%	9.04%	199502	201509
GRUPO BIMBO SA DE CV	200752	5,508	0.01%	1.43%	5.72%	199403	201812
ARCA CONTINENTAL SAB DE CV	250684	5,323	0.01%	1.38%	7.65%	200204	201812
GRUPO AEROPORTUARIO SURESTE	140045	4,695	0.01%	1.22%	20.38%	200207	201812
MEXICHEM SAB DE CV	201012	4,218	0.01%	1.10%	13.50%	199711	201812
GRUPO AEROPORTUARIO DEL PACI	166385	3,871	0.01%	1.01%	12.89%	200604	201812

Table A35: Wealth Creation by Country, Top 20 Nigeria Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
NESTLE NIGERIA PLC	204059	4,004	0.01%	25.91%	18.52%	199402	201812
GUARANTY TRUST BANK PLC	243344	3,998	0.01%	25.87%	20.96%	200103	201812
NIGERIAN BREWERIES PLC	204096	2,154	0.00%	13.94%	9.49%	199404	201812
BENUE CEMENT CO PLC	200704	886	0.00%	5.73%	25.46%	199510	201010
ZENITH BANK PLC	274186	879	0.00%	5.68%	3.29%	200511	201812
NIGERIAN BOTTLING CO PLC	204095	368	0.00%	2.38%	6.90%	199402	201108
GUINNESS NIGERIA LTD	202681	367	0.00%	2.38%	9.51%	199403	201812
11 PLC	205235	336	0.00%	2.17%	10.96%	199406	201812
UNITY BANK PLC	282784	266	0.00%	1.72%	81.01%	200703	200902
JULIUS BERGER NIGERIA PLC	202805	230	0.00%	1.49%	15.93%	199505	201812
TOTAL NIGERIA PLC	245833	227	0.00%	1.47%	7.47%	200107	201812
7-UP BOTTLING CO	243316	188	0.00%	1.22%	10.39%	200103	201812
PRESCO PLC	275651	170	0.00%	1.10%	15.75%	200604	201812
ASHAKA CEMENT	200260	166	0.00%	1.07%	7.70%	199404	201812
PZ CUSSENS NIGERIA	208695	165	0.00%	1.07%	2.08%	199402	201812
OKOMU OIL PALM CO PLC	282694	139	0.00%	0.90%	8.48%	200703	201812
EQUITY ASSURANCE PLC	285678	127	0.00%	0.82%	51.11%	200709	200905
GLAXOSMITHKLINE CONSUM (NGA)	246176	114	0.00%	0.74%	16.48%	200009	201812
UAC OF NIGERIA PLC	209271	110	0.00%	0.71%	4.79%	199508	201812
CHEMICAL AND ALLIE	282642	88	0.00%	0.57%	14.69%	200703	201812

Table A36: Wealth Creation by Country, Top 20 Poland Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
UNICREDIT BANK AUSTRIA AG	257360	24,894	0.04%	17.65%	40.62%	200402	200803
KGHM POLSKA MIEDZ SA	205812	11,661	0.02%	8.27%	18.22%	199708	201812
POLSKI KONCERN NAFTOWY ORLEN	232743	11,220	0.02%	7.96%	10.36%	200001	201812
SANTANDER BANK POLSKA SA	247552	7,439	0.01%	5.27%	17.89%	200110	201812
POWSZECHNA KASA OSZCZEDNOSCI	271187	7,207	0.01%	5.11%	5.91%	200412	201812
POWSZECHNY ZAKLAD UBEZPIEC	269546	6,770	0.01%	4.80%	8.81%	201006	201812
BANK BPH S.A.	206518	6,749	0.01%	4.79%	23.68%	199509	201609
ING BANK SLASKI SA	206516	5,233	0.01%	3.71%	10.98%	199509	201812
MBANK SA	200773	4,239	0.01%	3.01%	12.90%	199308	201812
LPP SA	254244	3,932	0.01%	2.79%	27.67%	200212	201812
CD PROJEKT SA	250358	3,641	0.01%	2.58%	32.06%	200203	201812
BANK HANDLOWY W WARZAWIE SA	205687	3,500	0.01%	2.48%	9.43%	199708	201812
GRUPA LOTOS SA	273130	3,103	0.00%	2.20%	11.61%	200507	201812
SYNTHOS SA	271560	2,658	0.00%	1.88%	23.11%	200502	201802
BANK PLSK KASA OPK GRP PEKAO	220161	2,650	0.00%	1.88%	3.01%	199808	201812
AMREST HOLDINGS SE	272820	2,297	0.00%	1.63%	21.81%	200506	201812
BANK MILLENNIUM SA	208261	2,159	0.00%	1.53%	8.49%	199308	201812
GRUPA ZYWIEC SA	209641	2,064	0.00%	1.46%	10.23%	199307	201812
CCC SA	271438	2,025	0.00%	1.44%	25.23%	200501	201812
POLISH OIL AND GAS CO	268671	1,996	0.00%	1.42%	2.13%	200709	201812

Table A37: Wealth Creation by Country, Top 20 Russia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
SBERBANK OF RUSSIA OJSC	221682	60,566	0.09%	19.33%	16.14%	199511	201812
MMC NORILSK NICKEL PSJC	248384	52,012	0.08%	16.60%	22.43%	200112	201812
NOVATEK JSC	273716	38,012	0.06%	12.13%	15.56%	201003	201812
OIL CO LUKOIL PJSC	206457	34,565	0.05%	11.03%	7.55%	201002	201812
GAZPROM NEFT PJSC	213127	32,001	0.05%	10.21%	21.03%	199910	201812
TATNEFT PJSC	064181	28,336	0.04%	9.04%	22.93%	200201	201812
SEVERSTAL PJSC	206702	10,478	0.02%	3.34%	7.54%	200507	201812
ALROSA PJSC	268692	6,670	0.01%	2.13%	11.08%	201201	201812
POLYUS PJSC	030725	6,486	0.01%	2.07%	5.09%	200606	201812
WIMM BILL DANN FOODS	147202	3,834	0.01%	1.22%	93.52%	201003	201108
COMSTAR UNITED TELE SYS	275350	3,773	0.01%	1.20%	5.00%	200606	201103
BALTIKA BREWERY	212062	2,981	0.00%	0.95%	21.08%	201003	201209
ACRON PJSC	212059	2,810	0.00%	0.90%	16.29%	201003	201812
VSMPO-AVISMA CORP	259738	2,760	0.00%	0.88%	15.66%	201003	201812
POLYMETAL OJSC	282731	2,651	0.00%	0.85%	24.47%	201003	201206
SILVINIT JSC	278065	2,295	0.00%	0.73%	30.51%	201003	201105
MOSCOW CITY TELEPHONE NETWK	206463	2,268	0.00%	0.72%	20.25%	201003	201812
GAZPROM NEFTEKHIM SALAVAT JS	289282	2,015	0.00%	0.64%	37.07%	201005	201305
PHOSAGRO PJSC	298317	1,252	0.00%	0.40%	3.71%	201108	201812
BANK VTB NORTH-WEST JSC	288962	1,238	0.00%	0.40%	43.56%	200607	200805

Table A38: Wealth Creation by Country, Top 20 Saudi Arabia Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
SAUDI BASIC INDUSTRIES CORP	248963	139,705	0.21%	35.59%	26.14%	200111	201812
AL RAJHI BANK	248858	42,036	0.06%	10.71%	13.76%	200111	201812
SAUDI TELECOM CO	259013	28,702	0.04%	7.31%	5.79%	200312	201812
RIYAD BANK	248868	15,800	0.02%	4.02%	11.30%	200111	201812
SAMBA FINANCIAL GROUP	248869	13,702	0.02%	3.49%	7.62%	200112	201812
SAUDI BRITISH BANK	248870	13,579	0.02%	3.46%	12.39%	200111	201812
SAUDI ARABIAN FERTILIZERS CO	248962	13,248	0.02%	3.37%	18.50%	200111	201812
ALMARAI CO	274337	10,123	0.02%	2.58%	11.55%	200509	201812
NATIONAL COMMERCIAL BANK (THE)	268696	9,332	0.01%	2.38%	7.47%	201412	201812
BANQUE SAUDI FRANSI	249018	9,202	0.01%	2.34%	9.55%	200111	201812
ARAB NATIONAL BANK	248860	8,296	0.01%	2.11%	10.78%	200111	201812
YANBU NATIONAL PETROCHEMICAL	282679	8,212	0.01%	2.09%	11.51%	200702	201812
SAUDI ARABIAN MINING CO	289361	7,647	0.01%	1.95%	7.94%	200809	201812
JARIR MARKETING CO	275074	6,038	0.01%	1.54%	24.91%	200408	201812
JABAL OMAR DEVELOPMENT CO	289305	4,671	0.01%	1.19%	11.46%	201106	201812
SAVOLA GROUP	248878	4,488	0.01%	1.14%	14.99%	200111	201812
SAUDI CEMENT CO	248872	3,665	0.01%	0.93%	14.93%	200111	201812
ALAWWAL BANK	249023	3,622	0.01%	0.92%	8.40%	200111	201812
MAKKAH CONSTRUCT & DEVLPMT	248863	3,100	0.00%	0.79%	9.71%	200111	201812
ADVANCED PETROCHEMICAL CO	282410	3,082	0.00%	0.79%	19.69%	200702	201812

Table A39: Wealth Creation by Country, Top 20 South Africa Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
NASPERS LTD	206059	81,143	0.12%	16.62%	18.84%	199609	201812
FIRSTRAND LTD	200019	30,676	0.05%	6.28%	10.41%	199206	201812
SASOL LTD	100465	29,482	0.04%	6.04%	11.77%	199002	201812
STANDARD BANK GROUP LTD	015781	27,050	0.04%	5.54%	13.24%	199002	201812
MTN GROUP LTD	211655	20,519	0.03%	4.20%	12.76%	199512	201812
SANLAM-STH AFR NAT LIFE ASSR	225136	15,336	0.02%	3.14%	13.81%	199901	201812
REMGRO LTD	240519	13,974	0.02%	2.86%	20.52%	200011	201812
ABSA GROUP LTD	016305	13,212	0.02%	2.71%	10.51%	199002	201812
VODACOM GROUP LTD	292072	12,429	0.02%	2.55%	12.07%	200906	201812
RMB HOLDINGS LTD	210803	12,093	0.02%	2.48%	14.60%	199409	201812
BIDVEST GROUP LTD	209955	11,383	0.02%	2.33%	11.90%	199403	201812
EXXARO RESOURCES LTD	146202	10,879	0.02%	2.23%	50.83%	200112	201812
NEDBANK GROUP LTD	015618	10,369	0.02%	2.12%	9.26%	199002	201812
KUMBA IRON ORE LTD	281544	9,992	0.02%	2.05%	16.92%	200612	201812
TELKOM SA SOC LTD	152649	9,761	0.01%	2.00%	36.61%	200304	201812
CAPITEC BANK HOLDINGS LTD	251924	9,087	0.01%	1.86%	43.49%	200207	201812
SHOPRITE HLDGS LTD	211166	8,971	0.01%	1.84%	17.73%	199403	201812
DISCOVERY LTD	240228	6,529	0.01%	1.34%	13.64%	199911	201812
FIRST NATL BANK OF S AFRICA	015695	6,243	0.01%	1.28%	25.83%	199002	199805
MEDICLINIC INTERNATIONAL LTD	205081	5,533	0.01%	1.13%	15.61%	199002	201602

Table A40: Wealth Creation by Country, Top 20 Thailand Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
PTT PLC	249399	53,095	0.08%	12.94%	30.49%	200201	201812
AIRPORTS OF THAILAND PCL	260558	28,697	0.04%	6.99%	22.94%	200404	201812
ADVANCED INFO SERVICE PCL	104558	26,823	0.04%	6.54%	17.92%	199211	201812
CP ALL PCL	258648	20,908	0.03%	5.10%	30.57%	200311	201812
SIAM CEMENT PCL	104125	18,498	0.03%	4.51%	8.88%	199002	201812
PTT EXPLORATION & PRODUCTION	208623	15,374	0.02%	3.75%	13.58%	199307	201812
SIAM COMMERCIAL BANK	025608	14,846	0.02%	3.62%	13.69%	199002	201812
KASIKORNBANK PCL	027833	14,563	0.02%	3.55%	12.07%	199002	201812
BANGKOK BANK PCL	025609	13,257	0.02%	3.23%	11.07%	199002	201812
BANGKOK DUSIT MEDICAL SVCS	200774	11,974	0.02%	2.92%	21.14%	199211	201812
CENTRAL PATTANA PCL	206379	10,352	0.02%	2.52%	16.00%	199508	201812
INTOUCH HOLDINGS PCL	104092	9,222	0.01%	2.25%	15.89%	199010	201812
CHAROEN POKPHAND FOODS PCL	103951	7,063	0.01%	1.72%	14.43%	199002	201812
INDORAMA VENTURES PCL	293933	6,934	0.01%	1.69%	19.61%	201003	201812
HOME PRODUCT CENTER PCL	250060	6,616	0.01%	1.61%	34.41%	200202	201812
GLOW ENERGY PCL	272788	4,894	0.01%	1.19%	19.61%	200505	201812
BIG C SUPERCENTER PCL	207185	4,819	0.01%	1.17%	13.82%	199208	201709
KRUNG THAI BANK PCL	025607	4,791	0.01%	1.17%	5.18%	199002	201812
SIAM MAKRO PCL	204588	4,501	0.01%	1.10%	9.02%	199503	201812
BUMRUNGRAD HOSPITAL PCL	200865	4,352	0.01%	1.06%	16.57%	199003	201812

Table A41: Wealth Creation by Country, Top 20 Turkey Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
YAPI VE KREDI BANKASI AS	209571	9,396	0.01%	19.62%	27.03%	199004	200007
AKBANK TURK AS	109299	7,286	0.01%	15.21%	37.85%	199312	200005
BIM BIRLESIK MAGAZALAR	273498	5,533	0.01%	11.55%	23.73%	200508	201812
TURKIYE IS BANKASI AS	203784	5,490	0.01%	11.46%	13.40%	199004	201812
ENKA INSAAT VE SANAYI AS	210725	5,211	0.01%	10.88%	14.21%	199004	201812
TURK HAVA YOLLARI AO	209265	3,251	0.00%	6.79%	34.18%	199312	200101
TUPRAS-TURKIYE PETROL RAFINE	211469	2,082	0.00%	4.35%	26.68%	199512	200108
BRISA BRIDGESTONE SABANCI	228796	1,002	0.00%	2.09%	43.80%	199312	201812
HACI OMER SABANCI HOLDING AS	210437	996	0.00%	2.08%	13.80%	199805	200007
KOC HOLDING AS	203442	561	0.00%	1.17%	7.23%	199004	200205
VESTEL ELEKTRONIK SANAYI TIC	209405	514	0.00%	1.07%	27.24%	199209	200011
VESTEL BEYAZ ESYA SANAYI VE	278666	504	0.00%	1.05%	13.65%	200609	201812
ANADOLU EFES BIRACILIK MALT	202055	486	0.00%	1.01%	3.41%	199312	201812
FORD OTOMOTIV SANAYI AS	220399	400	0.00%	0.84%	11.02%	199004	200108
PETKIM PETROKIMYA HLDG AS	208460	387	0.00%	0.81%	8.61%	199312	200105
ASELSAN AS	200258	384	0.00%	0.80%	20.26%	199312	200101
TURKIYE GARANTI BANKASI AS	203783	362	0.00%	0.76%	7.67%	199312	199905
AKSIGORTA AS	213019	318	0.00%	0.66%	36.95%	199701	200102
HURRIYET GAZETECILIK MATBAAC	203201	310	0.00%	0.65%	28.48%	199403	199905
TANSAS IZMIR BUYUKSEHIR BELE	215864	308	0.00%	0.64%	46.73%	199803	200101

Table A42: Wealth Creation by Country, Top 20 United Arab Emirates Firms

Firm Name	GVKEY	Wealth Created (\$Millions)	% of Global Gross Wealth Creation	% of National Gross Wealth Creation	Annualized Dollar Weighted Return	First Month	Last Month
ETISALAT GROUP	274234	54,150	0.08%	35.80%	17.66%	200303	201812
FIRST ABU DHABI BANK P.J.S.C	282949	25,035	0.04%	16.55%	14.82%	200603	201812
NATIONAL BANK OF ABU DHABI	251249	16,177	0.02%	10.70%	17.20%	200206	201703
ABU DHABI COMMERCIAL BANK	251138	13,359	0.02%	8.83%	14.56%	200205	201812
EMIRATES ISLAMIC BANK	283434	11,860	0.02%	7.84%	39.68%	200906	201812
EMAAR PROPERTIES PJSC	251248	7,433	0.01%	4.91%	8.77%	200205	201812
DUBAI ISLAMIC BANK LTD	251246	4,693	0.01%	3.10%	6.59%	200205	201812
ABU DHABI ISLAMIC BANK	251139	3,940	0.01%	2.61%	13.32%	200106	201812
DUBAI INVESTMENTS	274891	2,116	0.00%	1.40%	16.82%	200106	201812
NATL BANK OF RAS AL KHAIMAH	284899	1,963	0.00%	1.30%	9.61%	200602	201812
EMIRATE INTEGRATED TELECOM	279381	1,907	0.00%	1.26%	3.03%	200606	201812
WAHA CAPITAL PJSC	251233	1,293	0.00%	0.85%	15.52%	200106	201812
DAMAC PROPS DUBAI	319007	1,050	0.00%	0.69%	11.92%	201502	201812
NATL BANK OF UMM AL-QAIWAIN	284188	859	0.00%	0.57%	5.98%	200602	201812
AIR ARABIA PJSC	285482	842	0.00%	0.56%	5.83%	200708	201812
NATIONAL BANK OF FUJAIRAH	284425	690	0.00%	0.46%	3.94%	200705	201812
GULF PHARMACEUTICALS	282886	568	0.00%	0.38%	9.77%	200305	201812
AL WATHBA NAT INSURANCE CO	284859	562	0.00%	0.37%	16.05%	200707	201812
ARAMEX PJSC	274887	517	0.00%	0.34%	3.46%	200601	201812
AGTHIA GROUP PJSC	284156	395	0.00%	0.26%	5.57%	200602	201812